

So, good morning, welcome to the lecture on Math 2 for a very selected audience. What we're going to do is we continue our topic on integration, and we learn a few more techniques, how to solve not only double, but also triple integrals, and how to do change of variables, exactly. Last time on Thursday, we looked at double integrals, and I showed you that it is possible if it is useful to change your coordinate system from the Cartesian coordinates to the polar coordinates, so that your infinitesimal area element, DA , is not written as $DX, D-Y$, but in polar coordinates as $R, D-R-D-Teta$, and this R was an important factor. And I'm pretty sure that you remember. But the point is that this can be done in much more generality for general coordinate transformations. And here I assume that I have a transformation from the $x-y$ coordinates, the normal coordinates, to new coordinates, and I call them u and v . So what I think is that x is a function of the new coordinate u and v , and y is the same. And what you now have to think about is that you look here in the UV plane, and you are in a certain point $U-ZO-V-0$, and you think that you consider this part of the UV plane, a sort of partitioning in small rectangles. And now the point is what happens to these rectangles if you now look, if you apply these mappings, if you compute the corresponding X and Y coordinates. Now, this set here, this rectangle in UV space, then transforms to an other object in the X, Y space. Now, in this picture, it's like this. And we've also tried to, so this partitioning in rectangle is meant to this partitioning. So you do not get really rectangles, but you get, they are a little bit deformed. Now, the point is, if you now want to know what the corresponding correction term is, if you transform from X, Y coordinates to UV coordinates, then you have to think, okay, this such a rectangle here, that is $DUDV$, is mapped to this one, but because it's deformed, it's not exactly the UDV or the X, DY anymore. But there is a factor in between. Now, you can determine that factor by doing partial differentiation, because in that point UV , you are going to look, if you take a step in the U direction, how does your vector in the X, Y change? So that may be like this. And if you take a little step in the V direction, it might change like this. So that rectangle then becomes a sort of this shape you get something like infinitesimal like this and you have to compute then so a rectangle in UV is mapped to this and this $DX DY$ is then should be replaced by this area now you know what that area is because that you've done in linear algebra for such a shape a parallelogram you have that the area is the absolute value of the determinant of these two vectors that built this parallel graph.

Now, here that is written in this form, this DA , so the DX, DY that we usually have, if you have to transform it in UV coordinates, then you get here the UDV with this correction term, and that is the Jacobian, if you differentiate the X, Y , and the X, Y , and the X , partially to uv but I will I will show you in a minute you have to take a determinant that's because of that parallelogram and the absolute value and then you get this factor that is in between the area of such a rectangle and of such a parallelogram now I uh want to do that first in an example for the polar corners that you see what's happening um so let's first look at the situation we already know. There we have X , depending on R and $theta$, is $R \cos \theta$, and Y , depending on R and $theta$ is $R \sin \theta$.

Now, I consider this as X and Y as functions of R and $theta$, and then you have to know what the corresponding, what this notation means, and that is just the following Jacobian. You take the partial derivative of X with respect to, now here I have to do with R and $Teta$, with respect to R , the first, then the DX with partial derivative with respect to $Teta$, and then you do the same for the Y 's. So in the world you get every time the same X, Y, Z in later one variable, and then you get a partial derivative with respect to the first variable, R , and then with respect to the $theta$. In our situation, this is, the x_{DR} is $\cos \theta$, the x_{a-teta} , the x_{a-teta} is $-\sin \theta$, the y_{d-r} is $\sin \theta$, and if you differentiate this with respect to $theta$, you get, is it still, yeah, no. And if I compute the determinant of this, this is the Jacobian matrix, you get that this determinant, this determinant, is a Jacobian matrix, is equal to $r \cos^2 \theta + \sin^2 \theta$, because of this minus here, $R \sin^2 \theta$, is equal to R . And that means that in this case, we have in general that DX, DY , should be replaced by the absolute value of this determinant of the corresponding Jacobian, because that is the area of this

parallelogram times the UDV. And in our situation of polar quorum, you get the usual formula back. You get that DX , D , Y is the absolute value of this R , that's just R , because R is assumed to be positive, R , the R de Teta. You get exactly the formula of last week back, and that should be the case, but this is the official general way how you do it. Last time I did it with this picture, and this is the official way, because it really describes how this rectangle and this parallelogram are related now the nice thing is that if you know the general formula you can also do more general things so for example if you want to compute you want to solve this using a quarter transformation to find the area of an ellipse so x squared over a squared plus y squared over b squared is smaller equal to one how do you find that formula. In this case, you think, okay, an ellipse is something, looks a little bit like a circle, but it's a little bit deformed. So, you, A and B are fixed. I think, for example, A is 2 and B is 3.

And then what you do is you say, you have new coordinates, X , depending on R and θ .

I do not take the normal polar coordinates, but I take this. A times R times $\cos \theta$ and y depending on r and θ is b times r times the sign of θ if you look at the domain this thing so the domain the area of the ellipse is described by all x and y in r^2 such that x squared over a squared plus y squared over b squared is small equal to one now what I now have is I want to describe it in terms of R and θ . So it consists of the points $R \cos \theta$, $R \sin \theta$. And, no, in a different way. I'll not say that it is equal. If you now look at how it is described in these coordinates, you plug this into there and the formula for y into there. And you see that you get A squared times R squared $\cos^2 \theta$. and here you get b squared r squared $\sin^2 \theta$ this a square over a squared and b squared over b squared they drop out and you just get the equation r squared is small equal to one so I have a new domain d prime now in my new coordinates and this consists of all r and θ such that now θ is completely free because there is no, so θ is between 0 and 2π , and R should be smaller than 1, and of course larger than 0. So I get now a square or now a rectangle in our θ coordinates that describes that same ellipse. What I now have to do, question, you know R squared a small equal to 1, but R is assumed to be a distance, and therefore distance to the origin and for that reason it's always positive or non-negative. It can be zero if you are in the origin. Keep that in mind by the way, that's an important point that you always know that in these polar coordinates it's a distance and that it cannot be negative because you use that information quite often. Now in this case we are going to compute this Jacobian and that is in this case it's almost the same as the previous time. First we differentiate with respect to r , so I get a $\cos \theta$ and minus a $r \sin \theta$. And here I get $b \sin \theta$, and here I get $br \cos \theta$. And the determinant will be, you see that already, A times b times $r \cos^2 \theta$ plus $\sin^2 \theta$, is a times b times b , plus $\sin^2 \theta$, is a times b . times r . A and B are just fixed numbers. R is this new variable. A and B is just two and three or something like that.

That means now that if I want to have this area, what you want to is to integrate over D , the one function, DX , the Y , because it's just an area.

Now, then I have to replace the one function in terms of R and θ . That's particularly easy. That's still the one function. So this is equal to, and now I can write it in terms of D -prime, and then I get one times but this DX , D - D - Y , is now replaced by A - B times R D - D - $Tetor$.

I'm computing the area of that ellipse. So I'm thinking of an ellipse, and then the height of the function is equal to one, and then the volume that I'm computing is simply equal to the area. So it's a special case of that we had before. Now, in this case, you get A and B are just constants, so I get A , B , I can take out of my integral. R is not a constant, that's a variable. Then I have to integrate my θ from 0 to 2π , and I have to integrate my R from 0 to 1, but of course I have here R the R . Now, the primitive here will be a half R squared with 1 and 0. So I get A times B times, and here I get 2π , and here I

get a half. So this is simply π times A times B . And this is exactly what you expect. Think, for example, if the ellipse would have been a circle, if B is equal to A , then it's just a radius, then you get a normal BA squared. But for an ellipse, you do it like this, and if you do this, this corner transformation. It is quite straightforward. Okay, now I had here an other example, but I want to continue because I think the other things are more important.

Okay, so I skip this example.

I want to go to triple integrals, and I also come back to your situation. So until now, we've looked double integrals, so you have an area in the two-dimensional of space in the XY plane and above that lives a function and we were computing the volume below that graph. What we now want to do is to have, we have a solid so a three-dimensional object in three-dimensional space. Of course we're also interested in the volume of that. But for example, think of it as now here to pick it to Earth. The density will be depending on the radius, the distance from the center of the Earth. And And if you would be able to have that function, you would be interested in the mass. So you have objects where the mass density is not a constant, but it's depending on where you are. And then still you want to compute a complete mass. Now, what you then would do is the same idea as we did previously for these double integrals. You would take that object, divide it into very small cubes. For every cube, you would say, okay, the mass density. in that little cube is almost constant, so I take that value, I multiply that with the volume of the cube, so it's DX, D, D, X, Y, DZ now in this case, and times, of course, this local mass density, so that volume times that mass density will give the mass of that little cube, and then I have to add up all these things.

In that way, you get exactly the same idea as we had for these double integrals, but now for triple integrals. And in that way, you can compute also volumes and masses of solids in three-dimensional space and also the center of mass, for example, of an object. That's all possible, with help of these triple integrals. And in a way, when you know how this works for these double integrals, these triple integrals are not reasonable. that much more difficult, but you only have one additional integral because you have not to work over two dimensions, but over three dimensions.

Now, they are written like this.

This is what I already told. If you have a mass density, so that row, depending on X, Y and Z , the mass of the solid will be the triple integral of this function, whoa. If you just want to compute the volume of the solid, then we had to get the same thing as we had before, we simply assume that the mass density is everywhere one, and then the mass is equal to the volume. So this is an alternative way to compute the volume of such a domain that you consider it as a triple integral.

And if you can describe your solid in a nice way, for example, that x is always between fix those A and B , Y is between bounds only depending on X and set is depending on bounce only depending on X and Y .

Then you can write as triple integral as an iterated integral, so as a repeated integral in the same way as we did for double integration, but now with three instead of two variables.

Let's first do a simple example. And that is full. Oh, no, this is full. We look at a tetrahedron, very simple one. It's not about that it is difficult to compute this, but it's about the idea how this works. We think of three-dimensional space, and we have here the points $100, 010$, and $0-0-0-0-1$. And now we look at the theta-hedral, So with four planes. This one is a triangle in the XY plane. So in the XY plane, Z is equal to zero. Here you have this triangle in the plane Y is equal to zero. And then you have also here a background. And then here you have this plane. So that is on this front triangle that is living in the plane X plus Y plus Z is equal to 1. I know that for sure because it contains these

points 1-0-0-0-0-0-0-1. So it has to be this plane. Now, what I want to do is I want to compute the volume of this object.

And then I have to describe that as an iterated integral. And what I now think is, okay, I choose a certain order. It doesn't matter that much because I have lots of options. But you can think of it if I project everything here of this tetrahedron down to the xy plane so every point is projected down then I have a sort of a ground plane in the xy plane and that is this triangle here from it's one zero and zero one I have this and then my other plane is if the x x is coming out here that is like this now if I now want to compute that volume I say okay let's do this the following way my x runs from zero to one for fixed values of x my y once here from zero to yeah what is this this is the line y is equal to one minus x so to one minus x then i can get every point here in this bottom of or tetrahedron if i'm in one such a point or at the bottom then how is my z evolving yeah okay from that bottom from zero to a point in that plane z is x plus y plus z is equal to one so my z once from zero two and then i get here z is one minus x minus y what you now see is so this will be the volume like the one i will also do the mass but if you have a mass density an example but in the this case if you only want to know the volume you get the following thing i have to integrate here the one function over z so i get set so i get the upper bound minus the lower bound so this is the integral from zero to one d x and then here this last integral the outcome is simply one minus x minus y and now my z is completely gone when we would do this same volume computation with the method of last week we would have written down this expression immediately because you've said okay we have this bottom and we have a function on it which function the function f x is one minus x minus y and we compute the volume under the graph so i can express that both as in double integral but also as a triple integral but then i'm integrating the one function but i'm taking these bounds into account there is difference between that but it's only a different way of looking at it now this is now not very difficult to do I get here you can one I have to integrate you with respect to y so 1 minus x is a constant so I get here 1 minus x y minus a half y squared as the antiderivative with upper bound y is 1 minus x lower bound y is 0 the x and that means that i get the integral from 0 to 1 i get here 1 minus x squared minus a half times 1 minus x squared and for y equal to 0 i get here 0 and here 0 so then i subtract 0 that's pretty easy so what i have to do is i have to integrate from 0 to 1 a half times 1 minus x squared the x and that is not difficult that is something with 1 minus x to the power 3 then here minus 1 over 6 in front of it upper bound 1 lower bound 0 and then i get for x 1 i get 0 minus 0 minus 1 over 6 so i get 1 over 6 be careful if you get a volume it should be possible So if you know that you have a physical object, I once had a mechanical engineering student who computed in a test a mass of minus 8 kilograms, and then you certainly make a mistake somewhere with a minus sign. That's impossible. But here it works like this. Of course, there are many other different ways how you can do the same thing. If you say, okay, I do not like this bottom in the XY plane, but I like this to be the basis of the take. so the plane x is equal to 0 that last plane and then you do the same thing in a different way then you say y runs from 1 to 0 and then the z runs from 0 to 1 minus i and then for fixed z and y the x has to go from the 0 to this plane and that but x is there 1 minus y minus z you get this and if you would compute that then you would get the same result and you see that immediately because you have played a little bit around with the x and the ys but nothing really changes now i'm going to change something because i'm now going to compute a mass and i assume that the mass density is now equal to the function y so i put here now instead of the one function I replace that with the Y. Is that getting more difficult? No, not really because this Y is independent of X, so I can put it here. It's also independent of Z. So I can put it here. So I get the integral from 0 to 1, in this case, Y, DY, and then the integral from 0 to 1 minus y. And here I then have the 1 function, so 1 minus y, Z Z.

And then you can also continue. Then you say, okay, integral with respect to Z.

So this is the Y.

integral from 0 to 1. And then here I have 1 minus y, z minus a half z squared.

From z is 1 minus y .

From 0 to 1 minus y , that is 0 . I do the same thing. I get here a half times 1 minus y squared. So in the end, I get the integral from 0 to 1 of y times, uh, y times a half times 1 minus y squared dy . Then you have to compute this, which is not really very difficult, and I leave that to you. So in case of Cartesian coordinates, everything seems doable in this situation. But of course, this is a very easy situation. Um, what's name? Okay, other example. A more difficult one where you do not have that much of a picture. If I have to do these exercises, I always want to make sketches how this domain looks like to get an idea what I have to do. But sometimes you get questions where that is not so obvious. And then you sometimes also need the algebra to get your equations right. But that depends a little bit on the situation. Now, this is a typical example of the type of exercises that you have to do. you have to compute the volume of the region, the solid, under the plane Z is 3 minus $2Y$, and above the paraboloid z is x squared plus y squared. So to get some idea how this looks like, you have here a paraboloid. We have here this parabola that is around the, z -axis and then you have a plane set as 3 minus y something like uh like this and now you have to you have that volume that is below that plane and above that paraboloid and i want to have that volume so this then this cut off and then only that that part how to do that looks a little bit difficult but it's it is not not that difficult if you now use the algebra what is now the how is this described i need, so the condition is that if I look at, I'm standing on the xy plane and I'm looking where are points of my solid? Because not everywhere on the xy plane, there will be corresponding values of z .

They will only be there if x squared plus y squared, that's the z coordinate of this one, is smaller equal than three minus two y . you have to be above this white paraboloid and below the yellow plane. So there have to be z -in-between, and for that reason, this has to be smaller equal to that. Now, if you then look what this is, then you see that this is x squared plus y -square plus $2y$ is smaller equal to 3 . And if you see such a thing, then you immediately have to think about completion of the squares. this is x squared plus y plus one squared that's y squared plus two y plus one minus one smaller equal to three but then i can say quite easily i bring the minus one to the other side so this is smaller equal to four now we see that we have a circle the center is the point zero minus one radius is two so if i draw that Here's my x , my y plane. I have here minus 1 . I get such a circle, and it's above that circle. I have points of my solid. Okay, then it's not that difficult anymore to describe what this is. Of course, this is for x is equal to 2 and minus 2 . So you say that volume is, I have x running from minus 2 to 2 , from here to here. the x for fixed values of x my y once from the lower part of the circle to the upper part of the circle so for this value of x my y is here and of course i get here that y plus one squared is smaller equal than four minus x squared so y plus one should be between minus the square of 4 minus x squared and plus 4 minus x squared and if you then add minus 1 on both sides you get that this is the integral from minus 1 minus the square root of 4 minus x squared to minus 1 plus the square root of 4 minus x squared and then i get dy because that's how the y is changing and then I get an integral the last one over Z . So I'm now here in a point over here and my solid is here pointing into the womb. Now there it's easy to find the lower and the upper bound because it has to run from x -crat plus y spread to upper bound 3 minus $2y$ and you get this as a volume.

And of course the computation will be the problem here it is doable but This last integral is easy because the primitive is of course function z and you get simply the upper bound minus the lower bound. So I get here the integral from minus 2 to 2 dx and then the integral here from this terrible lower bounds this upper bound and then I get 3 minus $2y$ minus x squared minus y squared the y and this seems quite undoable but you can rewrite this in the following way you can say this is four minus y plus one squared minus x squared the y so this minus y squared plus two y white again with completion of the square as y plus one squared but you have a minus sign in front of it so that's why You have to add one, and for the rest, nothing changes. At this point, this is relatively easy

to... Did I do it right? You can integrate this, and you get... This $4 - x^2$ is a constant. So you get $4 - x^2$ times y , minus $\frac{1}{3}$ times y^3 plus $\frac{1}{3}$ to the power 3. With upper bound, y is $\sqrt{4 - x^2}$, and here y is $-\sqrt{4 - x^2}$. Now, that looks quite terrible. In the end, it turns out that it is pretty doable because this, if you do that exactly, you get that this is. but I now skip a few parts $2 \times 4 - x^2$ times the square root of $4 - x^2$ minus $\frac{2}{3}$ times $4 - x^2$ square root of $4 - x^2$ dx and then you see okay I get 2 times the same term but now with a factor 2 and a factor 2^{3rd} here you see already this if you plug this in that you get that almost this second term. So this is $\frac{4}{3}$, integral from $-\sqrt{2}$ to $\sqrt{2}$, $4 - x^2$, square root of $4 - x^2$ dx . And I say, okay, I always am afraid of integrals with square roots, because there are lots of times very difficult. In this case, I have a trick, because I see this $4 - x^2$, So what I do here is that I replace x by $2 \sin \phi$. So I do a substitution. Then $4 - x^2$ is $4 - 4 \sin^2 \phi$ is $4 \cos^2 \phi$, and dx is $2 \cos \phi$. If x is $-\sqrt{2}$ so then the sign is five should be minus one so I have for example $-\frac{\pi}{2}$ if x is $\sqrt{2}$ the sign of five should be one so five should be $\frac{\pi}{2}$ then I have $4 - x^2$ that's $4 \cos^2 \phi$ then I have the square roots of $4 \cos^2 \phi$ is $2 \cos \phi$, and then I have my dx , and that is $2 \cos \phi$. So what I get here is I take this four out and this square out of four also, so I get $\frac{32}{3}$, integral from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ of $\cos^5 \phi$. I'm now doing it very, in a very safe way.

You should not forget this absolute value, because sometimes it has consequences, in this case not. If I would have forgotten it here, nobody would have cared. And the reason is, of course, I know for sure that the square root of $\cos^2 \phi$ is the absolute value of $\cos \phi$. It can only be positive. But, not by accident, on the interval over which I'm integrating from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, this $\cos \phi$ is positive anyway. So if the $\cos \phi$ would have been negative somewhere, I had to split it up into two integrals, one where it was negative and one where it is positive and make it everywhere positive. But in this case, I can simply take, it's just $\cos \phi$, so I have to integrate $\cos^5 \phi$. And now I say, oh, this integral you should have had in calculus because that was one of these integrals, powers of this, a couple of powers of the sine and powers of the cosine. And the reason why you wanted to have this, be able to do this, is because these play a lot of force in these computations with these square roots and with these volumes and areas. And that's the reason. You remember how you did this? So this is from calculus. So if you have to integrate from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ of $\cos^5 \phi$, I always say if one of the powers of the cosine and the sign is odd it is easy, you do substitution if they are both even in this case the sign is to the power zero even and the cosine is to the power 4 is also even then you always go to the double angle but it's nasty work so what you have to remember is that, at least I know by heart, that $\cos^2 \phi = \frac{1}{2} + \frac{1}{2} \cos 2\phi$. And then it follows that $\cos^4 \phi$ is $\frac{1}{2} + \frac{1}{2} \cos 2\phi$. Then I write this as $\cos^2 \phi$ times $\cos^2 \phi$. So I write this as the integral from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ of $\frac{1}{2} + \frac{1}{2} \cos 2\phi$ squared, because it's cosine, square, five, squared. So this is the integral from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ of $\frac{1}{4} + \frac{1}{2} \cos 2\phi + \frac{1}{4} \cos^2 2\phi$. And I'm still not there, because I have this cosine squared 2ϕ , but then I use the same trick. So this thing is replaced by $\frac{1}{4} + \frac{1}{2} \cos 2\phi + \frac{1}{8} (1 + \cos 4\phi)$. And then I get here a quarter times a half is $\frac{3}{8}$. Integral from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$ of $\frac{3}{8} + \frac{1}{4} \cos 2\phi + \frac{1}{8} \cos 4\phi$ and the outcome of this is $\frac{3}{8} \times \pi$ I know this because I have to integrate this $\frac{3}{8}$ over an interval of length π so I get for that part $\frac{3}{8} \pi$ and the primitives of this cosine 2ϕ is something with sine 2ϕ and if I plug that in $\frac{\pi}{2}$ and $-\frac{\pi}{2}$ I get sine by minus pi that's 0 anyway So these things where I did all this trouble for, they disappear anyway. So it's only, this is the interesting part. The rest is rubbish. Now, and then I know what

this is. This is over 3 over 8.5. I did something wrong. Do you see what I did it wrong? I did it at home, so that's why I now miss a factor 2. Do you see it? have I made a mistake? I lost the two. And the dx is $2 \cos 5D$. So here I have 64 over 3. You get 8 times 5. So you see you have to be very careful because in every step you may lose such a factor as I did. And you should try to not to do that. But you see how it works. If you do it step by step, every time what you do is things that you learned either in this course or in calculus, but you can do it step by step by step and then you get where you want to get.

Okay, time for one break, I would say. We start again five minutes to ten.

So, welcome back. Let's start again. Before the break, I mainly did this, triple integrals in Cartesian coordinates. And instead of this question, with the volume, I looked for, I did another one, but that is not much of a difference. What I also want to be able to do is to be able to do these variable changes as we did for the double integrals.

And now you have to think about it in a sort of similar fashion.

Think about we are now in three-dimensional space and I have a function depending on x , y , and z . And I want to introduce new coordinates, u , v , and w . And then it's easy to change the function, because I simply get a function h depending on uv and w simply by plugging in f , this x , y , and z . And then I get it automatically depending on uv and w . But of course, the problem is that my infinitesimal volume element is dv , that was dx , dy , dz in the Cartesian case. to find out how this changes if you use different coordinates. And the idea is the same as what we had in the very beginning of this lecture. So you now have to think about, my drawings are too bad, you have to think about a UVW three-dimensional space, where you have all of these small cubes, and you map these little cubes back to the X, Y, Z space. Then they will be deformed. So we went from this rectangles to, in a two-dimensional situation, to a parallelogram. And in the three-dimensional space, you will go from a little cube to a parallel pipette, where you have three different directions. So you get, for example, things like this. this is in case of cylindrical coordinates. You take a small step in the T -direction, you take a small step in the R direction, and you take a small step in the Z direction. But then you get, of course, an object that is getting larger when you're further away from the Z -axis, when R is getting bigger. Now, the idea how you do, do this, how you compute the course, so you want to know the factor between these small cubes and these deformed cubes, and these deformed tubes may be larger or smaller, and you want to have that correction factor. And it works in exactly the same way as we had before, because you simply take small steps in each direction. You get these directions where you go, and now you have to compute the corresponding volume of such a parallel pipet. And that's also the determinant of such a Jacobian. So in the end, you get, if you want to change such a volume element, this should be the dx, dy, dz by something with du, dv, dw , you need now the absolute value of the determinant of the corresponding Jacobian. In the same fashion as we had in the two-dimensional case, but now you get a Jacobian matrix. That is, of course, a three-times-three matrix. Please imagine how this will work if you will want to go to four or five-dimensional intervals. It's getting worse and worse and worse because the computations get terrible. But in this situation, it's still doable and it's also not that difficult. I think I have it on slide. So why are you fighting down everything? So let's do it for... Is this readable for you? It's too small. We are going to look at cylindrical coordinates. So I have new coordinates, R , θ , and the old-fashioned Z that I had already before. So my X is $R \cos \theta$, Y is $R \sin \theta$, and Z is Z . What I'm going to do is I'm going to compute the corresponding Jacobian by computing all partial derivatives of this X, Y, Z , was specific to R, θ , and Z . Now, with respect to R , X , if I differentiate, with respect to R , I add $\cos \theta$. If I differentiate the respect to θ , I get minus $R \sin \theta$. And if I differentiate the respect to Z , there is no Z . I get zero. Now, similarly, for Y , I get this. And then finally, for the last variable Z , I get, if I differentiate it with respect to R , there is no R involved, so I get zero. There's also no θ involved, so I get zero. And if I differentiate a set with

respect to itself, I get one. So this will be my Jacobian, yes. Sometimes the audio is not functioning. Now I have to compute the determinant of this three times three matrix. And it's, of course, not very difficult because I see here this column with zero zero one. So I do the development with this element. And then you see that this return, this $R \cos^2 \theta + r \sin^2 \theta$ so you get again that r back so this correction term is here just r and my $d x d y d z$ is equal to $r d r d \theta d z$ in a way this is just what we had for polar coordinates but now we also have a $d z$ and in principle we do nothing special with the $d z$. Let's look if I have a nice example. I look at the cone, and that cone is given by the points X, Y, Z in R^3 , such that Z is larger equal than the square root of $X^2 + Y^2$, and it's smaller than a certain number H . and this age should be positive. So I have this cone, and then at a certain level, age, I top it. So you get your, think about your ice cream. If you get really this cones that you have that you buy, and they are also topped at a certain level. Now, I want to know what the volume is, and in this case, let's do it with slendicle coordinates. now we know that $x^2 + y^2$ is equal to r^2 so we know that z is between r and now we have to be a little bit careful because of course i have to look on this projection over here that is that big circle where i have my ice cream let's put it like that but you can also think about it in a different way you see say, okay, I get for my Z these inequalities, but in order to have values of Z between R and H , I have to take care that my R is smaller than H . Because if R is larger than H , then that is an empty set. It's impossible to do that. So what I get here now is that the volume I get here, so that is this, so my R has to be positive and smaller than age, and on my θ , I have no, I have no restrictions.

It's simply from zero to two π . I do not get, I see no θ in my inequalities, and it's simply this circular cemetery, so there's nothing, nothing. special about it.

Okay, now you see this H is of course fixed. That's my height of my ice cream. And now I have to take care that I write my integral in the wide fashion, because my R is here here between fixed bounds, but my Z depends as a lower bound on R . So what I do is I say, okay, I go from 0 to 2π the θ , that's the easy part about the θ is nothing very special. But now I first take the integral from 0 to H and I put here $R d R$ because I put here already my correction term R and then finally I have to say how my Z is behaving and my Z runs from R to H $d Z$ and you get this and I now play a little bit around at first I do this let's do it the safe way but this is the primitive is Z upper bound H lower bound R so I get the integral from zero to age of R times here I get H minus R $d R$ and what you now see is that h is simply a given number that this expression will be a number because it does not contain θ nowhere so the outcome of this expression will be a number age of course but age is also fixed and then that number i have to integrate from zero to two π the θ yeah then i simply have to multiply it with two π that's what's happening so now i'm a little bit doing something a little bit dangerous, and I say, okay, because θ is not involved over here, I can first do this step, and that's simply multiplying by 2π , because it's that number times θ , and then I have to plug in 2π and 0 . So I get simply a factor 2π . So this is now 2π , and now I integrate this. This is H times R , and I have to integrate it with respect to R , so I get a half $h r^2$ minus r^3 so minus $\frac{1}{3} r^3$ from r is h to r is 0 and now I get 2π times now a half minus a third that's $\frac{1}{6} h$ to the power 3 so this is $\frac{\pi}{3} h^3$ provided by three.

I hope you're still with me. Yeah. Okay, new question.

I have this cone. I have my ice cream and now I assume this is filled with ice, of course, and density is everywhere the same. But I want to know where the center of Mars is of my ice cream.

Yeah. Of course, because I only have this waffle at the bottom and then it goes like this, it'll be further to the top. It will not be in the middle, of course. It's of not at a half age. It will be higher. If you want to know where it is,

exactly the center of mass, then you think, of course, okay, I have this symmetry, but I have a complete circular symmetry. So I know for sure that this center of mass will be on the Z-axis, because everywhere it's the same around.

So I only need the Z coordinates. Now, what you then can do is for the center of mass that you are integrating, then instead of this, you integrate this. And in general, you get, if you want to have the Z coordinate of the center of mass, and you have a solid with a certain mass density, then what you do is you compute two integrals, one integral about the solid of $Z \cdot \rho \, dV$ and you simply compute the mass that's the integral of the solid of $\rho \, dV$ this is called the first moment if you multiply that with Z and then if you take the quotient of these two integrals that will be the Z coordinate of the center of mass if you want also the x and the y coordinate it, you do the same thing, but then instead of multiplying with Z , you multiply with X or with Y . This is how you get these centers of mass and these moments. In this situation, you see that, now I already do here 2π . This is a half Z squared, so I get the integral from 0 to H of R . This is a half Z squared, half H squared, minus a half R squared I now have done lots of steps in one step I hope you see what I'm doing but my advice is be careful yourself do not too many steps at once okay now this is I take the half out this half and this half against this two then I get here H squared times R so I get a half H squared R squared and here I get minus minus r to the power three minus quarter r to the power four from r is h to r is from r zero to r is and I get π times a half h to the power four minus a quarter h to the power four is a quarter no I did something wrong minus a half this has to be minus one over eight I get minus a half R to the power 3, so I get minus 1 over 8, R to the power 4, as it's, so here I get 3 over 8, half minus 1 over 8, 3 over 8, π , H to the power 4. And then you see that your Z , your Z quart of your center of mass, will be this one, 3 over 8, π , H to the power 4, divided by that one, in this case, the vol, volume by age to the power 3 over 3. That is... Right? No. What are you doing wrong here? I've done something wrong because here I get 9 over 8 times age and I should get something else. If you do it right, you get 3 over 4 times age. But I made some mistake. Here? Wait a minute. It is a half and then minus a minute. This is a quarter. I try to first to correct something. And here you see, also, if I try to do several steps at the same time, it doesn't work, so I have to be careful, and that same holds for you. I make lots of mistakes in these lectures. I try to do too much, too many examples. But then in the end, you make mistakes, so do not take too many steps at the same time. And I also should not do that. You see now a little bit how this works. There is this other interesting core transformation that are, of course, the spherical coordinates. Let's have a look at these. So I hope you remember them from the very beginning of this lecture. I had these spherical coordinates where a point in three-dimensional space was described by its distance to the origin, that was capital R . Then I had my Z -axis, and if I had a point, then I determined the angle between the way that connected the origin to that point with the positive Z -axis. That was a ϕ . And if you project the point in the X - Y plane, you also are in a certain angle-like for the polar coords, and that's called θ . And then you get this description, X is capital R , sine ϕ , cosine, θ , Y is capital R sine ϕ , sine θ , and Z is R cosine ϕ . And now you want to compute, of the term, that's correction factor. And if you want to do this geometrically, then it's a little bit difficult to imagine how this works. So now I use the algebra, but if you have done the algebra, you also understand how the geometry works. But now it's easier just to do the partial differentiation. This with differentiation with respect to ϕ I just get sine- ϕ cosine- θ . Then the derivative with respect to ϕ , I get here cosine- ϕ . And if I differentiate with respect to θ , I get here minus the sign of θ . Now, you have to do these other ones too. So for the Z -d differentiation with with two r is cosine ϕ , with respect to ϕ is minus r sine ϕ , and with respect to θ is zero. So I get this, that youcomian. And now you have to compute the determinant of that Jacobian, and that simplifies really a lot. And I do that with Saras rule in this situation. You had that for three times three determinants, that you had these products of these. diagonals minus this product of these antideagonals? I hope so. So the first term we get is this times this times zero, so that's gone. Then we get this times this times that one. That's R squared.

Then I get sine phi to the power 3 and I get sine squared t. Tta.

So this one squared.

Then this one times this one times this one.

Then I get R squared.

I have cosine five squared times sine five times cosine square theta.

Then I get minus this one times this one times this one. But with a minus I get here a plus.

So I get here plus R squared. Here I have cosine five, cosine five sine five sine five.

And then I have sine square theta. And then the last one, because, and I study with a zero, that sixth term will not bring anything, and get this times this, times this. Again, with a minus, so I get R squared, sine phi, sine phi, sine phi, sine phi, sine five to the power three, and then here, cosine theta, cosine theta. And now you see why I wrote it in this form on the blackboard, because you can add up this one and this one. the first thing is the same, the first factor, and you get your sine square theta plus cosine square theta, that's just one. So this is r squared sine phi to the power three plus. And here you get the same thing. This first factor is the same for both terms, times cosine square theta plus sine square theta, so that's plus one. So I get plus r squared cosine phi squared sine five, squared sine five. And now you can do the same thing because you write this as R squared sine 5 times sine 5 squared plus cosine 5 squared and that is again 1. So the end of the story is that this is just capital R squared sine 5. And for you is the story you just remember 4. Spherical coordinates I had this small R as this compensation factor and for spherical coordinates. I have capital R squared sine five because you can do this all over again but every time it will be the same and at a certain moment you know this by heart. So I get here this correction term. And then I can, for example, do this thing. This is a different... That was my ice cream from industry where this age is like this. cones from the supermarket. Now we do not get really them from the Italian ice cream, but now we have that the top is not fixed, but that it is like a sphere.

So what we now do is we have here, and we are inside the sphere, X squared plus Y squared plus Z squared is A squared, and inside the cone, Z is the square root of X squared plus Y squared. So this looks like a big sphere, and then I have this cone, and I have this part, but that rotated around the set axis. That is what we now want to do. Okay, what do we know? We know that x squared plus y squared plus z squared is smaller equal to a squared, so capital r squared is smaller than a squared, so we know that my capital R runs from 0 to A. And we also know that you should be in your cone, so your Z value should be larger than the square root of X squared plus Y squared. Now what is Z? That is R times cosine phi. Do I do it right? Yes. And so Z should be larger. And this should be larger equal than the square root of, and here I get R squared, sine squared, phi, cosine, squared, theta, plus r squared, sine squared phi, sine square theta. You see that you can add them up. It is the square root of r squared sine square five, and that is R. I assume that R is always positive, that's why I do not write an absolute value. but I find this, sign fire. And then you see this cosine phi should be larger. This R doesn't matter. So my phi should be between 0 and pi over 4. As you already see here, it has to go like this. So we could have seen this also from the picture. But now we have done it with the formulas. Not much of a difference. Now what do I get as outcome? Of course, my theta have this circles symmetry that's from 0 to 2 pi. So I get the integral from 0 to 2 pi the theta. Then my... The right way is from 0 to pi over 4. And my R runs from 0 to A, the R. but I should not forget my compensation factor and that we computed there on the blackboard that was R squared and 5. In this case, this is very, I can simply see this as a product of, this is integral from 0 to 2 pi de

theta. Here I integrate the respect to capital R and sine 5 is a constant with respect to capital R. So I take that out. So I get here the integral from 0 to $\pi/4$, sine phi phi integral from 0 to a r squared the r and this situation is very simple because this will give a number depending just on the number a this will give a number and this will give a number and i just have to compute the product of these numbers and this number is two pi this number here the primitive is minus cosine phi with upper bound $\pi/4$ and lower bound zero so that is minus a half square out of two minus minus one so one minus a half square out of two and then capital r to the power two primitive one over three r to the power three and replace this situation here it's very easy because such a cone in spherical coordinates is very easy because that will be rectangle in spherical coordinates that's why we we like to do in this situation this approach so how it works um okay now i think i've told you you can try this yourself you should try this yourself because you only learn this by doing it and for that the instruction is a good opportunity however you are with very little number of students and we decided there are only two instruction groups left and maybe we could also do with one but I have to look up so you can either go now to so group one still exists and that is in Atlas minus 1.715 and group 2 still exists and that is today in Atlas 4.215. If you were in Group 3 or Group 4, go to one of these two groups. Then it is...

Well, as long as there's more students and teachers, it's still not a private lesson. So thank you for staying. Don't leave, don't leave. Good. Good. Makes a lot of difference. Yeah, so my task, again, is to relate the things that are covered in math due to physics and also preparing for electromagnetic. Basically, today there's quite a bit of preparation for Electromagnetics 1, Q1 next year. But I'll talk you through that. Given that there's, again, delay, I think it would be good to, well, start with the gradient and directional derivatives and what that means. And if time permits, then I'll look at things that were still not covered. Well, I didn't finish last studio classroom. So there's two pseudo-classroom problems that you've seen before, this one, this one. I'll come back to these, if time permits at the end. But let's start with a gradient. This is an important concept that is, well, new. it's like a derivative but then in 3D and it makes it well it takes some time to get used to this and I think it was introduced it was just posed as being an operator at x on a scalar function right so gradient of f typically written with a upside down triangle in full written as gradient of f is an operator at x on a scalar field so it contains the x derivative in i direction the y derivative in j direction the z derivative in k direction which means that if you have a 3d surface no let's do a 2d surface then well which could be a landscape of hills then at every point on that landscape you can walk in say x direction or y direction and the gradient then states how much well derivative you have in in those directions whether you go up or down, or if you're at this point, if you go along, well, with minus a gradient, so we go down the slope, then you will walk into this hole here. So it's a directional derivative that states how a function of multiple variables changes with space. So this is never with time, this is always with space.

It operates on a scalar field, a function f that can be of x , y , z . I'll show later some examples of functions that depend on r θ z so cylindrical coordinates or r ϕ θ circle coordinates which are really relevant if you're talking electromagnetics and after operating the gradient on this vector field on the scalar field we get a vector field so this right-hand side here contains i j k so the unit vectors in cartesian direction so this is a vector field and the direction of the gradient of f is then the direction in which F increases most rapidly, so it's also the same as a derivative of a function of a single variable, but now we're talking 3D.

And it is perpendicular to level sets. So a level set is a contour, or say a height, cross-section of this landscape, where a level set would then be a contour at a certain height. And the gradient is then perpendicular to level sets.

So let's do an example. We have the function, Ψ of x and y that is $\sin 2\pi x \sin 2\pi y$ then please try yourself if we have the gradient in 2d if you apply this to that function so if you calculate gradient of Ψ what we get is you take the x derivative of this function which is only this part so we get $2\pi \sin 2\pi y \cos 2\pi x$ in front times $\cos 2\pi x$ and we leave the y part b in i direction and if you apply the y derivative we get $2\pi \sin 2\pi x \cos 2\pi y$ so that's here and we leave the other one b and that's then in j direction the 2π goes in front because that's a common factor so if we use matlab to make contour plots we get the 3d surface well 2d surface plotted in 3d we get that here on the bottom you see the level sets which are the cross sections at a certain height and then the gradient is always perpendicular to those level sets so if you're on the flank here The gradient would be in direction of steepest ascent, so it would be up here in this direction, so towards basically the point on the tip, so that would be this point, let's see, I was pointing here, that would be the gradient in this direction, so towards the tip, right? I hope that makes sense. There will be more examples of this one. So why do we know the gradient already? Well, you've all taken the physics course. I'm not sure if it was covered extensively in physics for AT course, but it was covered in the physics. for EE-E course. And the first example was covered in both because this relates to semiconductor physics. This was a part that was module one in the physics courses, and this relates to a gradient of concentration. So if you think about a semiconductor, then there's, if on one side there's more holes or

electrons, depends on what you want, then there's a push towards a lower concentration. So if there's more holes on one side, less holes here, and there's a net push towards the lower concentration. and this can be described, well, the amount of push can be described by the derivative of the concentration with respect to space. So if the whole concentration is, well, shows this profile with X , then there is a gradient, well, in this direction, basically. The gradient is the slope of this curve that leads to a push, a current, that pushes the holes in a certain area. So this is, you don't have to remember this, this is just what you've seen in the physics courses.

The diffusion then relates to minus the change of the concentration as a function of space. So if there's more holes here than here, there will be a push from here to this direction. So that's in the negative, along the direction of the negative gradient. So that's why there's a minus sign here. And this then results in a current, which is opposite for holes and electrons. But this is something you've seen in the physics course. The total current then depends on two concentration derivatives, both for electrons and holes. and you can find the current by integrating over the area. So this can be generalized to 3D as well. So semiconductor is not always a 1D device, it can be in 3D. And then we call that Fick's law. So if you have ϕ as a concentration of particles that depend on space and time, then there's a diffusive flux, so the push, the current, if you wish, in a certain direction, in X direction, which states how many particles are moving per unit of area per unit of time. So in one dimension, this is, well, a more stylized version of the diffusivity equations. So it's minus a diffusion coefficient times the spatial change of the concentration. And in 3D, you would have derivatives in both the x , the y , and the z direction. So if ϕ now not only depends on x , but also on y and z , you will get a 3D current density that relates to the gradient of ϕ . So that relates to the change of ϕ and x , y , and z direction. So you can have diffusion, so currents, current densities, in three directions. And the minus sign means that you push, the current pushes the excess holes or electrons to a lower concentration, right? So that's a physical reason why we use a gradient for this. Another example is conservative forces. These are also forces that you've seen in your module 2 in the physics course. a conservative force can be obtained by taking the derivative of the potential energy function. So if you think of gravitational potential energy, elastic potential energy, the force by gravity or by a spring is then minus the spatial derivative of this potential function. So if there's a region where the potential changes rapidly, there's large forces at play. And if F is in a positive direction, then if you push a particle or a point mass, in that direction, you will decrease its potential energy. Again, for physics, there's always a minus sign involved. For mathematicians, it doesn't matter, plus or minus. It's just a sign convention. For physics, it does matter. It's always towards lower concentration or towards lower potential.

Yeah, so that's what a conservative force do. It pushes the system towards lower potential energy. So how does it work? If you think of spring energy, $\frac{1}{2} kx^2$, if x describes the displacement from the nominal or equilibrium position, then the potential energy is $\frac{1}{2} kx^2$, its derivative, or minus its derivative is $-kx$, which is exactly the spring force. If you push towards negative x , you get a positive force pushing to the right, trying to push the mass back to its equilibrium position, and if you extend the mass to that direction, then it tries to pull back the mass to its equilibrium position.

Same holds for potential energy, so gravitational potential energy is mgy , which is positive, the higher you go, the more potential energy you have. Minus the derivative is $-mg$, which is a force always pointing downwards. So this is a constant slope, so a linear relation, which gives a constant force if you are close enough to Earth, then g is constant. This can also be generalized to 3D, so then you have forces in X , Y , and Z direction that depends on the X , Y , and Z derivatives of this general potential. function, as potential energy function, and this can then be written as the three force components in the three orthogonal directions, which is the same as the gradient of this potential function, right?

Was this covered in the automotive course as well? Physics course.

You have learned about spring energy and the potential energy, right? Yeah. So this is how they relate to forces. It's minus the gradient of the potential giving the force.

So let's try to apply this. Let's enough sitting and listening.

Let's try to apply this. Let's look at, all right, write a gradient down, I'll do it in 3D plus k. Let's try to calculate the electric field, which is also related to force. Let's calculate the electric field from the electric potential. So this is really an EM topic, electromagnetic topic, that has similar relations as the physics topic. Electric field is also a conservative field, so it should be possible to derive that from a potential, which is called the electric potential. So minus the gradient of V would give the electric field. So for these two cases, please take a few minutes and try to calculate the electric field and in what direction the field will be for a line charge and for a point charge in the origin. So I'll give you a few minutes to do this, so we have to apply the gradient to that, basically. Anyone been able to find the answer to the first one like this? Sorry, X here. Oh, sorry, numerous, sorry. It would be Y here, right? Yeah, yeah. So let's see if that's correct. there's a natural logarithm of a square root so this is $x^2 + y^2 + z^2$ to the power half let's put this one here to half so this becomes a four and we get rid of the root and then we take the derivative in each direction so it's minus times minus so plus so we get let me do like this d/dx of $x^2 + y^2 + z^2$ so we got row l over 4 by Epsilon, we get $1/(x^2 + y^2 + z^2)^{3/2}$ times, well, we have to apply the chain rule. It comes $2x$, an I direction. And for the J direction, we get the same. d/dy comes $1/(x^2 + y^2 + z^2)^{3/2}$ times $2y$ j. And then indeed you can get rid of this two and this two, make this a two. And then you get the final answer, row L over $2\pi\epsilon_0 k$, $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over $(x^2 + y^2 + z^2)^{3/2}$. We all agree i see puzzled faces please yeah that's fine please ask yeah so this is how an operator works so what i'm doing i'm applying this part to whatever is here so i'm taking i times d the x of what comes next i'm taking the x derivative of this part in i direction so the i is implied that i carry that over to to the answer yeah it helps to write them down especially if you go to different coordinate systems, you need to write them down. Yeah, in subscript notation, that's also, yeah. Yeah, so you could write this as $E_x\mathbf{i} + E_y\mathbf{j} + E_z\mathbf{k}$. So the electric field in X direction, which is just this part, and then the electric field in Y direction, which is just this part. Does that clarify? So this is the weird thing about an operator. It has both a vector and a derivative, but the derivative X on what follows, the vector is, well, to do the bookkeeping in what direction this happens. And that's something intrinsic to this course, well, on everything that builds on it. You have to, well, if you have multiple variables, you have to specify which derivative you need to take in which direction. And the direction part comes from I, J, K. Okay, let's do this one as well. I think you get the procedure now, and I probably also, I think that you probably also see that X, Y, Z derivatives will give similar expressions. There's the same denominator here. So it's basically $x^2 + y^2 + z^2$ to the power minus half. The power savings mode should be back. So this would give similar expressions. So it's minus and I take, let's do the i component only. First, x derivative of $1/(x^2 + y^2 + z^2)^{1/2}$ got $x^2 + y^2 + z^2$ to the minus half. So that would give minus half times minus i. so that's $1/2$ i had q over $4\pi\epsilon_0$ nought so the minus half goes the front i get $x^2 + y^2 + z^2$ to a power minus $3/2$ so i increase the power or decrease the power by 1 times what's left inside times $2x$ so the two cancels so what i get let me write that a bit more neatly q over $4\pi\epsilon_0$ naught i had times x over so to the minus three over two is to the power three over two in the denominator so this is the i direction and then for j and k we got similar expressions with one change that this becomes y or z so the total derivative then e is then q over $4\pi\epsilon_0$ not times $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over that denominator to the power three over two so then you have the electric field of a point source if you would draw that so if you have a point source somewhere it's a plus q then the electric field probably you know

from a secondary school physics the electric field would be away from that point charge and if you go further away the electric field gets weaker so then you draw smaller arrows basically so that's what this states it's hard to see that but once we get to spherical coordinates which we'll do after the break, this will become obvious.

Are there questions so far on how to apply this?

No? I want to keep to the regular break schedule. So this will be half an hour first and then we do 45 minutes.

So this we can do in Cartesian coordinates. Surprise, we can also do this in cylindrical coordinates and in spherical coordinates.

It gets more nasty, but this will be something you have to get used to. Show two ways to derive this.

One, formal mathematical one which I'll give just a bit of detail and then we'll do in a more intuitive one that hopefully shows how well how the procedure works at least so let's recap a bit in cylindrical coordinates we had instead of x, y, z so x, y, z to describe a point in space we had r, θ, z r is the length you go out from the z axis so that's $x^2 + y^2$ according to Pythagoras and then z is still the same and θ depends on where you are in the, what orientation you have with respect to the axis.

So these are the coordinates. You can also define unit vectors for these, which we derived last studio classroom. So $\hat{r}$ is the same. But $\hat{\theta}$ is the, well, has the angle. So also the unit vectors, the radial vector outward and the one perpendicular to that depend on the angle, right?

So if we want to do the gradient in cylindrical coordinates, then we can do two approaches, as I said. There's a very formal mathematical derivation. I'm not sure. Last year, this was covered in the math two lectures. I'm not sure if it was this year. I'll include a brief summary how that works. And then there's an easier, faster derivation. That's intuitive, hopefully. And I'll show in detail how that works. I'll do this for cylindrical only for spherical. I will not go through the bother of putting all the variables here. What you typically do is you use a formula sheet to find the gradient in spherical coordinates or cylindrical coordinates and apply that to your. This is not, by now I know them by heart, but we don't expect you to know these by heart, so that's something you will get from a formula sheet. So if we think about the gradient in Cartesian coordinates, then it had this form, so it has the $\hat{i}, \hat{j}, \hat{k}$ vector, unit vectors. So in Cartesian coordinates, you can also do derivatives in three orthogonal directions. So it should have a form like this. There's a dependence on R , there's a dependence on θ , and a dependence on Z . Well, the Z part will stay the same. So C will be one, eventually, if we want to calculate the gradient in cylindrical coordinates. But A and B , we have to find out. So it makes sense that it looks in the same way as we have for Cartesian, right? So there's the chain rule you can use for derivatives, and you apply that to the Cartesian components, and then you can rewrite your Cartesian components into cylindrical components while taking into account that you have these dependencies for the unit vectors on the Cartesian components. This gets a mess really quickly. So you would need several meters, square meters of blackboard to derive this. In the end, what you get at is the gradient in cylindrical coordinates, where A , so the scaling factor in front of this term, was one. As I said, C was probably going to be one anyway because $\hat{k}$ is the same unit factor. And then with θ , the B component, we had $1/R$ as a scaling factor. So basically, this is a very elaborate way to calculate A , B , and C to see how the gradient translates from Cartesian to cylindrical. I think this is a good moment for a break. Let's do a 15-minute break, and then I'll show you the easier, more intuitive way. And then we'll apply that to the same problems we had earlier. So let's start again in 15 minutes. Apparently I lost two students, that's a pity. Okay, so if we talk about derivatives in functions that depend on multiple variables, we can do multiple things, right? If that function is a scalar, you

like a potential function, then we can take the gradient. So that gives their derivatives in all the slopes in three orthogonal directions. But in 3D, we cannot only have scalar fields. We can also have vector fields. So if you think about the electric field itself, that's a vector field. At every moment, at every point in space, it has a magnitude and a direction. So field can be pointing here with this magnitude in that direction. Here it can be with a smaller magnitude in a different direction. Also there you can take derivatives. And that's something that I'll come back to in the next two-d classroom. You can then do two more things. You can take the divergence of a vector field, which is the same operator, but now with a dot product, but you can also take the curl of a vector field. This one looks at whether there's sources or sinks present in a vector field. This one looks at whether there's rotation present in the vector. All of these, the knob layer, so the triangle, is still the same operator. It's still in cylindrical coordinates, this one, or in Cartesian coordinates, this one. But it can operate also on vector fields. That's something I'll come back to in the next studio classroom. Let's now look at an more intuitive, hopefully explanation of what the gradient in cylindrical coordinate should be. For that, let's first do the Cartesian example.

Cartesian is nice and easy. We know how that works.

If you look at a change in the function, F of X, Y, Z , if you go from a position in space here to a different position in space, that is ΔX , ΔY , and ΔZ spaced from this original point. So we make a small step in three directions, basically. You can go in X direction, Y direction, Z direction, to go from this point to this point. So this is a 3D picture, going basically forward out of the plane of drawing. Then the change in the function relates to the derivatives of the function in X direction times a small step in X , the derivative in Y direction times a small step in Y , and the derivative in Z direction times a small step in Z . If you make ΔX , ΔY , ΔZ small enough, this is a good approximation, right? That's also in definition of a derivative. I can write this displacement ΔS also as a vector. So I go ΔX in i direction, ΔY in j direction ΔZ in k direction then I go from this point to this point in space and this means that we can write Δf so the change in the function as a product of well this thing in which you see the gradient already appearing dot product ΔS so if I take the dot product of ∇f with ΔS I get well $\nabla f \cdot \Delta S$ gives one so I get the Δf exactly this term, and the same holds for Y and Z , so I can write a small change in function as derivatives in three directions times displacements in three directions. Then I see what the change in the function will be. Well, this is not yet in a form in which I can use it. Let's normalize this. So let's write this displacement vector ΔS as a length times a unit vector. So the unit vector in that direction, let's call that U , and then ΔS is the length of that vector in that direction. So where I was ΔS previously, I now wrote U times the length, and then I can look at the limit of Δf over ΔS going to zero. So I'm looking basically at the definition of a derivative, but then in 3D. And that limit is then defined as the gradient times the unit vector. So this gives me an expression for the gradient, where I express that as the derivatives in three directions times the displacements in three directions. Is that concept clear, at least reasonably clear? There will be one more example, but now in cylindrical coordinates, because this is a nice and intuitive way to look at what happens in cylindrical coordinates. So now the function is not depending on x, y, z , but on r, θ , and z . So I knew that Luke made a drawing of a cylindrical coordinates with line segments here. On this blackboard, I think, line segments in cylindrical coordinates are Δr , are $\Delta \theta$ and Δz so line segment always needs well the unit of meters so just $\Delta \theta$ is not enough that's an angle that's dimensionless we need to multiply this by the radius at which you are because if I'm on a larger radius I'm making a larger displacement if I'm rotating right so the radius also plays a role for the rest the procedure is similar so if I'm going from a point here to a point that is here I can do that by doing three displacements Δr are $\Delta \theta$ Δz then I go from this point to that point actually this is not $r \Delta \theta$ this is at r plus Δr times $\Delta \theta$ but in a good approximation if Δr goes to zero this

well this is true and which I'll do in a second I take the limit of this going to zero so that allows me to write the change in the function so Δf as a well derivatives in three directions times displacements in three directions so So I'm doing the same as what I did in Cartesian. For that, I'm introducing a vector displacement. To go from this point to that point, I go ΔR in R direction, $R \Delta \theta$ in θ direction and ΔZ in Z direction, which is this. That's a displacement vector. And that allows me to rewrite this as a term with derivatives only times, well, this displacement factor. So I'm taking the dot product between something that depends on derivatives and the displacement factor. Well, here I have Δf times $\Delta \theta$ times $\Delta \theta$, but the displacement is $R \Delta \theta$. So to correct for this, I need to have a one over R factor such that this times this gives me exactly Δf times $\Delta \theta$. $\Delta f \Delta \theta$ times $\Delta \theta$. So this gives a way of intuitively seeing that I need a one over R , for R to compensate for the R that I have here. And this R comes from the line segment. Another way of looking at this is that the DDR has dimensioned something over meter. So this should also have the same dimension. If I just have θ and a denominator, this is dimensionless. So I need a compensation factor that has meters as a unit. So one over meter to compensate. And the rest of the procedure is analogous. So I'm again introducing a unit vector times the length. to describe this Δs as a length times, well, the unit vector, and I take the limit of Δf , of a Δs going to zero, before Δs going to zero. And then I end up with something which are called a gradient times that unit vector. And this has this nicely, this factor, this factor $1/R$ in there. And you will have to trust me that you can do the same for spherical coordinates. What we end up with, and this is typically what you would find on your formula sheet, is the gradients, I'm not sure if you'll give this one. but these two should be given on your formula sheet. Spherical coordinates has this one over R , like a correction factor if you wish for the line segment. In spherical coordinates, there's also $1/R$ here and a $1/R \sin \phi$ here, which relate to, well, the line segments in spherical coordinates because it's $r \sin \phi$ times $d\theta$. That's the line segment in cylindrical coordinates. Sorry, in spherical coordinates. So what is useful of this is that you learn how to work with this, You don't have to repeat the derivation. I hope this gave some intuition for why this happens. These are tools in a toolkit that you should be able to apply to a function that we give. Are there questions on this? If not, then let's apply this to a case, so it's through a classroom problem. Two made-up functions, one is cylindrical coordinates, one is spherical coordinates. $F(R, \theta, Z)$ is $R\theta Z$, just having the product of all coordinates. It made no sense physically, but it's just to practice. and also here $R^5 \theta$ as a function to practice and try to apply these derivatives, so the gradient in cylindrical coordinates and spherical coordinates to these functions. Let's see if you can do that. I'll give you a few minutes time for this. This is first order derivative that should be normal. We're ready for the first one? So DF , the R , times R head. So the FDR , so that's just θZ , R head. And then we have DF , $D-D-\theta$, times $1/r$ so $df/d\theta$ becomes $r Z$ divided by r so this is Z $\hat{\theta}$ and then we have df/dZ in Z direction so that's $R \theta$ in Z direction so you simply apply this operator to the function f so everywhere where there's $r \theta$ here on this slide you put $r \theta Z$ and then you do the derivative then you end up with this expression and the same holds here and now we will get a sign term for free so it's df/dr the r times $r \hat{r}$ so that's $\phi \theta$ so r head then we have one over r times the $df/d\phi$ so we get $\theta \phi$ head and then we have one over $r \sin \phi$ times $df/d\theta$ so we have $d/d\theta$ θ is $r \phi$ and then we divide by $r \sin \phi$ so we get ϕ over $\sin \phi$ θ head so that's how this operator works once you know the trick i'll just do differentiation and leave the unit vectors $\hat{b}$ then it should solve itself. Sometimes, well, also just in a break, there was a question on, well, sometimes people learn different notation for vectors. That is all nice and fine if you're thinking Cartesian. If you're talking about cylindrical coordinates or spherical coordinates, then the preferred way and also the way we will do this in the electromagnetics one, if you write the unit vectors also with that. In that way, you never make a mistake, and that also makes grading easier. Questions on this? So let's look at the same problems we had. in Cartesian. So we again have the line charge and the point charge, but now I've given the potential in terms of their, well, cylindrical coordinate system here

and a spherical coordinate system. So let's see if we can do that here as well. Let me copy these, and then I'll bring up the gradients again.

So we have $5aV$ equals minus row L over B , V equals point charge in the origin. So for these, you should now also be able to calculate the gradients to find the electric fields. Now you apply these to these expressions. And these are examples you will see a lot in the electromagnetics 1. So much, in fact, that by the, well, you will have nightmares on those. But these that puts them in muscle memory. So we want to calculate E . Well, from a description you have to deduce, okay, this is lowercase R . This has to be cylindrical coordinates. And here is gradient V in spherical coordinates. And this should not be too difficult. Bless you. So what's the answer to A ? If the applies gradient and cylindrical coordinates to that expression. That's more than I would think. And are there other components as well? Are there also Θ and Z components? Yeah, true. No, you don't have to use them here. So let me clarify that. This is only a function of R . So this is V of R in cylindrical coordinates. And R does not depend on, well, this is not a position vector. This is just a scalar field that describes what the potential is at a point in space. So this only comes down to DV of R , $D.R$, $D. R$ hat.

That's the only term you need with a minus sign. So let's do that then. So it's minus times minus.

So that's row L over $2x$ or not times the R derivative of L and R .

So that's one over R in R head direction.

That's it. So that's why we use these different coordinate systems in the electromagnetics.

Because if you have a line charge, so an infinitely long line carrying a charge, then the electric field is only radially outward. It does not depend on the angle. at which you look at, or on the Z direction. So it depends on R only, and the radial direction. And if we think about a 2D vector, so let's see, this still doesn't feel like it's the same answer as we had in Cartesian, right? So let's try to reverse-engineer that. If you have a position vector in Cartesian coordinates, oh, sorry, Y , J , sorry, just in 2D, because we are invariant in Z direction, then the length of this vector would be R -HAT, would be, that would be the vector itself. I'm sorry, sorry, sorry, the unit vector, I forgot. The unit vector, let me rewrite this as R -hat, is the unit vector is R over its length, and that is indeed, well, vector R divided by its length, which is x squared plus y squared square root, and this we can put in here. So this then evaluates to row L over 2π absolute not times, okay, I had R over 1 over R , so I get x , I , plus, y j i'm filling in this one now first divided by this one but there's another r here so i get x squared plus y squared so i get exactly the expression we had earlier because r so that's the length of that factor is x squared plus y squared so indeed this cylindrical description is the same as what we derived in cartesian by analogy the one in spherical coordinates also depends on r only so i have to only use this term.

It does not depend on the ϕ direction or the thing. So I have to, well, this comes down to minus dv , which is a function of r , times r hat.

I have one over r . It's derivatives, so it's r to the minus one, it becomes r to the minus two times minus one. So I get with another minus, so I get minus, two over four π epsilon, one over r squared, times r hat.

And this, this becomes a plus.

So we get Q over $4, 5, \epsilon$, R squared, R -hat.

And this is also the same as what we have in Cartesian coordinates. But I will not go through the hassle of writing that down.

So for some cases, like really symmetric cases, line charges, point charges, using a different coordinate system makes life a lot easier. And the same holds for integration, which Luke has started to cover today and we'll continue to cover next week. That also makes life easier if you go to a different coordinate system. The questions on this so far. Okay, so that concludes the gradient part of the lecture. Let me finish the unfinished business from a studio classroom two. I think I would suggest we go through this quickly and we call it a day early. Please have a look at this problem. This relates to actually week two, but it's not a reason why you should have forgotten it already. So this is really a parametric problem, a symbolic problem. r , v and a are given as a function of time and we know that at every time t the acceleration is perpendicular to both r and v the question now is to show that this vector given here has a constant length what's a well just the problem so we know that r is perpendicular to a v is perpendicular to a in other words if we think about vectors $r \cdot a$ is zero and $v \cdot a$ is zero so the dot product between two vectors that are orthogonal has to be zero. So what is the length of this vector? T minus TV .

The most general form we know is that you take the absolute value, right?

If that needs to be constant, length needs to be zero. So it's constant in time.

So it has constant length. It doesn't change up time. So I have to show that T minus TV is zero. So this needs to be constant. then also the square of this expression needs to be a constant. It needs to be zero. So I can also write this as a square. And this makes my life easier for doing the processing. It's not needed, but it makes it on again. So I have to look at the length of this. Another way of writing the length of this square, I can write that as a dot product between the vector and itself. I can write this as r of t minus t v of t . Dot product, r of t minus t v of t . This is the same. This is the length of the vector, is the dot product of the vector in itself, that gives the length squared. If I take the derivative of this expression, I can equate that to zero, or I can show that this has to be zero, and then I have an answer to the problem. So if I take the time derivative of this expression, I have f times g , so I have to apply the product tool to do the derivative. But the function is the same. So I get, just as I have d , d x of, sorry, x squared, I get two x . I get the same expression here. I get twice this differentiation, well, the first term and the second term. So I get two times minus t v . I'm omitting the arguments now for sake of writing times the time derivative of this expression. So r , d d t of r becomes v , and d d t of t times v . well this is a product so I can differentiate this one let that one be or the other way around so I get minus v that's a time derivative of t explicit here d d t of this term becomes d d t of this term and this is what I'm working out now and I get minus t times a where I use that d r d t is v and dv t equals a right so if I take the time derivative of this function I get d r d t which is v and the d d t of this term so that's one times v minus t times a so that gives me these two terms so what do we get this is just two times r minus t v dot product a hey that's that's cool we already established that r is perpendicular to a and v is perpendicular to a so this gives zero and this gives zero with that i've shown that this has constant oh sorry yeah uh t a good point but that does change the answer but indeed it's correct t v dot a is zero then also t squared dot v dot a is zero yeah and the other one relates to the length of the curve so this was also a topic in the midterm test to calculate the length of a curve a given curve and you should hopefully now be know how to do that set that up here so please have a go at this yourself first if r is given you can calculate v and a right that's something you should be able to do then i'll look at the the answers I'll write out on the answers. Two, this is just the time derivatives. It's Cartesian, so these never depend on time, these coordinates, these unit vectors. So this gives me $3T$ squared i plus $2T$ j , and A is then, the VDT is, well, $6Ti$ plus j . So also the acceleration depends on time. So what was the formula for calculating the length of a curve? Hopefully you know that this is $DR-D-T$, and we have to evaluate this from T is minus 1 to T is 2. That's the length of the curve, so the distance the object has traveled. And there we had to take into account that, well, this is the absolute value. The length is always positive, so if

there's a sign change, we have to account for that using the absolute value. So let's first look at $\sqrt{9t^2 + 4}$, so that's the length of this vector, right? $\sqrt{9t^2 + 4}$ is the length of V , is the length of this vector, so that's $\sqrt{9t^2 + 4}$. t to the fourth plus four t squared and then square root so we can write this as t squared times $\sqrt{9 + \frac{4}{t^2}}$ so that's t times absolute value of t so the square root of t squared is the absolute value of t don't forget this is not just t this is the absolute value of t which we need times well what remains $\sqrt{9 + \frac{4}{t^2}}$ squared plus four and then the length would be let me do that here minus 1 to 2 $\sqrt{9 + \frac{4}{t^2}}$ and then because of the absolute value I have to split this into two intervals I have to split this from minus 1 to 0 then I write well t is now negative so I write minus t is $9t^2 + 4$ plus well from 0 to 2 the integral from 0 to 2 of t times $\sqrt{9t^2 + 4}$ plus t right is it clear that we have to split.

Okay, well, I see a square root and I see the derivative, something that looks like the derivative of what's in the square root in front, so this will hopefully work out.

You can use a change of variables if you want, so you can use U , use, introduce U as $9t^2 + 4$, and the U is, well, $\frac{1}{18} \frac{dU}{dt}$, so $\frac{dU}{dt}$ is $18t$ $\frac{dU}{dt}$ is $18t$.

If I would substitute that into this function, I would get square root u times $\frac{1}{18} \frac{du}{dt}$ which is easy to integrate but while doing that I have to take into account that my bounds are changing t is minus 1 I get u equals $9 + 4$ is 13 t equals 0 I get u equals 4 t equals 2 I get $9 + 4$ is 13 so the bounds will also change I have to not use these, I have to go to a new variable to find L equals from minus 1 to 0. So that's from 13, 4, $\frac{1}{18} \sqrt{u}$, square root, $\frac{du}{dt}$. So I've now substituted $t \frac{dt}{dt}$ with a minus sign, $t \frac{dt}{dt}$ times this expression with u . So square root du over 18 plus we go from 4 to 13. U , this is a 4, sorry, for my handwriting, 4 to 13 of $\frac{1}{18} \sqrt{u}$. U to U . So I can get rid of this minus sign by flipping the bounds. So I get the primitive $\frac{1}{18} \sqrt{u}$ from 4 to 13, plus $\frac{1}{18} \sqrt{u}$ from 13 to 4. And this gives me the final answer. So that gives me $\frac{1}{18} \sqrt{13}$, 1 step in between, $\frac{1}{18} \sqrt{13}$. and 4 here there's minus 4 here oh yeah sorry yeah this was I have integrated it should be yeah indeed so I'm dividing by 2 over 3 multiply my 2 over 3 so I get $\frac{1}{27} \sqrt{13}$ yes that was my was zoomed in too much this is what in the end we get so the length of the curve and the 4 part $4\sqrt{4}$ is 8 so I get twice that term so minus 60 we will try to make sure that if you end up with a formula like this in the in the exam, then you have at least something that can be solved by substitution or by knowing that, okay, the primitive should be like this. We do not want you to learn all the intervals by heart, but this builds on what you have learned in calculus, right? Are there questions on this? If not, then, well, exactly in time. See you next to your classroom. Well, have a nice weekend already.

So good afternoon, welcome to this lecture on mathematics too. I hope you're all doing well, and that your grade for the midterm was not a disappointment. I hope. For me, it was quite disappointing because it was only 25% of the students passed, and I think almost 40% has the three or less. On the other hand I think you also have to do something for this course and I have the impression that I'm pretty sure that you are working on it and then I hope you pass it but students who do not do anything about it they also should not be surprised if they do not pass we just start with the second part of the contents and that's about integration I made already the introduction last time, but today I will mainly show a lot of examples how you do this integration for this double integration, so integrating over an area. The idea was very simple in a way, so what you did was you have an area in the XY plane, a domain, and you want to integrate a certain function of two variables on the domain that is depicted here by the graph here in green and then what you want to know is what is the volume under that surface and the idea is to do this by approximation split this up in very small squares for every square you take the corresponding function value that is an approximation of the height of such a bar then you can compute this volume and you add up all these volumes to get the integral and if you then make this division into small rectangles fine enough, you get a reasonable approximation and this approximation is described as the integral, the double integral over domain D of a function of two variables F_{XY} and then we write DA for this small parts of the small area where you take a small step in both the X and the Y direction. Now, if you do so, you get several interesting properties. If your function is always positive, the integral should also be positive because you always have this function value is positive, also it can, of course, sometimes be negative, then you count that volume and it gives a negative contribution. If you integrate the one function, you just get the area of your domain. So also areas can be computed in this way simply by integrating the constant function equal to 1. And these were the more interesting properties that you also know for integrals in the one-dimensional case of linearity. So if you have a couple of times the function f plus a couple of times, the function G and you integrate that linear combination, then the outcome is α times the integral of F plus β times the integral of G over D .

And I also discussed these two properties that if F_{XY} is on the whole domain small equal than G_{XY} , then if you integrate on both sides, you keep the inequality.

This was a triangle inequality if you have an integral and you take the absolute value that is always less than or equal the integral of the absolute value and the reason is that here you first take for any XY you take the absolute value of f_{xy} so you always get something positive and that you integrate so the contributions of all parts count positively if you first integrate you might have function values that are negative and they give a negative contribution and you first do this addition and subtraction and then determine the absolute value, so this will be lower. Main thing was about, and that's where we, what is the main topic of today, is how to compute these things using this idea of slicing that I already explained a bit last time, but I hadn't time to do any examples. Now, the idea is you look at specific domains with specific properties and it is relatively easy if you have x simple domains or y simple domains and what does that mean in this case you have in the $x y$ plane xy plane you have in uh domain d where the x value is always between a and b and if you have a fixed x your y value goes from in lower bound that depends on x x to an upper bound dx so it can be described in this way so x is simple because that's between a and b the y is between some functions depending on x and it's called y simple if it's just the opposite way around in this situation your y values are always between C and D , and for a fixed level of Y , your X values are between A of Y and B of Y . So it can be described in this way.

In case such a domain is X simple or Y simple, it's possible to solve that integral by iteration, by slicing in the right direction. You also have domains that are, a lot of domains that are both X simple and Y simple, and then you can do both, of course. doesn't matter.

A domain is called regular if you can split it up into five number of parts and every of these parts has to be X simple or Y simple. So there you can do the slicing and everything will work out and then you have to do it for all domains and you can also do integrals like this.

Now how does this slicing work?

Now I think this figure from the book is very easy. We have here a domain in the X-Y plane that is this probably is a sort of circle. And we have a function that depends on X and Y. That's the sort of surface that is living above this cylinder, I would say. And now you're interested in the volume of the solid that is above this domain so that you get a sort of silender, on the top, this surface cuts off a certain shape. And then the integral over the set D will be the volume of this object. So you have a sort of cylinder, let's say you have a sort of candle, and you have a very strange shape on the top of it where you have cut it down, and you want to know the volume of that candle. Now, the idea is slicing. So what you say is not by accident, but on purpose. this domain is both x simple and y simple and now i first use the first approach i say okay i take a fixed x between a and b so this x is now for a moment it's fixed then i intersect here with the plane that x is this value so you get such a vertical plane and then you cut out this sheet i would say and you can compute what's the area of that sheet is x is fixed so what you have to do is you have to integrate from this value of y that is cx to d x you integrate a function f xy d y then you have that area of that sheet and then you say okay i need a volume so now i multiply this with d x with such a differential in the d direction so now my sheet gets some infinitesimal thickness because for every x i get a different area of course because that is changing i make it instantaneously thick and then so i multiply that with d x and then i integrate, that that is then an approximation of the volume of that thin slice, and I integrate that over all possible values of x. So I integrate that from A to B. So I have all these slices. First I have computed the volume of one slice, and then I integrate to get the volume of all slices. And then you get this integral from A to B of this thing, DX, and note that this expression will be a function of X because you integrate here with respect to the variable Y.

So if you do the integration, you substitute Y is CX and Y is DX in your expression. And then in the end, this expression will be something completely in X. Y is gone. After integrating with respect to Y, you have no Y's anymore. So you get a normal integral of new function of X, DX, and that you can do in the calculus way. Now, if your domain is also Y simple, you can also do it the other way around. And this is the same picture, but now the same stuff, but now done in the opposite way. You say, okay, let's take a fixed Y between C and D. So this one, and now I slice in the other direction. So I look at all points in this solid, where the Y coordinate is the same as this one. And then you get this purple sheet, and you can compute its area. How do you do that? You have to integrate from the lower bound of x to the other bound of x, the function. So you get the integral from the x runs from A depending on this value of y to bY of fxy, dX. That's then the area of this sheet. You multiply it with a differential divide, and you get the volume of such a very, in volume element and that you integrate then over why for y from c to d now the nice thing is if your function is continuous and this is possible so your domain is both x simple and y simple then whether you do it this way or that way doesn't make any difference so that's the same the computation are not the same i will show you that with help of an example a very simple one to start with. So, simplest example. I have my domain, D, are all points, X, Y, in $\mathbb{R}^2$, for which X is between 0 and 2, and Y is between 2 and 3. And if possible, it's always a good idea to make a drawing sketch of that domain. And in this case, it's very easy. That is maybe a little bit over the top to make a sketch, your x is going from 0 to 2 and your y from 2 to 3 so you get this rectangle that's my d and you see it's this is even rectangular so this is very simple and i have a function and i also do not take a difficult one Z, so is FXY, is 8 minus 2x, minus Y. So if the Z-quant is pointing inward to whom, I have such a, this is a sort of plane, and

it cuts off this bar in a strange way. On strange, it's like a plane. And you want to know what is then the volume of what is left. Now here, the computation is not not difficult, but I will want to show you that when you do both these techniques that are on the slide, you get the same result, but in a different way. That's the important thing. So first I do it the following way. I say, okay, my X runs from zero to two, and now I use my other notation, that is also the notion that is used in the book. I write immediately a differential behind the integral sign. In fact, they should have been really on the very end, but then it's so difficult to see that this integral from zero to two belongs to the X . So I like this notation from the book better, because then you see me, okay, these bounds from zero to two, they belong to the X quadrant. If X is fixed, you have to say what the Y coordinate is doing, and that's here very easy, because it's always the same. It goes from 2 to 3. So what do I have to integrate? Integrated from 2 to 3. Then my function, $8 \text{ minus } 2x \text{ minus } y$, $D.Y.$ And now you just integrate, and this second, now you work in that direction in a way. So you start with this inner integral. And you see here this differential DY , so I have to integrate with respect to y . So I consider this as a function of y , and I integrate. So then I get integral from zero to two. Now I do this preventive. I have $8 \text{ minus } 2x$. It's not dependent on y . So that's a constant $8 \text{ minus } 2x$ times y . Minus, I have to integrate this, half y squared, from y is 3 to y , from y is 2 to y is 3. like this, and this expression I have to integrate the respect to X . I think it's easy to write down what I have to plug in. Y is 2 and Y is 3, otherwise you may get confused. We have to use the X or the Y . Now, let's just do it. The integral from 0 to 2. Y is 3. I get $8 \text{ minus } 2x \text{ times } 3$. Yeah, times 3. So $3 \text{ times } 8 \text{ minus } 2x \text{ minus } 9 \text{ over } 2$. minus and then I have to plug in two times eight minus two x and then y squared is four minus two the x and you see indeed I get now an expression out of that inner integral that is just an expression in x and the y is completely gone that should also be the case when after carrying out the integration perspective to y the y has to be gone now what do I get here I get three times $8 \text{ minus } 2x \text{ minus } 2 \text{ times } 8 \text{ minus } 2x$ that's $8 \text{ minus } 2x$ and here I get minus $9 \text{ over } 2 \text{ plus } 4$ so minus $9 \text{ over } 2 \text{ plus } 4 \text{ over } 2$ so it's minus $5 \text{ over } 2$ the x now this is I did I do this hope I did this right oh yeah yeah I was worried about it because I was not computing much um $8 \text{ minus } 5 \text{ over } 2$, the $16 \text{ over } 2 \text{ minus } 5 \text{ over } 2$ is $11 \text{ over } 2$, so I have the integral from 0 to 2 of $11 \text{ over } 2 \text{ minus } 2x$ the x , and that is piece of cake, $11 \text{ over } 2 \text{ times } x \text{ minus } x \text{ squared}$, upper bound 2, lower bound 0, is $11 \text{ minus } 4 \text{ minus } 0$ is equal to 7. Of course you can do it the other way around, and that is here very easy because these bounds are fixed. So you can also say I integrate my y from 2 to 3, and then I integrate my x from 0 to 2. For fixed y value, my x is still from 0 to 2. But this is only so easy because we have a rectangle. Normally it is the difficult part. And I have the same function, $8 \text{ minus } 2x \text{ minus } y$ the x . Now I have to integrate first with respect to x , so I get here $8 \text{ minus } y$, that's a constant in x , so I get times x , minus $x \text{ squared}$, from x is 0 to x is 2, so I get here the integral from 2 to 3. If x is 2, I get here $2 \text{ times } 8 \text{ minus } y \text{ minus } 4$. the 0, I get $0 \text{ minus } 0$, so I get minus 0, I will write it, minus 0, dy . So I get here the integral from 2 to 3 of $16 \text{ minus } 4$ is 12, minus $2y$, I'm still the, yeah, and that is, primitive is $12y \text{ minus } y \text{ squared}$, upper bound 3, lower bound 2, 2, That is $36 \text{ minus } 9 \text{ minus } 24 \text{ minus } 4$. So that's $27 \text{ minus } 20$ is 7. So indeed you get the same thing. You see you get different integrations, but none of the steps are difficult. And the reason is this is a simple function and also a simple domain. Very simple, not in the way of X sample Y zone, but this is an example that is simply too easy. Okay, now I want to do a more difficult example, and that is on a triangle. So I have now a domain that is a triangle in the xy plane, with vertices the origin. This is the point one zero, and this is the point one-one. So this is my domain. And I have a function f_{xy} is equal to $x \text{-squared } y$. and I want to find the integral of this function on this domain D . Note that this line, that is the straight line from the origin to 1-1, that is the line y is equal to x . This is y is 0, this is x is 1. Now, I still, this domain is both x simple and y sample, so I can do both techniques, but I only will compute one of them and then show you just the expression of the other. you can say okay this is the integral from 0 to 1 the x 0 to 1 and then I have to walk if my if my x is fixed I go here with my y from 0 to x because the largest value is then x here on this because this is the line y is equal to x and then I have $x \text{ squared } y \text{ di } y$ in this form now also the integration is easy. Here I

integrate with respect to Y , so I should consider X squared as a constant. So what is here the primitive? That is a half times that constant times Y squared, because this is simply a constant times Y , so I get a half times that constant times Y squared. If I differentiate this back with respect to Y , I get indeed X squared Y . With as an upper bound y is equal to x and as a lower bound y is equal to zero and now you see that it is important that you write down what your value is y is what is what this I have to integrate the transfer to x so I have to replace the y here by x so I get here the integral from zero to one and then I replace I place I plug in y is x so I get a half times x to the power four minus I plug in y is zero and a zero the x that is pretty easy that's uh the primitive is one over 10 x to the power five of about one low about zero it's one over 10 I think this is the easiest you have a question I now want to do it the other way around and then you see that you get the same result but in a more difficult way so it's not always the same. So now I say my Y runs from 0 to 1, and then for a fixed Y , I have to say how my X is evolving. Now that X starts here, that is at this value Y , and it's once until upper bound 1. And now I get here X squared $Y \sqrt{X}$. So you see if you change this order of integration and you use this theorem of Fubini that both the things are the same, it is not that you change here these integrals and these values. You have to look at a picture and derive what the low and upper bounds are. And it is not sort of moving around with integral signs. Said that it was in the first example because everything was fixed. But if you have these more interesting domains, you have to watch out how this domain looks like, and you have to change these bounds. Now, in this case, I keep the outer one. I have to integrate this with respect to X . Now, this is a constant times x squared, so I get 1 over 3, y times x to the power 3, with upper bound x is equal to one and with lower bound x is equal to y like this and uh the y sorry now you really have to use the y difference otherwise you get confused and I made a stupid mistake just on the blackboard um now what do I get I plug in x is equal to one then I get one over three times y minus I have to plug in here x is equal to y so I get 1 over 3 times y to the power 4 y times y to the power 3 dy okay this is integration I get 1 over 6 y squared minus 1 over 15 y to the power 5 with upper bound 1 and lower bound 0 now the 0 doesn't give anything but my outcome will be 1 over 6 minus 1 over 15 and minus 0. Now you say, oh, is that the same? Yes, because I can make this is 5 over 30 minus 2 over 30. That is 3 over 30, and that's indeed 1 over 10. So you get the same result in a different way. And in this case, I would prefer the first measurement. Because I think that that is easier, easier computation. So the choice you make in which order you integrate, so whether you slice in this direction or in that direction, that makes a difference. And sometimes it's even worse, and that is, for example, in this situation. You are asked to compute the following iterated integral, the integral from 0 to $1DX$, and then here the integral from the... square root of X to 1, E to the power Y to the power 3, DY . Now you think, oh, it's simple. I don't have to do this iteration myself. This is just already given like that. So first step I can forget. Unfortunately, it's not the case, because if you try to do it like this, E to the power Y to the power 3 and you want to find a primitive, then be my guest. But I'm not able to find one, and I'm sure you're also not able to find one. Because it doesn't exist like a nice formula. Yeah, there is a primitive, but that's written like such an integral. So you do not get a closed formula for a nice primitive function. They say, okay, you can think, okay, now that's it. I cannot go on any further. But then you have to think, okay, but I can change the order of integration. And that may help. That may help because I get different integrations that we have already seen. And maybe we get some integration steps that are easier to solve. And by the way, if you e to the power y to the power three, if you have to integrate that first with respect to x , that is easy because e to the power y to the power three is just seen as in x . It's just a constant. So there I know what a primitive with respect to x is. That's simply x times e to the power y to the power three. And then I can continue and hope for the best. So that's the idea. So if I see this function, then I do not like to integrate it with respect to Y , but I like to integrate with respect to X . And then I say, okay, then I take care that my inner integral is with respect to X and my outer integral is respect to Y , because then I have hope. But now I have to do the other way around. I have to look what this domain of integration is. And what do we have in this case? X runs from 0 to 1. So this is my X , so from

here, here. And my Y runs from the square root of X , that is $2, 1$. That would be Y is equal to 1 would be this. So here I have this from 0 , and I have this domain. So this is y is equal to 1 , and this is y is the square root of x . I have that domain. And what I'm now going to do is to change the order of integration the other way around. So, and what has changed the order of integration? Instead of looking at it like this, you look at it like this. That's the bed for my back to do it. But on paper, it's just rotation. I cannot do that with these blackboards. Okay, what do I then do? I say this is, now I like the integral with respect to X on the inner side. So on the outer side, I like an integral respect to Y . I want to integrate the Y from 0 to 1 . And when I have this Y from 0 to 1 , I take a fixed Y , and I have to describe what my bounds for the X are. Now, the lower bound is 0 . And you write here, Y is square out of X , but now my Y is given, so what is that my... This line Y is square out of X . It can of course also be witnessed as Y squared of X , is equal to x then you also have this part but don't worry it's okay and now you say if y is fixed what is my upper bound for x that's then y squared so i get here the integral from zero to y squared e to the power y to the power three d y is this clear yeah of course it was struck to x because i i said x had to go from that way so i made a stupid mistake So is it now clear? But now the integration is, this inner integration is easy because I get there, if I integrate this with respect to X , I get X times E to the power Y to the power 3, with around X is Y squared and lower bound X is 0 . And that is the integral from 0 to 1 . I have to replace X by Y squared, so I get Y squared. E to the power, Y to the power 3, minus when X is zero, I get zero. So now I have to integrate Y squared times E to the power, Y to the power 3. And this is easy because Y squared is, it's not directly the derivative of Y to the power 3, because that would be $3 y$ squared, but it's only a difference of a constant. So should I do the substitution, or shall we do it in one step? Who is for in one step? Who is for substitution? I'm sorry. So what you do here is you say the derivative of y to the power 3 is $3 y$ squared so e to the power I get something like E to the power y to the power 3 times 1 over 3 because if I differentiate this I get this E to the power y to the power 3 times the derivative of this thing that's $3 y$ squared and I've compensated that 3 with 1 over 3 .

Y is 1 . Y is 0 . And then you get 1 over 3 times E minus 1 over And once at a sudden you can solve this. So you see, and here you have an example where it does matter in which order you carry out the integration, because in one order it is undoable, and in the other order it is doable. And then of course you get a question, yeah, what should I do then? That depends a little bit on the situation, and mainly you have to get some experience with it, so that you recognize, okay, I have here, here you should recognize, oh, I have a function that is difficult in Y , but easy in X , so I first like to integrate it with respect to X . But sometimes you have a lot of times X squared with Y squared in it, and you have some circular symmetries or something like that. Then you make use of that, so it depends a little bit on the situation. Okay.

Finally, for the break, when doing integration, you should also watch out not to do steps, difficult steps that are not really necessary. And this is still from the previous part.

Sometimes you can use simply integration by inspection, that you know how this thing looks like. An example would be this. This is the double integral over the circle with radius 2 around the origin of the square root of x squared plus y squared DA . If you would try to do this in the way we just did this with x, y , then it will probably be difficult. After break, I will do it in a different way where you can see, oh, it's pretty easy. But here, we can still already do this. because you have to think about how does this look like. In this case, I have this circle in the X, Y plane, with radius 2 , and I have here, let's draw that cylinder. But now I have to function, and if X squared, if it take a circle here, then the corresponding function value here is simply the same as the radius. So what I have is the graph of this thing is this cone. You have this cone from the origin. And what part of the volume that I want to have is the part over here. So I had this cylinder and I make this cone into it. You have to think about a cylindrical tone. And you make such a cylinder. in it and what

remains that is the volume that we have to know yeah now what do i do i compute the volume of the cylinder i compute the volume what's inside the cone what i take out with my uh and what remains that's the volume i have so to compute this volume i take the volume of the cylinder and that is a this is area is a times h now here my r is equal to 2, 8 times h, so that it is πr^2 times h. And my r is 2 and my h is 2. So I say this is $2 \times 2 \times \pi$ is 8π . Then I look at the volume inside the cone. So that is my cone and it's filled with ice cream and the amount of ice cream. That's what I want to know. What is that? There is also formula for that. $\frac{1}{3} \times \pi r^2 \times h$. The only will see how you get this. But this is a formula that is also given in the book. So that is here $\frac{1}{3} \times \pi r^2 \times h$ is $\frac{8}{3} \times \pi$. So what is this? This is 8π minus $\frac{8}{3} \times \pi$ is $\frac{16}{3} \times \pi$. So when you know these volumes of these standard things, then life is getting a lot easier because you can in this case do this integral simply by observation how this geometry looks like and do it that way. So sometimes you can also do this integration by inspection. After the break we will look at a different method that is particularly useful for these volumes where you have circular symmetry and where you can use polar coordinates to do it in a different way, but I will discuss that after the break. So let's continue. In the break, I got a very good remark, oh, this works nice for planes and for lines and for triangles, but what about if you have circles or balls? That's a subject now, and I want to make the clear with an example. suppose you have a domain in the xy plane and that is simply the unit disk so all points in the planes such that $x^2 + y^2$ is small equal to one and i have the function $f(x, y)$ is $1 - x^2 - y^2$ minus y^2 squared That's from a paraboloid. And I want to compute the volume of the object which above this circle. Here my unit circle. But now I do not have a ball, but I have a paraboloid. So it looks a little bit like this. You might think that it's a ball, but it's more like a sort of this shape. And then rotate. you can do this in the way we did before the break then we get the following i first take the picture of the domain x, y and here is my so that's the volume i want to compute this is my domain and that's simply here the bottom of that thing and uh yeah that's x simple and y simple so what i could say is uh okay my x runs from minus one to one for fixed x my y once from here to there now this is y is the square root is minus the square root of $1 - x^2$ squared and this is y is plus the square root of $1 - x^2$ squared because i know that $x^2 + y^2$ squared is equal to 1 So I get then here minus the square out of $1 - x^2$ squared to plus the square root of $1 - x^2$ squared. And then I have to integrate the function $1 - x^2 - y^2$ squared DY . Now you might think that this is difficult. It's possible. Let's say it that way. Here you can, the first step would be here to say this integral from minus 1 to 1. and I integrate this, so that's $1 - x^2$ times y , minus $\frac{1}{3} y^3$, y to the power 3, with this upper bound and this lower bound, and you have to integrate that with respect to x . In the end, you get an integral that is computable, but you have to do a few substitutions to get there. It's possible, I've done it, but I skip it, because the more important things to do. because we can do it much more easily so you can do it this way with a lot of theory from calculus and then it works but we want to do it in a different way which is much easier okay now the idea is as follows we want to do this because of the circular geometry and with help of polar coordinates so you I remember this for sure if you had this circle and you write this as the x as $r \cos \theta$ and y is $r \sin \theta$ then this domain can be written as the points $r \cos \theta$ where r is between zero and one and θ is between zero and 2π or between minus π and π and i don't care That's exactly the description in the polar coordinates of the same unit circle. And these bounds are much easier because here in our sense is fixed between 0 and 1 and the θ sense is between 0 and 2π , or between minus π and π . If you do it in a different way. So I would prefer to use these polar coordinates, but how to do so. And now you see here the following thing. You have to look at what happens with such an infinitesimal. area element.

And if you look, if you take Cartesian coordinates, if you take a little step in the X direction from X to $X + DX$, then the length in the X direction of the such small element is DX .

And the height goes from Y to $Y + DY$ is DY . And that was the reason that this

D-A is written as the product of the differentials DX and DY . That was the easy situation. Now we look what happens if you use polar coordinates. Now, then you describe your points in the plane in a different way because you say on what way they are with angles. So these are, and on what circle they are. I do not take the complete picture, but this is the idea now. Now, what you now see is this here is a step, a small step, in the θ direction. This is a small step in the R direction. Now, if you now want to look, if you now do that here, and you look what is now the area of such an infinitesimal element, then with the same $D-R$ and the same $D-\theta$, I get here this little thing, But if R is larger, I get this thing. The same d , the same $d\theta$, but a different thing. In the X, Y , if you take the same step, DX, DY every time, then every time this DA is the same size. But now it's not. And what you have to say is, okay, approximately, this length, this is approximately a rectangle. Of course, it's not a rectangle, but we pretend it is. this length is DR , but this is times $d\theta$. On the larger circle, if you dare take this same small $d\theta$, the distance that you cover very close to the origin is very small, but for very large R , that is already quite large. So think about my arm, the movement over here and my face. and that is simply $r d\theta$ so this area is approximately $r d\theta$ and that is then my da so what I do in this case is the following I have a function simply of x and y I want to integrate that over certain domain now I describe that domain in terms of polar coordinates with r and θ and I also describe the function in terms of r and θ simply by replacing x by $r \cos \theta$ and y by $r \sin \theta$ and then I get this function h depending on r and θ and what's now the trick is that this is not a good notation if I have a certain domain in x and y corn so the domain $f(x,y)$ da this should be a d prime and not a d because this is the description of the domain in polar coordinates.

And then I have this same function, $h(R,\theta)$, but the DA is now written as $R-D-\theta$. So instead of $D-X-D-Y$, you get $R-D-\theta$, and this correction factor R comes from this fact that far away for large R 's, yeah, that area is simply bigger, and it describes this factor. I think it gets more clear if I do this example over here. In this situation, my D -prime is, are the combinations $R-\theta$, where R is between 0 and 1, and θ is between 0 and 2π . So that's a nice rectangle in polar coordinates. So circles become rectangles. And I had here my function f_{xy} is $1 - x^2 - y^2$. and that becomes the age of $r \theta$ is $1 - r^2 \cos^2 \theta - r^2 \sin^2 \theta$ because of pythagoras. So if I want to compute the integral over my circle of my disk D, F, X, Y, D, D, A , I can do that by taking this d prime and then I have to replace my f_{xy} by $h(r,\theta)$ so $1 - r^2 \cos^2 \theta - r^2 \sin^2 \theta$ times da I I I I simply might still da because I want to describe also what you now have to do as an iteration so this da and that is $r d\theta$ and that means I have an integral for θ running from zero to 2π and I have because I had a circle it becomes a rectangle in polar coordinates the r runs from zero to one and then I get one minus r^2 times $r d\theta$ and I have to compute this iterated integral and now I'm doing something dangerous because I always have to compute it from, yeah, in the right direction. So I first have to compute this thing. But I see this integral has fixed bound, zero and one. And it's just a function in R that I have to integrate with respect to R . So if I do this, I get a number, not something with θ involved, but it's certainly a number, fixed number. And then I have to integrate that fixed number from 0 to 2π $d\theta$. that is simply 2π times that fixed number. So now I did it the opposite way, which is dangerous because it is not always possible. If you have here, if the bounce depends on θ , you cannot do this because you have to take care that everything works. But now because that inner integral just gives me a fixed number, I'm sure, yeah, okay, then it's just a fixed number. So that fixed number I can put in front of the integral and I'm done. so I could have written it like this this fixed number times the integral from 0 to 2π $d\theta$ now you know that this is 2π this integral and that's why I write this as 2π times this number doing it like this saves often a lot of writing because you see oh this is simple just do it already but be careful because it is not always allowed the way I explained it from back That is the safest method, but sometimes a little bit, a lot of writing. Now, in this case, you get your 2π times, now this is $R - R^3$. So I get a half R^2 as a primitive, and here minus a quarter, R^4 , with upper bound 1, lower

1, 0. So this is 2π times a half minus a quarter, minus zero so this is a quarter times π times two is π over two yeah other example okay volume of a ball $x^2 + y^2 + z^2$ is equal to capital R squared yeah and that volume is two times the integral over the units of over the disk with radius r the r so r are the points x, y in r^2 for which $x^2 + y^2$ is small equal to capital R squared i take this football i cut it in the middle and i take the two halves i compute the volume for two times and the i what is then the height the lower bound is zero the upper bound is yeah z squared is equal to capital R squared minus x^2 minus y^2 so uh i get here the square root set runs from zero to height is $R^2 - x^2 - y^2$ d a this is a cartesian i think almost impossible to compute at least with a lot of fuss to do this but if you now go to polar quons it's very easy so i say x is $R \cos \theta$ y is $R \sin \theta$ i have my d prime and that is or r where r is between 0 and 1 and θ is between 0 and 2π and then I can say okay this is two times the integral from now my θ runs from 0 to 2π small r runs from 0 to capital R my function is now the square root of capital R squared minus small R squared.

My notation is okay, but to say small R in capital R is a little bit, it's not the right choice. Times I have my DA , and it was R the R the θ , so times R the R . Are you still with me and convinced that this is okay? Yeah? Because this is the important step. Now it's just calculus B, although it looks a little bit more difficult, maybe, but that's not the case. This is, now let's do it on the safe side with, and now I have to integrate this one. So I have to integrate the square root of, so I have to integrate capital R squared minus small R squared $R dR$. And what you now do is I replace now for a moment. I take U is capital R squared minus small R squared. Capital R squared is simply fixed. That's that radius of my football. And R is my polar coordinate. My DU is now $-2R dR$. So I get that $R dR$ is minus a half DU . So what I get here is that this... Oh, wait, wait a minute. I made here a mistake. This has to be to the power a half. Because I made, it was a square root. So what I get here is the integral over U to the power a half, that's the square root of U times $R dR$, that's minus a half dU . I can do that. This is, to integrate this, this is minus a half times, this is U to the power a half, so that's times 2 over 3, u square root u plus a constant so this is minus 1 over 3 times R^2 capital R squared minus small R squared square root of capital R squared minus R^2 plus a constant and this R helps me enormously because it's the derivative of this R^2 and that's the reason why this works without this R , this integration is much more difficult. I have to do a lot of dirty tricks to try to solve this, but here this helps me a lot. So I get here two times, and now I take this primitive, so that's minus 1 over 3, capital R squared minus small R squared, times the square root of capital R squared minus small R squared. From R is capital R to R , is 0 and that is and that times the θ and that is 2 times the integral from 0 to 2π if you plug in capital r i get 0 minus when i have r is 0 i have minus 1 over 3 times R^2 times the square root capital R squared day θ and then you see what you get you get this 2 this is just a constant in θ so it's just this constant times 1 over 3 times R to the power 3 times 2π is 4 over 3 πR^3 which is known to be the formula for the volume and now you see why it is okay. So this can be done with these polar coordinates and without these polar coordinates it would be much more difficult just because the integration is difficult. And of course the question is, yeah, which coordinates should I use? That depends on the situation, but here I have a lot of circular symmetries, so I try to do this. In other situations, something else is maybe important. Yeah, how should I know what I have to do? Yeah, you have to practice. You have to look at several examples and see what is there, and sometimes you have to go back a few steps because you may, in this kind of exercises, take an approach, and then in the end you get an integral that is terrible, and you don't know how to compute it and nobody knows how to compute it. And then you simply get stuck, that happens. But then maybe you should have done it in a different order or something like that. You have here several possibilities to compute something because you can play around with these orders and with the kind of coordinates you use. You have to be a little bit versatile in using a choice that makes life. easy and how to know that before it i also do not know that beforehand but if you do a lot of exercises you get some inside oh

i try here this because i've seen that before um now difficult one but this is this this this could be an exam exercise nice exam exercise it's the volume inside circular silender $x^2 + y^2 = z^2$ is equal to y and inside parabolic silander $z^2 = y$ now it's getting a little bit crazy first i want to understand what is going on that helps a lot this looks a little bit difficult but it's not the case. First, I look at this equation, $x^2 + y^2 = 2y$, but I write it in this way. I bring the y to the other side, and what I now can do is completion of the squares. I can write this $x^2 + y^2 - 2y = 0$, that's $y^2 - 2y + 1 - 1 = 0$, and then I can hopefully erase here, because I bring the minus 1 to the other side. So I get this is equal to 1. And then you see here that in the XY plane you have a circle with center $(0, 1)$ and radius 1. Plain 1, 2 is also 1. And you have this circle. And it touches the axis. Now I'm considering this in three-dimensional space because it's spoken about a silander. So I have here my Z corner to put it out, and then I have this tube here. This is half the tube, but you can imagine now how it looks like. This is a tube that's going like here. And we cut something off because this $z^2 = y$. Then that is if you would have here the Y axis and here the Z -axis, then $z^2 = y$, then $z^2 = y$ is a parabola that goes like this. y is equal to z^2 . Yeah? The parabolic cylinder, then the X is free. So I have this surface, which you have to imagine that. Z is coming in here, and now you have here this, in this cylinder, I put here a cylinder through this. That is what's happening if X is going here in this direction. Then you have this cylinder that is also. over here and the vertical cylinder takes out here a solid. Between these two, in this powerbolic cylinder, and you have here your circular cylinder, your tube, and it takes out that part, and I have to compute the volume between these botish parts of this powerballic cylinder. That's what I have to do. Okay, now, what do I do? Let's take this as the ground part, so then I have my integral. over D , this is my D , my X, Y , in R^2 , such as $x^2 + y^2 \leq 2y$, is small equivalent to y . And what do I have to integrate? I have this thing here in, and that goes from, for fixed X, Y , this goes from minus the square root of z to plus the square out of set, so I get a square root of y to plus square root. I get two square root of y because my z runs here from minus the square root of y to plus the square out of y from that lower part to the upper part and the height is therefore two times the square out of y . Yeah, you're still there? Yeah, you're still there. Okay, how do we going to do this? What kind of coordinates? It seems here a little bit strange because I have this over here, but you can do it. So I use polar quads. So X is $R \cos \theta$, Y is $R \sin \theta$. But now I have to check what my domain is, because it's not that it is, and for that I use this equation, $x^2 + y^2 \leq 2y$. And I simply plug in, x is $r \cos \theta$, y is $r \sin \theta$. Then I get here. $r^2 \leq 2r \sin \theta$. r squared is smaller equal than two times $r \sin \theta$ in this case you know for these polar coordinates that this r is always you can assume it to be non-negative and i can here divide by r which is a little bit tricky but i get out of this my bound for r r is should be positive always because it's a distance and it's smaller than two sine theta follows from this So my bound for the R is now depending on theta. That looks a little bit, maybe a little bit tricky, but it's the same we had before in the first hour with these triangles. Then we also had that the bound for X was depending on Y and the other way around, so that can happen. But what is now the bound for theta? Now, you have to look here. There has to be, have to be ours between zero and two sine theta. this would be negative, there is nothing in between. You get empty, empty set. So to make this happening, I need that the sign of theta should be non-negative. So theta is between 0 and pi. So you get now your D prime, your domain D , but in terms of the R -teta coordinates, R -teta in R^2 , such that theta is between 0 and pi. and for fixed theta, my r is depending on tida. Is this strange? No, it's not, because you have to think about this drawing over here of my circle. And I start here with theta is zero, and then my lower bound for r is zero, and my upper bound is also zero. And then I'm going with different titas, that r has to go from here to every time to a different value. And you see there are only parts where this theta runs from 0 to pi. If you weren't from pi to 2 pi, there is nothing. So this is just a description of this circle in terms of polar coordinates. You can also understand what's happening if you think about these ways. This enforces us to do the integration in a certain order because my R depends on theta. I have

to keep that in mind. Now, what do I get now? My volume is. I had there my description. So the integral over. Now I get D' . And I have now two times the square root of y . That's in our case, two times the square root of $R \sin \theta$. And that's now R , the R , the θ . But we have to do it in a very specific order. We have to say my, this is on this thing. So I have to say my θ runs from 0 to π . And for fixed θ , my R runs from 0 to $2 \sin \theta$. And then I have here 2 , square root of R , square root of $\sin \theta$, R in the R . Note that our $\sin \theta$ is always positive. for θ is from 0 to π so this is no nonsense if i would if sign θ would be negative i would take the square root of a negative number and then uh big course and forget about it so then so here you see that's what you are there should make sense and it still makes sense now um this is i take this this this is an integral in r and the square root of $\sin \theta$ is simply a constant in terms of r so i can take it in front of that integral so i write that as the integral from zero to π and i put the square root of $\sin \theta$ here and then here the integral from zero to two $\sin \theta$ and then i have here two r square root r dr oh that is dual because this is simply two times r to the power three over two so this is this i have to copy and here i get is r to the power of r squared square with r that's r to the power 5 over 2 to i think it's 4 over 5 uh yeah but i made a mistake no i 'm still i 'm still fine with here r is $2 \sin \theta$ as upper bound and r is 0 as lower bound now that lower bound if i plug that in that is zero so that's easy so i get here the integral from zero to π and then here the square root $\sin \theta$ and here i get times four over five and then i get here four $\sin$ squared θ times the square root of two $\sin \theta$ so i write it as the square root of two times the square root of the $\sin \theta$ the θ and this looks terrible but now it's the main thing is is gone but the the main problem is solved. This is, now I have to check everything what I can take out. This one, 4.05 , then 16 over 5 , square out of 2 . So this is 16 over 5 times the square out of 2 . And then what remains is the integral from 0 to π square out of $\sin \theta$ times the square out of $\sin \theta$. So I get $\sin \theta$ to the power 3. Now, these are integrals you solve. in calculus and there I hope they taught you that if you have these powers of the sine and cosine that's the way I do it if you have a combination of sign times cosine and these powers are all even then it's difficult but if one of them are odd it's easy and in this case I do this separately and go from zero to π $\sin$ to power three θ it is easy because this is a three that's an odd number. And what I do is I write that in this way, $\sin$ squared θ , $\sin$ squared θ , $\sin \theta$, and now I do something strange. To get this part, I introduce u is cosine θ , then du is minus $\sin \theta$, and I get, if θ is zero, u is 1. If θ is π , u is minus 1. Sign squared θ is 1 minus cosine square θ , so I get 1 minus u squared. That's this thing. And $\sin \theta$, the θ is minus du . With that minus sign, I change here the order of integration to get it normal again from minus 1 to 1. So I get the integral from minus 1 to 1, 1 minus u squared the u and that is u minus 1 over 3 u to the power 3 from 1 to minus 1 that's 2 over 3 minus 2 over 3 that's 4 over 3 so this is 16 over 5 squared of 2 times 4 over 3 so this is 64 over 15 So here you see what was on this slide but now applied.

You see an example of this situation that you have such an area like this where the R runs in this case from zero to an upper bound and here you have this R is now a function of θ . If it's only from this part you have also a lower bound for your R values that depends on θ . So that can happen, integrate over such a sector. So it doesn't have to be, it does really have to be complete circles, but this helps. Question, do you want to know why this is the case?

Then here, that's somewhere I end, because I'm far too late, but here, this is a trick. How you can compute this integral from minus infinity of infinity of e to the power minus x square the x . It has to be the square out of π . And you can prove that now, at this moment, with theory we have. Although we cannot find a primitive function of e to the power minus x squared. And the trick are polar coordinates. And the trick is as follows. You define y to be this integral from minus infinity to infinity of e to the power minus x squared dx . And now I'm going to compute i squared. What is that? Yeah, I times i . But the second one I write like this. Yeah? It's e to the power minus y squared divide. That's the same number of course. What I do is say, okay, I combine this to one integral

over the whole R^2 plane, this product, I consider this as an iterated integral, but I see it as one integral of this function, the integral over R^2 of E to the power of $\sqrt{x^2 + y^2}$, the x - d - y . If I would do this, then I would split it up in this way, and you would get this thing. But now I can also compute this in a different way with help of polar coordinates. So now I go to polar coordinates, And my θ runs from 0 to 2π . My r wants from 0 to infinity. So over the whole, R^2 . E to the power minus x squared, minus y squared, is e to the power minus r squared. And then our trick is I get this compensation factor R . So I get my RDR here. And this is easily integrated. So this will be a number. and that number has to be integrated from 0 to 2π so I get 2π times that number so I get here 2π times the limit for let's say k tending to infinity of now I do here the integration and I get here a half times E to the power minus square with a minus sign because if you differentiate this you get E to the power minus r squared times minus $2r$, and you get exactly this. So this R helps me to make the problem easy. Upper bound K , lower bound 0 , that's 2π times the limit for K going to infinity of minus a half, e to the power minus k squared. Minus, for 0 , it's minus a half. If k tends to infinity, this term goes to zero very, very quickly. So you get just simply plus a half. So the outcome is π , and I've shown that π is equal to i squared. And this is the reason why you know, and this is what people from probability theory used, because this is something that has to do with normal distribution. And this is how this sometimes can work. But you need here the transition from one-dimensional to two-dimensional integration. But I think it's a nice application. On the slide, you will find more information also about general corner transformations. I will say a few words about that next time, but don't worry. In the exam, I usually ask the standard ones, and I put much more emphasis in my lectures on the standard ones that you often use than the most general situation. So don't worry about that, but I will get back to that. next week.

So I stop, I hope that Hott Mestrom is here because he will continue with a studio classroom, I hope.

Okay, good afternoon. Welcome to this lecture on calculus. The lecture will be completely different than the test because the test is about the past, and this is about what we still have to learn. And last time we talked about linear approximations and about differentiability, and I also wanted to do one topic, and that is higher order approximations. But I had no time for it, so I will try to do that now before the break and then switch to the really next subject that's about integration. But let's try to do this in a quick way, although it is technically quite heavy, but the ideas are very simple. I want to remind you the situation for functions of one variable that I've already wasn't down. If you then want to have an higher order, a better approximation of a function, G of X , around X_0 , you want to approximate it with a polynomial. You get, of course, this Taylor polynomial, and this is the Taylor polynomial of degree N . And I've done this for X around X_0 . So X_0 is fixed. For example, in lots of cases, X_0 is just zero in these standard cases, but it can also be 2 or 5 or whatever. And then the idea is that a difficult function, you want to approximate by an easy function, an easy function would be a polynomial. And then it turns out that's the function value of G in that point, fixed point X_0 , and all its values of the derivative in X_0 play a role in this way, $G(X_0)$ plus the derivative in X_0 times x minus x_0 this was the linearization as you know but then the next other term would be one over two factorial the second derivative of g in x_0 times x minus x_0 squared and so on and so forth this is the general term one over n factorial times the n th derivative of g in x_0 times x minus x_0 to the power n and if you now stop here the approximation so you say i instead of g of x i look at this polynomial in x then around x_0 that is a very good approximation and that is written by this term you say then that means that around x_0 the error that you make when you take the difference between $g(x)$ and or approximation is of this form and you have to think if x minus x_0 is small and that is the you are approximating around the point x_0 so if that is these close together then of course the more terms the higher terms you get the smaller they get and that's because these powers grow and if x minus x_0 is less than one then if you take it for an high power then that number gets closer and closer to zero so that's what you know from calculus in a way i want to describe that in a little bit different way x_0 so x_0 is fixed that you should remember and i now write x as x_0 plus h so x_0 plus a small step then h is just x minus x_0 and then my formula becomes that the g of x_0 plus h is approximated by now here i get still these terms $G(X_0)$ plus and now I write this as H times G' of X_0 then here I write this as 1 over 2 factorial H squared times the second derivative in X_0 because this is H squared but on purpose I put it in the beginning and the general term will be 1 over n factorial h to the power n $g^{(n)}$ of x_0 derivative times x_0 this thing nothing different than this other expression only h is now x minus x_0 however this helps us this form to generalize the situation to functions of more than one variable so and what we do is the following so you look at more than one variable.

And now we want to do something similar. We want to have a function of two or three variables, and we want to approximate that around a certain point. So I have now this x_0 that is now played by a fixed vector A , and that consists of A_1 until A_D .

So I have a function f from $\mathbb{R}^D$ to $\mathbb{R}$ i take a fixed point in that space just think about these two so point in two dimensional space point one two for example and i also have a vector h that describes how far i go from that point so that plays the role of the h in the previous part but that now consists of D components H_1 until H_D . And now what I'm now going to do is I define now a new function, capital F from the interval from 0 to R by the F of T is the small f evaluated in A plus T times H . So what you think, okay, this A is just a vector in $\mathbb{R}^D$. take a small step of length T in the direction of that other vector H , and depending on the T , I get a new value, and for that reason, this capital F will be a function from the real numbers, so from T 's to the real numbers, because it only depends on the T . This is fixed, and this is fixed. Now, what I now try to do is to find out what the Taylor polynomial of this capital F is. And for that I have to compute its derivative first, so the derivative of F depending on T . And now you have to use the theory from the last time, because I have a function here, small f of several variables, and then I have to use the

chain rule. So I say this is the partial derivative of this small F with respect to the first component times the derivative of the first component with respect to T , and that is H_1 , because the first component is simply A_1 plus TH_1 , and then I have A_2 plus TH_2 . So I have the partial derivative of F with respect to the first derivative times this first part, and then the part of the derivative with respect to T , and that's H_1 . Plus, okay, I also have to do this for the second component, for the third component, and so on. So I get here the partial derivative of F with respect to the second component times the derivative of the second component with respect to T , and that is H_2 . So I get this H_2 in front of it. And this continues, and then until you get to the end, and there you have this. We have D , A , plus T , H . What I now do is I'm going to compute what's the... I assume for a moment that this function small F , in the point A , it should exist, in the point A plus H , it should exist. On the line that connects these two points, it should also exist, but I also assume that there is a sort of neighborhood around these points where everything is okay. So that is the assumption. I have to assume that in a neighborhood of this line from A to A plus H , this function is nicely continuous and differentiable. whatever you like. So it's neat in that name-word. Then I can compute, so as a function of T , the Taylor series of this capital F . And the first component, yeah, the first component will be the F of 0 , but that is, of course, the F of A , so that is not very exciting. But then I get the derivative of F in 0 , and that is then age times F_1 , in A because I plug in T is 0 plus H_2 times the partial derivative of small F with respect to the second component and then plug in A and so on and so forth.

And now the trick is that you can write this as the dot product of H with the NABLA symbol applied on F in A .

I hope maybe I should that this first description that is clear. That is just writing out everything. And the only thing that I want to do is to use shorthand notation for this terrible law expression. And this will be shorthand notation for that. But it's not very... It is also easy to understand. You have to think what is the meaning of this H factor that is H_1 times the first unit factor plus H_2 times the second unit factor until H_d times the last unit factor so the factor zero zero zero and then one what is the nabla now the nabla of the nabla you can say that that is the d d x_1 times the first unit factor plus d d x_2 times the second one now i write it the wrong way it doesn't matter plus y if i would now apply nabla f so i compute the gradient of f in a point a that is a vector and the first component is the partial derivative of f with respect to x_1 in a in the first component so e_1 and then until the last one and that's exactly what we last time showed what the gradient was simply you you collect all these partial derivatives of this function f and you put them in a vector. That is, and this is a sort of difficult notation for that. Now, what you now should think about is that what does now the inner product from age with Nabla, what would that mean? Then I have to take the inner product of this with that. So what I get is that this inner product would be age 1 times taking the partial derivative with prospect to x_1 plus h_2 times the partial derivative with x_2 and so on and this is a little bit difficult to understand because what is this that this is an operation on a function because these partial derivatives have to apply on a function if i do that on f so i apply this on f let's do this on f then i get here this f f it's easy how to interpret that and now you see that if i then do that in a so if i evaluate these partial derivatives in that fixed point a you get exactly that this thing is h_1 times the first partial derivative in a plus h_2 times the second derivative in a and so on and so forth.

So exactly, you get this shorthand for this expression.

Yeah? Still okay. Now, I want to have also the higher order derivatives.

I will not continue with this because you get completely crazy of the notation.

But what I would like to know is what the sixth derivative of T would be, of F , of capital F would be.

Now, then I have to differentiate with perspective to t this terrible expression and that consists of such a sum so I have to apply the sum rule and now you get now it's getting really ugly because I first have to do that for this first term now that is h_1 times and then the derivative of this one but the derivative of that one is general the f_{-1-1} in a plus $t h$ times h_1 so the derivative with respect to the first component of this f_1 times the derivative of this a plus t with respect to t so of a_1 plus $t h_1$ so then you get this h_1 this this h_1 plus now and then I have to do it with respect to the second component h_2 f_{12} a plus $t h$ and so on and so forth plus $h d f_{1d}$ a plus $t h$ this is only the derivative of this term so now it's now you need really a full blackboard to write down everything I will not do that I only want to write down the second then you have to go to this term so that is plus h_2 and then I have to do the whole thing again h_1 now f_{21} a plus $t t h$ f_2 f_2 a plus $t h$ plus $h d f_{2d}$ a plus $d f_{2d}$ a plus $t h$ and third component and so on and so forth so in the end you get this $h d$ times and then now I will not write down everything I think you understand what I mean you get n squared terms you really get sick and tired of this if d is two you can still do it and only four but if d is five you have already 25 terms and you get completely crazy okay now what happens if you plug in t is zero then everywhere this thing this t plus age is zero times age so that disappears so you get in the first first line H_1 squared F_{11} of A plus $H_1 H_2$ F_{12} of A until $H_1 H_1 H d$ F_{1d} of A plus and then the next one H_2 , F_{21d} of A plus and then the next one H_2 , F_{21}^2 , F_{21} of A plus H_2 squared, sorry, that's why this is $H_2 H_1$. to 2 of A , and so on. And what you see is you get n -square terms. For every line you get n -terms, they're all quadratic. But you see, in particular, a lot are the same. For example, this one is exactly the same as this one, because this mixed partial derivative, whether you take it in one order or in the other order, you get the same result. And you also have the same factor in front of it, H_1 times H_2 . So this seem to be n squared terms, but in fact it is less because you can collect the same terms and then you can diminish this. Now, also for this I want to have a shorthand notation, and that is $H d$ squared of F in the point A . What does this mean? Yeah, this $H \nabla$, that was this operation. here.

Yeah, I've put here in the F . And now $H \nabla$ times $H \nabla$ is this $H_1 d d X_1$ plus until $H d d d X_d$, and you have to square that. Now, what do you do as you get, you have to compute that product so you get again these n -square terms, and you have to apply that on F from every of the terms, and you get exactly the same as I had in that scheme, but now written as a polynomial.

So in the end, And what this is, instead of writing this until the very end, I was too lazy for that, you get that this is the, you have to apply the operation H times ∇ , H dot product with ∇ , on F , and then evaluate that in A . So all these higher other parts of derivatives you take in the point A . Now if you have done so, and now I can get back, hopefully to my... You say, okay, now I'm writing down the Taylor series of that function capital F around zero, but I will place now that's by these operations, by this notation, these things that I have computed, and you can guess how this works for the third derivative and the four derivative. What do you then get? And that's why I wrote also in the beginning that in a different way. You get that X is now a fixed factor that plays the role of what I there had the A . Age is also a fixed factor. That's what I had also as an age. That is now, now the capital F_0 , that is this term, F of F of X , it was F of A , but now it's L of X , plus, and now I have this term H times ∇ applied on F in the point X .

That was just this term that I computed it here.

Then the second derivative, I have 1 over 2 factorial, of course, because it's table polynomial, so I have this 1 over 2 factorial. And then I have to apply this $H \nabla$ twice on F and plug in X , and so on and so forth. And here I have 1 over K factorial H in a product with ∇ applied k times on F and evaluated in X . And here you get that an error term, and I don't care that much, that small.

That is of the size that will be small.

So this looks very much like this expression that I had here, but with it in a

different way, because I have here this G_X , and here I have H , 1 over 2 factorial, H squared, and then the second derivative of G , and then H to the power three times the third derivative, but now you have to replace that with h times this ∇f and then this $h \Delta h \nabla^2$ on f and so on so instead that you have a h times h to power n times the derivative that end order derivative you now have this operation $h \nabla$ applied on f so it looks very much the same but the difficulty is interpretation because now you see that formula and you think okay, this looks very much like a Taylor polynomial, but how do I have to interpret it? Now, the point is that because of the normal Taylor series results, this will be the Taylor series of a function in several variables. And if you then say, I want to have that of order two, then you only want to continue until this part, for example, and then you forget about the rest. And then you get already a quite good approximation of your function around a certain point. To make everything a little bit more alive, I hope that this is, is this somewhat clear? No, or? I want to do an example, because then you see what you have to do. It's an awful lot of work, but it's not difficult. Okay, look at the following example. I have a function of two variables. $F(X,Y)$ is the square root of X squared plus Y to the power three. so that is x squared plus y to the power three to the power a half. And I want to do that around the point x or a is $1-2$. So x is 1 , y is 2 . For that, I need partial derivatives. Of course, it's obvious that the f of the point $1-2$ is 1 squared plus 2 to the power of 3 . That's 1 plus 8 is 9 . taking the square root is 3 , and then I have to compute the partial derivative of F with respect to X . Now, that is a half times x squared plus y to the power 3 to the power minus a half times $2x$. So that is x times x squared plus y to the power 3 minus a half. And the partial derivative of f with respect to y is also changing half x squared plus y to the power three to the power minus a half times the derivative of this with respect to y that's three y squared if you evaluate that in the point one two i get here x is one so this is one and this is one plus eight is nine so i get one over the square root of nine so that's one over three and if i do that here get something different because I get 3 over 2 and then y squared is here 2 squared is 4 so I get 3 over 2 times 4 times and then here 9 to the power minus a half so that's 1 over the square of 9 times 1 over 3 so if I'm correct then this is 2 which I will quickly check this is still okay now then you have to go to the higher order with this and now I now change notice and now I now change notation because I do not like this so f_{1-1} so this differentiated with respect to x is you have to differentiate this with respect to x so i get the derivative of this one that's x squared plus y to the power 3 to the power minus a half the this times that plus i keep this and i differentiate this times minus a half x squared plus y to the power three to the power minus three over two times the derivative of this thing to x so that is x squared plus y to the power three to the power minus a half minus x squared i hope x squared is y to the power three to the power minus three over two continue f one two $x y$ that's a little bit more easy because then i have two different say this with respect to y and get only one term because this factor does not depend on y so i just get x times minus a half times x squared plus y to the power three to the power minus three over two times the derivative of this thing with respect to y is three y squared so i get minus 3 over $2 x y$ squared x squared plus y to the power 3 to power minus 3 over 2 did I do this right and similarly you can also do the f_{22} but I just give it to you because you can do this yourself so the $f_{22} x y$ and you have to differentiate this with respect to y and if you do that correctly you get $3 y$ times x squared plus y to the power three to the power minus a half minus nine over four y to the power four times x squared plus y to the power three to the power minus three over two you see that doing this further you get completely crazy from the computations so you like to avoid this if possible again i need now the evaluation of these parts of derivatives in our point one So the F_{1-1} in $1-2$ is, then I have to plug in 1 here, so this is 9 to the power minus a half, so that is 1 over 3 minus, and this is 1 , this is 9 to the power minus, so it's 1 over 9 square root 9 , so it's 1 over 3 minus 1 over 27 , and that is 9 over 27 , and that is 9 over 27 , is 8 over 9 . 27 yes this one is easy $f_{1 2 1 2}$ is minus 3 over 2 times 1 times 5 squared so times 4 times 1 over 9 times 3 you get this is gone this is gone I get minus 4 over 2 that's 2 over 9 and if you do the last one correctly and I can assure you that I made mistakes when I was computing it, then you get here 2 over 3 . But, okay, we have now all partial derivatives. What

we now want is to get this Taylor polynomial of degree 2 out of this. Now, what do I then do? I get now, so the Taylor polynomial, and I'm evaluating it in the point $1 + H$, comma, $2 + H_2$. Now, $1 + H_1$, $2 + H_2$. Let's use the same notation as before. Often you use 3 values, you use H and K , but the book they do that, but let's have the same notation. What is that? That is, now, the function value in the point $1-2$ that we have computed in the beginning, I hope. Yeah, that was 3. And then plus, and then we get this H nabla, and that is we get this H_1 times the first, the part of the derivative is respect to the first component. So that is, in this case, where is it, over there? So I get plus 1 over 3 times H_1 , plus parts of derivative with respect to the second component, that was 2 times H_2 . And this together is the first other component. plus and then I get the next term for my 1 over 2 factorial and what do I now get I have now this age nabla squared so instead of having these two terms I now seem to have four terms but the term with the partial derivative once with respect to 1 and once with respect to 2 they are of course the same so what I get here is this one this is two times with respect to x_1 so I get get 8 over 27 times H_1 squared plus, then I get a term with H_1, H_2 . But I have that twice, because in this scheme I have this one and this one. So, times two times this, this was minus 2 over 9. So I get plus minus 4 over 9. So let's say 2 times minus 2 over 9. It's maybe more clear. but you should not forget is two. And then plus, and then only one times, that's this one, 2 over 3 times H_2 . Now, if you then want to continue, you can do so, but you get, I didn't do the computations. But let's first write this down what this is. This is 3 plus 1 over 3, H_1 , plus $2H_2$ over 2, plus 4 over 27 H_1 squared, minus 2 over 9, H_1, H_2 , plus 1 over 3 H_2 squared. And then you have already around that point 1, if H_1 and H_2 are small, this is a very good approximation of your function, second order approximation. What you see if you do these computations, that it is in principle the same as the Taylor polynomial in the one value case, but you have to think about what these higher terms are, Yeah, that is not one term, not one derivative, but it is a combination of all these partial derivatives, and there are also different numbers in between, because the next term would have been, if I would have done so, 1 over 3 factorial, and then I would have had the third derivative of F with respect to 1, for a component, times H_1 to the power 3. I didn't compute this because I get completely sick of tired of that. Plus, and then I have the term $1-1-2, H-1-2, H-2$. But how often do I have this one? Now, this 2, you have to count of how many places can this 2 be. These different orders do not make any difference. So I get this term, once for the 2 on position 1 and once for the rest. also another copy if the two is on the second position and the ones are on the first and the third and this one so I get this three times because in our scheme it then becomes even worse you get three times this same high order partial derivative now then you have a term with 2 times H_1, H_2 squared.

F.1.22 in 1. How many times? Do I have this one? Yeah, three. One can be on position one two or three.

And plus F22, here you have only one possibility.

One two times H_2 to the power three. If you would have computed all these high order partial derivatives, once more, then you could have also got these numbers. So you see that in the end, it is not very difficult, I would say, but it's a lot of computation and bookkeeping, really a lot, but not very difficult, I would say. It looks very strange with this notation, but you have to understand what that notation is, and that is what I've now written on the blackboard, and that is, I think, much easier than what is written there. So you have to interpret this formula in the right way. That is what What's up to you? And if you have done it a few times, it is not rocket science, I would say. Now, I think I skipped this for a moment. This is, and I will give the definition. And that helps to rewrite this in a different way. But what you should remember is we've introduced the Jacobian matrix, D of F or Nabla F , you know it, for function, and also for a vector field. This only works for a function of several variables to R . And this is, you can think of this as the second order derivative, but what is this? In this case, this is a matrix of size D times D , and on position I, J , you differentiate, to take the second partial derivative of F , where you differentiate one time with respect to x_I , and one time you get

x_j and then you get such a scheme if you would have two variables you get this $f_{11} f_{12} f_{21} f_{22}$ and that's called the hessian matrix and with help of that hessian matrix you can rewrite this in a different way but let's skip that I think it's more important to discuss the following thing because now you will get a little bit frustrated, I hope.

Not long, but for a moment. I want to look at another example.

X, Y is equal to E to the power X minus $2Y$, also around the point $1-2$.

Of course, you can do what we did before. All these high order parts derivatives, you differentiate until you get sick and tired of it. fill out everything and you think of quite a lot of work although differentiation of exponential functions is really easy what i do is the following thing i write this first in terms of x minus 1 and y minus 2 so i write this as e to the power x minus 1 minus 2 times y minus 2 but then i have x minus 2 i have here minus 1 plus 4, so minus 3. I write in this way. It seems a little bit strange, but I do this because I can now write this as e to the power minus 3 times E to the power X minus 1 times E to the power 2 times Y minus 2. And E to the power X minus 1 is just a function of one variable. And E to the power 2, I minus 2 is just a function of one variable. question.

Of course, of course, of course. And now I remember if I have E to the power T around zero, that you write as one plus T , plus T squared over two factorial, plus T to the power three over three factorial and so forth.

Now I do this here. This is E to the power minus three remains.

And here I look at the Taylor series of this part around one.

So then if X is almost, uh, uh, 1 then x minus 1 is 0 so i can just take the the standard 1 so i get 1 plus for t i have to take x minus 1 x minus 1 plus x minus 1 squared over 2 factorial now and i know it and here i do the same but now the t is minus 2 thank you for your correction times y minus 2 so i get 1 plus minus 2 times 1 Y minus 2 plus 1 over 2 factorial minus 2 squared, Y minus 2 squared, and so on. Yeah? The dots, we know how we have to fill them. What you now get is if I now work out this product, I get the full Taylor polynomial. But you're only interested in lower order terms because you are interested in approximation and you not want to go to the very end because then you will never speak. stop. So what is this? Now I keep this constant factor E to the power minus 3, and then I'm going to compute what this product is, and I do that term by term. So first the zeroed order term, that is 1 times 1.

Then I have the first order terms, and that is this x minus 1 times 1, and that is that 1 times minus 2 times y minus 2 so this is the first order term you have a question now this is this is minus simply so it is this term minus this term i multiply this 1 with this minus 2 why minus because now i have to look at the second order terms so it's of degree 2 i put here these brackets just to you do not need them in the description but okay for me, then, okay, my first order term is complete. Now, how do I get terms of degree two? Now, one possibility is this one times this term, and that is 4 over 2 times y minus 2 squared. That is this times this one. Then I have this one times this one, so I get minus 2 x minus 1, y minus two and then i have this times this one so plus x minus one squared over two that's it if you now would like the third order term then you have to have to do this and you then go for the terms of degree three you will get four terms if you write out that product what you see is that in this situation, if you have the standard tailor polynomials from our formula sheet, let's say, and I use these, then that is, of course, in this case, much faster than doing all these partial derivatives and plugging in and using this formula.

So you use that formula that we derived in case it is necessary, but if it's simple, if you can use these tail of polynomials with one component and do multiplications or addition or things like that, then you You prefer to do that. So in simple cases, you can, that's called a serious substitution, that you do

it in a way in the one-dimensional way, the one-variable case way instead of the two-variable case way, because it's just in this case a product of functions of one variable. Now in the exercises on this subject, you will see both of these things, some where you really have to do the differentiation. I think it's good to do that once or twice, more than once at least. And also of that other type. But there are these two different approaches to this. Questions at this moment? Or is it relatively clear? It looks more difficult than it is, I would say. It's a lot of bookkeeping, a lot of, and you forget always terms. At least I do. But you have to be very quiet. You have to step by step, and then it is not really. It's not really difficult, but you tend to forget terms. If there are no questions, then we have a break until, let's say, now, let's say, a quarter to three. After that, I begin, so I now stop with chapter 13 from the book. We skip chapter 14, and I go to chapter 15, and that's about integration. So after a break, I will start with integration, and that will be continue the next weeks, I would say. we make a start after the break. Can I start again? I don't know whether we miss somebody, but then it's only for one minute. So let's start again, and I now jump to the beginning of chapter 15, and that is about integration, and we start with integration of functions of two variables, and later on we will extend that also to three variables, but let's start at the beginning. so I want to give a meaning to this expression so I have a function F of two variables X, Y and I want to integrate that over certain domain D and you first have to think what does this mean and we find our inspiration in the way how we have defined integrals of functions of one variable so that's why this picture is here in case you have an integral of a function of one variable you have an interval from A to B , and then if you have the graph of the function F , and then if that F is always on the positive side, so if Fx is always larger equal to zero on the interval from A to B , then the integral from A to B of Fx , $D X$, X , is simply equal to the area under the graph. So above the x -axis and below the graph, and between A and B . If the function would go like this and gets below the x -axis, then the area that you would have here is counted negatively. So you count it positive if it's above the x -axis, and you count it negative if it's below the x -axis. That was the way how we did that. Now, we want to do the same thing, also the same approach here, or a similar approach, for functions of two variables. And now first you have to think, what is this? Here we always at an interval from A to B and now if we integrate our function of two values we need a sort of domain or part of a domain D and that should be a part of the X, Y plane in such a way that this thing has some area.

Also in this case if A would be equal to B then there is nothing to do. Then there is no then there also would be no area. You can only do it if B is really large and in this case we need also that we here have some area so that the area of d is not equal to zero and on the slides they call it it should not have measure zero so this d you could think about it instead of writing it as measure zero in the book they write it has not area zero but it's it's the same it would be the same thing now now i want to do the similar thing as here and here you have this green thing should be you should imagine that this is the sort of servers like this and that is the graph of the function F so then you get solid let's try to take this here I have here my D and then I look at it doesn't have been elipsed but I look at a sort of cylinder and here is my my graph and then you are interested in the volume of this solid so the bottom is d on the xy plane and then you get this cylindric form so that goes up until it reaches this surface that is $f xy$ and then you stop of course in the same way as you had before if this surface goes below the xy plane, then where $f xy$ is negative, you count that part also in a negative way, adding to the volume. So then you would get an integral. It can be zero if the volume above the xy plane is exactly the same as the volume below the xy plane, but then count it with a negative sign. So that is the meaning. So just you think about these geometric objects related to this domain d or this integration area d and you want to compute volumes or the integral should depict or should represent these volumes and now how to do so now we used the same approach as we did for function of one variable there we partitioned this interval from a to b in small pieces and then we said, okay, on every small interval, the function is almost constant, not exactly what, and then we took that piece of paper and we got some bars where we took one of the function values on such a small interval, and then we added the areas of these

small bars. That was the idea. Now, we hear with you the same, and we first consider the situation that you have a rectangular domain that is not necessary but I will show you quite soon so it can be much more general but this is easy to explain how this works so I assume now that this integration domain is interval X between A and B Y between C and D and now I split that up I do two partitions both of the interval from A to B and from the interval from C to D . So from A to B , I take a partition X_0 , small than X_1 , small than X_2 until this one, so that these values here.

And I also do that for the Y interval, the Y interval from C to D . I have Y_0 , Y_1 , until Y_D , and then you get these partitions.

Now, the blue part is the whole domain, and that is now split up, in small squares and the idea is now similar to what we did before now to get an approximation of that volume we just take an x_{ij} star so point in every of these little rectangles we compute the function value there and then we take here a bar out of the until that function value we compute the volume of that little bar and we do that for all these rectangles and you get an approximation. Of course, it's not exact because we simply did for all points in such a little rectangle. We use the same function value, but you get an approximation. Now, the idea is if we make our partitioning finer and finer, so more and more rectangles that are smaller and smaller, then this should converge, should get to one certain very. value.

So what you do is following. You first look at, so we have this rectangle, x_i minus 1, x_i , y_j minus 1, y_j .

This has an area. That's simply the product of these two differences, area of that small rectangle. It also has a length of a diagonal. So this is with Pythagoras, the square root of the sum of squares of these differences is the length of the diagonal from one, corner to the opposite corner. If you have here this rectangle, then this length that is then this Δ_{ij} . Now, you have a sort of measurement, how fine your partitioning is, so how good it is. If you take small and smaller rectangles, you get better and better approximations. And a measure then is, what is the largest possible rectangle that I have? Now, you measure that with this thing, and you say the... largest diagonal that occurs. That is a sort of measurement how good, how fine my partitioning is. Now, if you have one partitioning, you can compute a corresponding Riemann sum to it, and that you do in this way. You choose, and you can do it in a very different way, you choose an element X_{ij} star, Y_{ij} star, out of that rectangle ij , you then multiply the, this is the area of that rectangle, that is Δ_{ij} , that's this one, times, yeah, I take just the function value in one of these corresponding points. And if my function is continuous, then it will hopefully not be that bad. They just take a point. And then I sum up, and it's a double sum, because I have this key, I have all these rectangles, so I have to sum up i from 1 to n and j from 1 to m because I have to take all these n times m little rectangles into account. Now, what you now hope is that if you take your, these norms, so if the largest diagonal becomes very small, so if I get smaller, small and smaller and smaller rectangles, then the mistake that I make here will also get smaller. And you hope then that the limit for when this gets finer and final of these remand sums, this is a mistake, of these remand sums goes to a certain value, and if that is the case, that is then the value of the integral. Now you have to be a little bit more precise in this situation. So what you could say is the following. thing.

f is integral. And I just give this definition, but you will not have to prove things like this, but I just want to give a definition that you know, okay, there is a formal way of writing this.

f is integral over rectangle D and has double integral is, and they write it as the double integral over d of $f(x,y) dA$ and now i get something like with limits so it's getting confusing if for every small number epsilon positive there is there is a number delta that is positive that depends on epsilon such that the difference of the remand sum with respect to partition p so what i did with

these this addition of these volumes of these bars minus this value I gets less than epsilon now for every partition P of D satisfying that the norm of this P is less than δ so my partition has to be fine enough and every choice X_{ij} in this rectangle i, j so if you say I want to get to i less than 10 to the power minus 3 then there is a δ maybe 10 to the power 10 to the power minus 5 and if you take a partition that is fine enough so this diagonal is really small less than 10 to the power minus 5 and it doesn't matter which points that you take always you get that the corresponding remainder is less than epsilon away from that eye so in a way you always whatever you do if your partitioning is fine enough you always get that your remainder sum gets as close as to this integral as the value as possible you have a question now because if there's an asymptote then I have to assume here that my function is bounded on that interval D on that domain what I usually assume that it is continuous because if it is continuous on that whole set on that whole D then that's one of the results that I will not prove you can do this then this is defined so for continuous functions this definition that I gave here works but F should not be too strange but if F is nice so for example continuous on D it's even less necessary then you can do that And it is in this theorem, if F is only discontinuous in a set, which has measure zero, so on a set that doesn't have an area, so maybe on a line it is discontinuous, but a line has no area in $\mathbb{R}^2$. Then the function F is integral in D , and the integral does not depend on the choice of P and the choice of this point. So then it's just integral, then that definition works. so if f is completely continuous fine it is not even necessary that's completely continuous if it is discontinuous in sets that have no area it's also okay yeah and then it works in the same way as we did before it is easy now to generalize this to more general domains and I will do that here so I look now I want to have an integral of a function of two variables And this red part is this area on which I want to determine the integration. And now I have no rectangle, so that approach of the previous slide with all these rectangles doesn't work, because I do not have a rectangle. Now, what do you do? You take a larger rectangle, the blue rectangle, for example, that contains this area. And I extend my function simply from the wet part D to the blue part, r and I do that on the following way I define a new function f tilde defined for every xy in r in such a way that f tilde xy is just the function value f xy when xy is in d so here nothing changes so I feel here's this surface above that doesn't change but if I'm not in d but still in r so on the blue part the function value is zero so here it's just on the screen the function is zero and the rest is out of the screen now of course if you then determine the volume of the blue above the blue rectangle that's the same as the volume above the domain D because the function is defined zero on this point that belongs to the rectangle and not to D that doesn't give any contribution So you get then that this integral is simply the integral of the function F -tilde over R . And then we can just define, use the definition of the previous slide, where we do everything with little rectangles. Of course, this function F -tilde will not be continuous, because it will have discontinuities just on this boundary of D . There you will make a jump here at so on a value, and then you jump here to zero. But you see here, the area of this boundary, in this example, is zero. This boundary is just a curve and has no area. So that my function is discontinuous there. It doesn't matter. This integral still exists, and you will get the same result. So that's the trick, that's how you go from general domains simply to rectangles. I would say this is sort of cheating, because you say simply I define zero, And then I don't mind. The point is, of course, that the assumption is that this boundary, where the function is discontinuous, that that has no area. Otherwise, you cannot do it. That is the point. Now, I think I do examples a little bit later on. I first go to the properties. and I want to go, yeah. Now, important properties, if D has zero measures or has no area, then the integral is zero. That's pretty obvious because you have no volume. Then you have linearity. If you have two functions, F and G , and you integrate α times F plus β times G over the same domain D , then you get α times the integral of F plus β times integral of G and that is quite logical because your function description is simply satisfies this condition and then it follows directly from all the previous parts so it is not and it's very long you do it all the time then there is something about that inequalities go over and it says assume for a moment that on d the function value of f is always smaller equal than the function

value of g . So then we must have that the integral over D of F is also smaller than the integral over G of F . That's quite logical. That surface that bounded from above, one is lower than the other than I have additional volume. So that is pretty obvious. This is an important one. It's a triangle inequality. It says if I take the integral of F and I first compute the whole integral. that volume, and then I take absolute values, then the outcome is small or equal than first taking the absolute value of this function have and integrating that function. And this is similar as for the one variable case. The situation is if you have that function, it can have both positive and negative values, and negative values contributes negatively to that remand sum, so also to that integral. Now, if you first take absolute values, then everything where this is, that has a negative contribution is first mapped to the positive side and then counts positively in the integral. And that's this one. And in this one, you first take that volume with plus and minus part and something will cancel out. And then in the end, you do the absolute value, and that's, of course, less. So that's why you have this. And finally, this one that's also, suppose, I have. disjoint domains so d_1 d_2 until d_k and you want to integrate for example let's do it for 2 d_1 and d_2 over the union of d_1 and d_2 and d_2 and d_1 and d_2 are disjoint so they do not have points in common then if you integrate over d_1 union d_2 that is simply the sum of the integral over d_1 and the integral of d_2 that's just like that you had four functions that if you integrate from 1 to 2 and then from two to three you just integrate from one to three. It's nothing. Okay, what is now the trick? Hopefully I can explain that a little bit. I will come back to that later on, but then you can do lots of exercises. Oh, I have to go to this other slide. It's now about the computation. I will give easy examples also later on, so a lot of easy things will follow, but it turns out that computation is possible and relatively easy. your domain is X simple or Y simple. And what does X simple mean, that are domains like this, where X wants from A to B , and with a fixed X , your Y wants from a certain value C to a certain value D , but it depends, of course, which X you have. So there are continuous functions C and D from A to B , such that the domain can be within this, from x between a and b and y between Cx and Dx so between this function and this function then it's called x simple on the other hand you can have y simple and then it's just the other way around so here you have i'm not that large so now you have a value y once from c to d fixed and for fixed values of y your x value once from a of y to b of y and it's called y simple in the book they use the exact opposite definition of x simple and y simple so what i call x sample because then x is simple they call y simple and the other way around but i think that this is more useful because x between fixed values and y is then a little bit more difficult they use the opposite terminology which i find a bit strange now finally a domain is called regular if you can split it up in x simple or y simple domains for example if you have this domain with such a hole in it and you split it up in these four parts then this is for example x simple with this upper bound for y and this lower bound for y so you can sometimes split up in an domain in accent y simple parts now the idea and that's what i want to give you is if you have an x simple or y simple uh domain then you can do the following thing so here i want to compute this volume and my domain here is both x simple and y sample by the way and i look at looks the fact that it's x sample so x runs from a to b and then y runs from Cx to Dx and now you have to think that you have a piece of meat for your sandwich and you're at a butcher, and then they will slice this thing. Slice your, this... And if you come at this situation, you get this slice out of it. Now, the area of that slice, when X here is fixed, is AX is just the integral from... I go here from there to there, so that is CX to DX , F , X , Y , D , Y . That is the area of this slice. Of course, if you have given your, you think your X is to be fixed, slice at position X . Of course, your piece of ham is, of bacon, has some small, very small. So what you want to do is then the amount of bacon that you have is then the integral from A to B , F , X , So such little slice has volume AX , DX , this is, of course, very small, and then you have to integrate from A to B , $AXDX$. So in this way, that integral that you want to compute, in case that your domain is X simple, you can do in this way that you first compute this. This is called the inner integral, and then you compute this auto integral. And then instead of a double integral, you get an iterated integral. This is the way how we will solve, how we will be able to determine these integrals function of 2D.

And this can also be generalized to higher dimensions. So in this case you get that the integral is and you write as A to B and then you would have to write this because now I replace AX simply by its definition but it is easier in my opinion and the book gives that possibility to write this AB and to this integral sign belongs the variable X . So I put here the DX and then the function that I have to integrate over X and that is a function of X of course and that is this function and this is then the same rotation you see to this integral sign belongs to DY so here you have to integrate with respect to Y you get such a slice and then you have to integrate all slices to get the volume now in order not to make you So this is the idea that we will follow, this idea of slicing and going at iterated integrals. If you want to continue with the material, then you can read now yourself. I did at least section 15.1. And I made a very start with 15.2, but there's much more to tell, and I have to give you lots of examples to make it clear. But this is the idea, this idea of slicing that you really can imagine. I always see, or you can also do it with cheese if you don't like meat, if you don't like bacon, you do it with cheese when you slice it. If you keep that idea in mind, it is not very difficult. So I sometimes think that you have to envision things to understand it. All technicalities will come next week on Thursday because Monday is in Pentecost. That's why I stop here. So you can work on these exercises on the beginning of 15 yourself, but then you have to do some reading. I wish you now lots of... We start quarter to four with the midterm test.

I'm not sure this room is also used. I will stay here. Nevertheless, I ask you all to leave, also if you are designated to this room, to leave for a moment because I want to make the room ready for the test, and I will call you in. And I wish you lots of... success with the test.

So good morning. Welcome to this lecture on calculus. I have a problem with the beamer because it seems to work. And there seems to be a problem with the connection. So maybe I can do something about it in the break. But let's first just work on the blackboard. What we'll be going to do today is we try to finish the sections from chapter 13 from the book, mainly about differentiation of functions of several variables and I want to get back to the gradient of last time and then I will go to linear approximation and Taylor polynomials that are than the last parts of chapter 13 now what will we do first now I have to take them what did we do last time we looked at a function of several variables, so it was from a domain D to $\mathbb{R}$, where D is some subset of, I call it $\mathbb{R}^D$, D for dimensioned number of variables, and in case D is equal to three, we usually look at functions of two or three variables, but of course you can do it for more. Then we defined the gradient, so if I have a point P is point A, B, C now in the three-dimensional space, then the gradient of F in the point ABC was, now what did they have to do? I had to compute all partial derivatives, in this case three, because I have three variables, X, Y , and Z .

So ∇F , the X , in that point times the first unit vector, plus the partial derivative of F with respect to Y of the, times the second. unit vector plus the partial derivative of F with respect to Z in the point ABC times the third unit vector. And shorthand notation for this, if I write it as a column, and if I write the partial derivatives in a different way, you get a column vector with the first component, the partial derivative of F with respect to the first component in the point ABC , then partial derivative.

With the second component in that point and here you get the ∇F so computing gradients was not very difficult we note some very important properties and I want to say a few things more about it so what were the important properties first of all and the nice thing is that with these so please note that the gradient maps a point to a vector so if you had the function f it mapped points into three-dimensional space to a value, but now this gradient maps a point, is point A, B, C , through a vector.

And for that you get, for that reason, you get also geometric properties.

So the gradient in the point A, B, C is perpendicular to the level set through point A, B, C .

Now, what is that level set, that are, of course, in the point ABC , the function has a certain value, and that level set are the set of all points where the function takes the same value. So that level set consists of all points in three-dimensional space, in this case, for which we have that the function value is equal to a constant number, a constant, and that constant is the function value of F in the point A, B, C . And the easiest example was with functions of two variables, there are these level curves. Were these curves, for example, in a region with hills, where we have this hilly landscape, and here's the top of our hill. And then you know that, so these are points on the same altitude. And then you know, if you want to go upward, you have to go perpendicular to that level curve, and that means that in this point here, if I have now the point A, B , my level set is simply this level curve, and if I compute the gradient in that point, it is pointing perpendicular to that level curve. But it also works in three-dimensional, four-dimensional, or whatever space, although you can then not... In the three-dimensional situation, you still can consider it geometrically because then your level set will not be a curve, but it will be a surface in three-dimensional space. And of course, you also know if you have a ball, for example, what is perpendicular to a ball. Now, so this was the first thing. And the second thing that was important is the things that follow from the directional derivative so if you have a vector u and that is a unit vector is a factor of length one so that you always have to assume so it's it's just a direction and you have normalized that so you can always if you have a factor divided by its length and then you get a vector in the same direction that has length one and that you have to always do. And then we had that the directional derivative of F in the direction of U in the point A, B, C , is equal to the gradient of F , and then the dot product or the inner product with the vector U .

That was the second important thing. And directional derivative, you simply have to think we did these partial derivatives that we said, okay if I have a function I only look in one direction what the derivative is so if I take a step in that direction and we did that for the x direction and for the y direction and now we take an arbitrary direction you but we take care that this vector has length one and we just take well how does the function change if we take a step in the direction of u now from this property two things follows.

First of all, the gradient is pointing in the direction where F increases the fastest. And the second thing is the length.

So this is only, it says something about the direction of the gradient, and the length of the gradient is the maximal weight of increase. So if you have two points and you have to compute the gradient, and by accident, one is just a scalar multiple of the other, then where the gradient factor is only a small vector, then pointing in the gradient, you go up the fastest, but that is very slow. And if you have a very long vector in the same direction, if that's the gradient, then you go up there fastest in that direction still, but you go up very quickly. And this all follows from the fact that if you look, look at this inner product, then you can write that as the length of the gradient of F times the length of U , times the cosine of the angle between them. This number is always one. The cosine of α is maximal 1. So the largest rate of increase is this length, and then you have to take care that this α is zero, so that you, the points in the direction of the gradient. Now, now I want to, so this we did last time, but now I want to emphasize something that I didn't tell last time. It follows from everything, but there's always quite some confusion about it among students. And there was a difference, so note, difference between, first of all, the tangent plane to the graph of F , and secondly, the tangent plane to a level set. of f these are two different things and you have to really so if you see tangent plane or tangent line you always have to think yeah two watt and a level set is something different than a graph and i want to explain that with help of an example because still they have something to do with each other but then you have to work with two different functions and i want to look at this function $f(x,y)$ is the logarithm of the plus y squared, and you can also write it as the half times the logarithm of x squared plus y squared. Domain of this function is everything except the origin, because if the logarithm of zero, I don't know what it is. And if you now look at the graph of f , this is a function from $\mathbb{R}^2$ to $\mathbb{R}$, and the graph of f is then an object, a two-dimensional surface in three-dimensional space. So the graph consists of all points x, y , and then a third coordinate $f(x,y)$ in $\mathbb{R}^3$, such that x,y are in the domain of f , and that is everything in $\mathbb{R}^2$ except for the point zero zero. You should not, because the function is not defined. You get that, now if we want to compute now the tangent plane at a certain point, so I take as an example, For example, the point P is $1-2$. And if I now want to compute a tangent plane to the graph of F in the point $1-2$, and then I have to take, of course, also the third coordinate, the logarithm of $5, 2$ squared plus 1 . No, for half, mistake. And then you know what to do. You can simply do that with help of the partial derivatives. And that's pretty easy. the partial derivative of f with respect to x is a half times 1 over x squared plus y squared times $2x$ so that is x over x squared plus y squared and in a similar way you get the partial derivative of f with respect to y and that is then y over x squared plus y squared works in exactly the same way What you compute now is the partial derivatives in this point, in the point $1-2$, and that is here 1 over 5 , and the partial derivative with respect to y is in this case 2 over 5 . And then you get the following tangent plane to the graph, and that is, of course, that z minus the corresponding function value, that's in this case a half times the logarithm of 5 is equal to 1 over 5 , that's the partial derivative in the x direction times x minus 1 , plus 2 over 5 , partial derivative in the y direction times y minus 2 . Note that this shows that if you do it now in a general way, if you have simply a question, this has to be a y . I do stupid things, so you should correct me if I'm wrong. This function works in general. So if you have, and you can write it in this way, also, z minus the function value in $1-2$ is equal to the gradient of f in the point $1-2$. Dot product with x minus $1, y-2$. Because this is the vector 1 over $5, 2$ over 5 , and then you have the same formula. Now, in the book, you will see a general

point, A , so A is now the point $1-2$, and then they write Z minus the F of A is equal to the gradient of F in the point A , and now I have to write this column as a vector, and they write that then as the vector X , that consists of the component X , Y , minus $1-2$, so minus this element, A . And this is a formula that you see in the book, but it is exactly the same as what I did before, and if you do it with numbers, then it's much easier to understand. But just, so this formula, you should not be afraid of the formula.

It just means this. Now, now it will be getting a little bit confusing, because I'm now going to look at a different function, G , and that's not a function from R^2 to R , but from R^3 to R .

And I define this. as a function as the G of X , Y , and Z is equal to the FXY minus Z .

If you now compute the gradient of this function G in a certain point, and now I have, of course, this depends now on three coordinates, A , B , and C . What do I get? I get the partial derivative of F with respect to X , and if I differentiate this with respect to X , I get 0 . And they have to plug in A , B , C . So the first component will be F_1 , AB , because in this function, this derivative is not dependent on Z .

Then I do it with the second component. Of course, you get then F_2 , AB .

And then with respect to the third component, that is Z . Okay, that is always minus 1 .

So I see that for this particular function, the gradient of G is simply the gradient of F followed by a minus 1 , always. Now I look at the level surface GX , Y , Z is equal to 0 . The solutions of the equation FXY minus Z is equal to 0 . Now what is that? That is just the set we had here. So that is the level surface. Other points X , Y , Z in R^3 . such that Z is FXY . But you can see immediately that this is the graph of F . So you see, I define here this G as a function from R^3 to R . Then I have this level surface, and this level surface is exactly the same as the graph of F , where F is a function from R^2 to R , the same set. So this is the same object. So such a surface, I can consider that as, So if I have the tangent plane at a certain point of the graph, I can consider it as a graph and then I have that tangent plane. But of course I can also consider that same surface as a level surface of a function of three variables, namely of the function G that is fx_y minus z . If I do it this way and I want now to compute the tangent plane to that level set, tangent plane to a level set, that's now not this. difficult, so, and now I do that with, or function F over there, so I now have the function G , X , Y , Z is equal to a half logarithm X , squared, plus Y , squared, minus Z , set at level zero, in the point one, two, logarithm 5 , what do I get?

I know the In this case, I have a level set, so I can work with the gradient, and I know the gradient is always perpendicular to the level set. So, now, I have to work, of course, with the gradient of G . So I get the partial derivative of f_1 in the point $1-2$ times x -minus 1 , plus the partial derivative of f with respect to 2 , of f with respect to the second component in the point $1-2$ times Y minus 2 . minus 1 times z minus a half logarithm 5 is equal to 0 because in this case I know that the normal vector of this plane consists of these three components so that is exactly that gradient of G and I take care that a point 1 , 2 , half logarithm 5 is on the plane. So this is a plane through that point perpendicular to that gradient vector. So in this case, you get, in our situation, 1 over 5 times x minus 1 , plus 2 over 5 times y minus 2 , minus z minus a half logarithm 5 is equal to 0 . And if you then bring this term to the white-hand side, then you get exactly the same expression as you had before on top. So there is no difference, but you consider now, instead you have a situation where you are living in R^3 , and if you have to have a surface, you can consider it as a level surface of a function of three variables, that is the G , or in some situations, as the graph of a function of two variables. And then the formulas look differently, but what you're computing in the end is exactly the

same. So you have to be aware, of this difference and these two different points of view that give the same result. Is this clear?

Because I think it's important.

I hope it's clear. Okay, I do not have my pictures, but hopefully after the break.

I want to look at the next subject, and that is linear approximation. And for that, I want to go back to the situation you know from calculus, and that is how if you have a function of one variable and you have a point on the graph of that function, then you know how to compute the tangent line, and you say, okay, that function that describes a tangent line is locally a linear approximation of that function. So I have a function F from $\mathbb{R}$ to $\mathbb{R}$. I have a point x_0 in $\mathbb{R}$, and then the linear approximation of F around x_0 , is, now then I write it as the L of x , it's a new function, but it's a function just that describes a line, that is just $F(x_0)$ plus the derivative of F in x_0 times x minus x_0 . So in this way you take care that the derivative in x_0 is the same. And if you plug in for x x_0 , you get here zero. you go through the same point.

So if you think about the graph, it's simply this is our graph of F and then we have just this, let's this point here.

So this is the L , just as you know from high school.

What does it mean that this is a good approximation, a linear approximation, that is a good approximation of F around that point. Now you can describe this as follows. if you take the limit for x going to x_0 of $f(x) - L(x)$ so divide by $x - x_0$ you know of course when x goes to x_0 then this gets small and around x_0 this is also almost the same and it's a good approximation if this numerator goes much faster to zero than this denominator so when this limit goes to then you know okay the difference of this numerator is much smaller than the difference of $x - x_0$ now what is this this is the limit for x going to x_0 of now $f(x) - L(x)$ and now I write it $f(x_0) + f'(x_0)(x - x_0) - f(x_0) - f'(x_0)(x - x_0)$ divided by $x - x_0$ that's just the same and now you see that because of function I've assumed silently that f is just differentiable in x_0 and then you see that this quotient converges to the derivative of f in x_0 and here this caution that's simply $x - x_0$ divided by $x - x_0$ that will be just one so the limit here is the derivative of f in x_0 minus the derivative of f in x_0 and that is indeed zero so you get indeed that if you subtract the linearization from the function that's if you take a limit for x go to x_0 you go to this quotient goes to zero that means that the mistake you make is much smaller than this term because that is a multiple $x - x_0$ and that difference is much small now what we want to do I repeated this because we want to do the same thing now for functions of two or more variables, but let's do it for two. Same idea of linear approximation. And what does it mean? Now, again, I look now at a function G from, let's say, $\mathbb{R}^2$ to $\mathbb{R}$. And I want to do a linearization around a point A, B . That's just what we did before on that other blackboard for that particular function because that's just computing the tangent plane around that point. So what I now have is that the L of X, Y , is equal to the function value of G in the point AB plus, and then the partial derivative in the first direction. so I write this as $G_1(a,b)$ times $x - a$ plus $G_2(a,b)$ times $y - b$ exactly the same what I did on the blackboard but then I call this z as the third variable and if I call it z then I have the graph of this function of two variables and that will be then a plane okay so but now I do not write this with the z I look it as a function of x and y and now I want to know whether it's a linear approximation. I do the same thing as we did before. So I now look at the limit for x going to x_0 of x, y going to ab now, of the g of x, y minus its linearization so minus $L(x, y)$ divided by and now I have to take care I have to take the length here and I have to take the difference from so instead of that $x - x_0$ that was also the difference in length but now I have to divide by the length and that's in this case the square root of $(x - a)^2 + (y - b)^2$ it's just the length of that difference so I take a point x, y a little bit different from a, b and I want to know what this uh quotient is and

then here I get this $\mathbf{g} \times \mathbf{i}$ minus $\mathbf{g} \times \mathbf{j}$ minus and now I write this as the $\mathbf{d}f$ that is just a differential that so $\mathbf{d}f$ is in this case in under notation for the gradient of f in the point $\mathbf{ab}$ of the range of $\mathbf{g}$ of course in the point $\mathbf{ab}$ times the vector $\mathbf{x}$ minus $\mathbf{a}$ $\mathbf{y}$ minus $\mathbf{b}$ divided by $\|\mathbf{x} - \mathbf{a}\|$ now say the length of the vector $\mathbf{x}$ minus $\mathbf{a}$ $\mathbf{y}$ minus $\mathbf{b}$ that's the same thing and here I had to take the limit for $\mathbf{Y}$ going to $\mathbf{AB}$. G_1 is to be supposed the partial derivative of G with respect to the first component and G_2 partial derivative with respect to the second component. Now it looks very much like this, but I have to put here lengths of vectors because otherwise I don't know what such a quotient is if I cannot divide the vector by a vector. Now, if this limit goes to zero, so if this limit is zero, then we say, then G is called differentiable in. So I, until now, I only spoke about differentiability in the X direction, the Y direction, but I only looked at partial derivatives. And what I now do is I speak about differentiability of a function, and that means that That's the same thing as it means here, that if you take the difference of the function minus its linearization divided by the difference of the length of the difference of these two vectors, that that goes to zero. So it's simply this idea, but now two functions of two variables. If that limit is zero, then it's called differential. That was not difficult here because if you had a function of one function, we only had one direction in which we could take a different. So that was the ordinary derivative. But now we have something more difficult because in this situation, you can go in any direction in a plane and you only know, if you have computer partial derivatives, how it is changing in that direction and what changes in the other direction. Now, in a lot of situations that are interesting, if you only have these two directions, you also know the thing in other directions. But that is not always the case. can imagine that you have that you go on like this if you think about this about these strange walks that I showed you if you would have these walks where you go on a point or you are on the top of a mountain where you have here a ridge and there are which and you walk in that direction then the derivative is zero and if there's zero but if you walk in that direction you fall down but that means that in that point your function is not differentiable because if I then look at this linearization that I made, that is then an horizontal plane, but of course that is a very bad approximation if I do not go along such a ridge, because then I fall down. So the condition that this is the case is that everything is smooth, that the direction, so how much the rate of increase in the X direction and the Y direction is, that that also gives an impression how it is in the direction, for example, of the vector $\mathbf{1}-\mathbf{1}$. and that it is, for example. So that is indeed the condition, and now you get here, and normally so, for mathematicians like me, this is interesting, but for you, it's not that interesting, but you should remember this. By the way, this is interesting, because this tells you how good such an approximation, linear approximation is. But you get the following theorem, and that you have to, If all first order partial derivatives, so or G_1 and G_2 , of G in the neighborhood, this case of a vector of A exists and are continuous, so you have to exist and you have to be continuous, then F is defensible in A , and this condition holds. So if I then take the linearization, then it's a good approximation in any direction. But you need these additional things. It does not only have to exist in A , but also the neighborhood, if you think about that ridge. As soon as you're on that top, as soon as you go in a little neighborhood of it, there is not a different because there you fall down. You have this fall down from the cliff. So maybe they still exist, but they are certainly not continuous. And then you do not have this differentiability. And then you also do not have that this linearization. is a good approximation. So this is a difference between differentiability and the existence of partial derivatives in the X and the Y direction. What we usually do is that we simply assume that these partial derivatives that we compute satisfy that condition that also in the neighborhood they exist and everything is continuous. For engineers, that is the usual situation. But you have to keep in mind that if there's something strange happens, it can be one of these very strange situations where this condition is not satisfied. But usually we do not worry about it. Now, in our example here, you see that it's also not a problem. In this, here we compute that graph and that linearization. And you look at these partial derivatives, and only the point zero zero does not belong to the domain. So, of course, that I have to keep out. But as long as you take a point outside

0-0, then you see that this is everywhere defined. So it's nicely, these partial derivatives everywhere except in the point 0-0 exists. And you also can see quite easily that they are continuous. So there is one special situation. In case if the gradient of G in a point A is the zero vector, then the linearization or the tangent plane, then the tangent plane is horizontal, is a horizontal plane. Z is some constant, simply plain like the horizontal piece of paper. If you look at this situation, then you see that this cannot occur because the only case where these partial derivatives, so you use this, for example, to compute the maximum and minimum function, you want to know where the top is, and their derivatives, partial derivatives should be zero. In this example the only way that to get both zero is that you take X is zero and Y is zero, but they do not belong to the domain.

So there is no point in our example where this is the case. I think this not happen, but if you then think how this looks like, if X squared with Y squared is some constants, then the corresponding function value is always the same. So If you have a circle in the x - y -plane around zero, then the corresponding function values in the three-dimensional space are also the same. So then you have there also a circle at a certain level. And what you can see is that if you would do this, try to compute this graph, you get something like this. and this turned around. So in a way you have to think about the function of the logarithm of X simply in the XY plane that goes like this, and it is always increasing and it goes to zero. And now you rotate that along the y -axis, and then you get such a sort of shape where you never have a vertical, an horizontal tangent plane you have a sort of shape that goes like this on all sides.

So an hyperball that is rotated around one of the axes. And then you can also understand, okay, there will be no maximum or minimum because I can go as low as I want and I also can go as high as I want. Okay.

I think this is the right place to have a short break. I hope that after break I can solve this problem. Let's continue. I'm sorry that I do not get the beamer working. I tried it with a different connection, but not sure what is going on. I want to do a small application of what we did before the break. So I have this function, F_{XY} , a function from R^2 to R and that is the sign of πX , y plus the logarithm of y and i want to approximate i want to know approximately the function value in the point 0.051 1-1 or not exactly but i want to approximate it now what i then do is i say okay this is very close to the point 01 and to get impression how it is the the function in the neighborhood of 01, I replaced the function by its linearization around 01. So I compute the partial derivative of F with respect to X , that is the cosine of πxY plus logarithm I times the derivative of this with respect to X , and that is π times y . The partial derivative of f with respect to y is then the partial derivative of f with respect to y is then the cosine of πxy plus logarithm of y . And now, according to the chain rule, times the derivative, departure derivative, with respect to y . And that's in this case, πx plus 1 over i . And now I'm going to compute this linear approximation. Now, the function value in the point 0.01 is, if x is 0, then this will become 0. if y is equal to 1, the logarithm of 1 is also 0. I get here the sign of 0 plus 0, so that is 0. Then the partial derivative in the X direction, then I have to plug in 0 1 here. Now, if I now plug 01 in here, I get here a 0, but in the cosine of 0 is equal to 1 times π times 1, so that is π , and the partial derivative in the Y direction, then here I get again the cosine of 0. 1 times here I get π times 0 that's 0 plus 1 over 1 so I get 1 times 1 is 1 so my linearization becomes $l \times y$ is the function value is 0 plus π times this is in the x direction and I have here x minus 0 plus and then the partial derivative in the y direction times y minus 1 and then I get an approximation of my function values simply by computing the linearization in the point 0.05, 1.1, and that is 0. I get here π times 5 over 100, that's 1 over 20, plus 1.1 minus 1 is 0.1, it's 1 over 10, plus 0.1.1. Now, that is approximately. π is approximately 3.14, half of it is 1.57, and then I have to divide by 10, so it's approximately 0.157 plus 0.1 is approximately 0.257. And of course, this won't be the exact value, but it is a first nice approximation. So here you see how you can work with these linear approximations. Okay, now I go to a new subject, and now it is of new. It is a pity that I cannot show you

anything, because I repaired something on slide. But I want to look at the notion of vector field, and you can think about that just as a generalization of what we did before, we have a domain that is a subset in a D -dimensional space, R^D , and now a vector field is a function from this set D , not to R , but to R^N . So a vector field is just a mapping. If you have a point in some D -dimensional space, it maps every point in that domain to a certain vector in R^N . So not a value, that was just R^1 . That was what we did, functions of several variables. But now also the image consists of several components. And how you often see that is that they are written in the following way. You have simply, this is Y_1, Y_2 until Y_N , that is, and then I have a function. I get trouble with my notation because I get now I have this is not partial derivative this is just function F_1 of X_1 to X_D and I have a function F_2 from X_1 to X_D until F_N X_1 to X_D so if somebody gives me a point in your domain with D components I have here a function of I compute this real number, this real number until the last one. I put them in a vector and then I get the image. So you can think of this as a function of several functions of several variables. So it looks a little bit complicated. To depict this, and that's why I, hopefully I can show it next time, it is difficult to depict this. But you've seen these pictures of vector fields every day in the weather forecast. because what's then happening you see this so the east example is the wind at a certain point you think about the Netherlands and then in every point in the Netherlands you can at a certain time you can measure wind speed and wind direction so they are determined by a vector small or a larger one in a certain direction and then every if you do that then you get a picture of the with all these arrows in it of the wind. And you see them every day in the weather forecast. You see then these arrows going and they get larger. And you see them then as a function of enough time that it changes during the day. But this idea of an wind field, you know, of course, very well. An other example where you can see a little bit what a vector field is is if you have a magnet, and I had a picture also, cannot show it, and you have this iron, I think it's called filing. If you put this magnet here, and then you put, you've seen that on high school, this probably in physics, you look at, you put this iron filing around it, then all these little iron parts are getting in a certain direction according to the magnetic field of that magnet. So that magnetic field is also a field, because in, every point you get that this iron is always put in a certain direction and that is then sort of it depicts in a way how this enrich direct direction this vector points so although this seems very abstract from from your own experience and certainly from forecasts you know this and you know also how to interpret these things so that's in that sense not not new now um What you now have is the following thing. If you have such a function of several variables, you also want to do some kind of differentiation. And that looks now very difficult, but the derivative is described by a certain matrix, and that's called Jacobian. And so Jacobian of F . I write this with a bar here because it's a vector value thing. and what is this, it's always denoted by this $D_x F$ in a point X , and what you do is the following. You make a matrix, and this matrix, we will use this several times. The first row, you write the partial derivatives of the first component, F_1 , with respect to all variables. So the $D_x F_1, D_{x_1} F_1, D_{x_2} F_1, \dots, D_{x_D} F_1$ until, $D_{x_D} F_1$ and of course then you have to plug in the value because this um then the second row you can guess then you take the second component function F_2 and you differentiate it with respect to the first component with respect to the second component and so on and this you do for all components.

So you end here with the $D_x F_1, D_{x_1} F_1, D_{x_2} F_1, \dots, D_{x_D} F_1$. That means that all these partial derivatives, if you evaluate that in that, in a certain point, certain point X , you get a number, and in this way, you get here and this is, no, I have to put it here, this is an n times d matrix so this is F maps from R^d to R^n and then $D_x F$ is an n times d matrix and that's not strange because if you have an n times d matrix that also maps vectors from R^d to R^n If you have a matrix, 2 times 3 matrix, then it maps factors from R^3 to R^2 , A times X . This also happens in this situation. So in a way, how you can think about the Jacobian as the derivative of a vector field, and it consists of all these partial derivatives, but you have in every point, this Jacobian is a matrix that you have to be aware of. Special case is if n is equal to 1 . So I have a function of only that maps to R , the old situation. And then you get the

df of x . So f is a function from $\mathbb{R}^d$ to $\mathbb{R}$. And that is then the df, dx_1 in x , dx_2 in x until dx_d in x and this is just the gradient the gradient of f so in a way you can think about the jacobian to be a generalization of the gradient that instead of a function of several variables you have a vector valued function of several variables so you have more components and for that you get instead of an row vector, this gradient, you get a whole matrix. Let's do an example. The f of xy is equal to x to the power y plus $\cos y$ by i , x squared, x minus e to the power y . So f is a function from $\mathbb{R}^2$ to $\mathbb{R}^3$, because it maps a point. point in $\mathbb{R}^2$ to a vector in three-dimensional space. Then you can compute what all the partial derivatives are. So I get DF_1DX is E to the power Y . Df_1DY . Is X times E to the power Y minus $\pi \sin \pi - y$. Then I get DF_2DX $2x$ $2x$. I have to differentiate this function. And if I differentiate F_2 with respect to Y , I get zero, because the second function does not depend on Y . And then the DF_3 , DX , that's just one. And the DF_3 , DY , is just minus E to the power I . What I'm going to do is now to write the, of course, I can write this as a Jacobian, and then I have to make this 3×2 matrix with these components. I have them already here in the right order, but I do not want that write it down once more. So the Jacobian of our function F in the point $1-0$ is, now what do I get? Here I have e to the power y , and y is 0 , so I get 1 . Then here I get 1 times 1 minus π times 0 , so I get 1 . Here I get $2x$, it's 2 times 1 , it's 2 , and this one is always 0 , and here I get 1 , and y is 0 in this case, or here I get minus e to the power 0 , so that's minus 1 . So my Jacobian in that point, $1 \ 0$, is this 3×2 matrix. And of course you can compute it in any other points, and usually you write that depending on x and y . So that you write this. Df in $x \ y$ is, but then it's just e to the power y , x , e , to the power y , minus e , e to the power y , minus y , minus y . E to the power minus E to the power minus Y . In this way, and then you plug in all these values. But that doesn't matter. Now, in this case, if you look at value of F in the point $1-0$, yeah, that's a vector with three components, and that is the vector. Now, x is 1 . So I get 1 plus $\cos$ of 0 . 0 , so that's 2 , then x squared, that's 1 , and x is 1 , y is 0 , so I get here 1 minus e to the power 0 , that's 0 . So the f of $1 \ 0$ is the vector $2 \ 1 \ 0$, just that's not difficult. Now you can use this Jacobian also to get approximations of your function around that point $1 \ 0$.

1. So in this case, so I want to know something about the F in 1.02 , 0.01 . So I make a step 0.02 in the X direction and 0.01 in the Y direction.

Now then the Jacobian describes the F in 1.0 times this difference factor 0.02 . $2.0.01$ that is in this case let's do it here $0.02 \ 0.01$ and that is now just a matrix vector multiplication so this is easy peasy first component 0.03 then here 0.04 2 times this 1 plus one and then one times the first minus one times a second so I get here 0.01 now if I then want to know but from the linearization the function value is in the points $02 \ 0$.

0 . now 1.01 from the linear approximation then you do the same thing as you did before you have the function value of course so that is that $2 \ 1 \ 0$ plus the factor that you get the in this case the Jacobian times this may factor x minus x_0 so in this case 1.02 minus 1 first component 0.01 minus 0 second component and then you get exactly this factor but I have now to write it as in as in row $0.0 \ 4 \ 01$.

0.01 and that is 2.03 , 1.04 and here 0 . You have a question. No, you take the step length.

You are in the point 1.0 . You have a vector and now you take a small step. And you, just as we did here, for the tangent line, we had this linearization of a function of just of one value, you get as the $f \ x_0$ plus the derivative times x minus x_0 . In an all situation, you have the same thing. This is now a vector value thing. This is what we had as a derivative, but now it's the Jacobian. And then you do x minus x_0 , so you do the point minus. In our situation, it was a point one zero. And then you get these differences. You have a question. I had here 1 plus 0.02 for the x and 0 plus 0.01 for the y . And I plugged in this minus the point 1.0 . And then I get this vector. So that difference, just as I explained here, this is simply a difference between the vector minus the point to which I make to the nearest. And then you have to multiply your Jacobian with

that difference. So then it works just in the same way. And that way you can see that this Jacobian describes how this function changes after small steps. But now it's not only one value is changed, but the whole vector has changed a little bit. And it is important that you have this matrix because in some directions, in some components you have more changed than in other components, but that depends on these partial derivatives. Now, you have to try this yourself, and then this is easy. You only have to know what the definition is, and you have to keep in mind on every row. The function component remains the same, and you differentiate with respect to the different $x_1, x_2, x_3, \dots, x_n$, and then you do that for every row. Now there is a nice feature of this, and that is the following thing. Once you know the chain rule, once you know this, the chain rule gets a very easy form. So suppose I have two functions, F from $\mathbb{R}^D$ to $\mathbb{R}^N$, and I have a function G from $\mathbb{R}^N$ to $\mathbb{R}^M$. And I'm looking at a new function. The H is G after F . So the H of a vector X is I first apply F . I get a vector in $\mathbb{R}^N$. And on that $\mathbb{R}^N$, I apply G , and then I get a vector in $\mathbb{R}^M$. So it's simply the composition of these two functions. So this is much more general than what you had in matrix multiplicity. because there, this relationship is always the same, but here it can all change. In every position, it's different. Now, what you now want to know is the relationship between the Jacobian of F , G , and H . At least that seems to me a logical question. What is now the Jacobian of H depending on, now, what you can prove? I will not prove this. that this is and it's just a chainable as you know it the DG so the Jacobian of G evaluated in $F(x)$ times the Jacobian of F evaluated in x so it's just what you did in for functions of one variable now if you had the G of $F(x)$ for function of one variable then the derivative would be the g' of $f(x)$ times f' and now you have it for vector fields then you get the jacobian of g in the point $f(x)$ times the jacobian of f in the point x and then you can check whether this works h is a function from $\mathbb{R}^D$ to $\mathbb{R}^M$ so the jacobian of h of h in a point x is an M times D matrix the jacobian of f in a point x is an N times D matrix and the jacobian of g in the point $f(x)$ is an M times N matrix and if I now use these sizes just in that chain rule then you get that everything works, hopefully. This is the DH , so this is M times D . This thing is M times N , and the DF is N times D . And you see indeed out of the point of an M times N with an N times D matrix that exists, and it's an M times D matrix, and that's exactly this one. So in the end, then you get this very general chain rule in this way, because it works for several functions and you can do quite a lot with it. It works in the end with help of these Jacobians. But in most cases where you have to do it also earlier, then you could use these chain rules much more easy and then you do not have to do all this technical things. But in the more channel situation, you would get this. And then it looks very much like the normal chain rule. You have a question? It's a good question if the Jacobian is an invertible matrix. That's very often in the condition. So we will use it quite a lot, and this Jacobian, it plays a wall in a section that I skip, 13.8 from the book, and that's about implicit function theorem, that you have an description like this. You have a sort of equation, $x^2 + 2y + e^{xz} = 0$, and you want to know can I write this as a function from x and z as a function of y and then you have to look at jacobians and certain jacobians not of the whole thing but of something else I will not explain it and then the condition will become that this jacobian has a certain jacobian has to be invertible so it is an import condition. If you want to know all the details or a little bit more information, then you have to do, see section 13.8 about your implicit function theorem, but I will not cover that in this course.

Okay, it's about time. Then I didn't do Taylor polynomials yet, so I will do that a little bit next time, but it's quite technical. It's also a little bit because I didn't have slides, so I had to write everything on the blackboard. So after this, please go to the instructions to practice with the material also from now from 13-6, and you can also try, you just ask your instructor to say something about how to compute Taylor polynomials, and he or she will explain, so that's not a problem. The material of today, what I cover today, does not belong to the material for the mid-term test. after the lecture I will put an old midterm test on canvas but please note that the material for that test of

last year was more than what it's now that's because of all these holidays in the beginning so you only had a few lectures so the last ones you do not last exercise do not you do not have to be able to do them but you get some impression how such a test looks like. I will also publish, but maybe not today, the formula sheet that is available to you, but that you have information what's on this. And next Thursday, we first have two hours of lecture, and after that you have the midterm test, and yeah, you still get information on this where you have to be and so on. Oh, did you already have this email? Okay, you already have it. Then it's okay. If there's still any questions, ask your instructor or ask me. There's also some little time next Thursday. Okay.

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Good morning, welcome to this lecture. Very select audience, but welcome. What are we going to do today? I will continue in chapter 13 of the book, mainly about differentiation of functions of several variables. And the first subject that I want to do, and that will come back also later on in the lecture, is the chain rule. And you know the chain rule from calculus for functions of one variable. Then you had a function, let's say h of x , and that was the composition of two functions, the f of the g of x . And if you then wanted to differentiate the function h , then you know you had the chain rule, the derivative of h was the derivative of f applied in g' times the derivative of g in x . What we want to do is to generalize this to a situation where you have functions of several variables. And I would say it is not much more difficult, but the bookkeeping gets more and more terrible as you get more variables. First, why do we want to do this? Suppose that you have, here is a graph of a function of two variables, and this is simply the landscape of something, of where you are hiking. And you are hiking on this path. Then what is interesting to know is, at a certain point, how steep the descent is. You're going down from a mountain. If you take the other way, you hike up to the top of the hill. And then you want to know how large this ascent is. And that has something to do with the derivative of that curve, of course. And now the question is, how can you compute that? So in that example, we have to think about a function that is the call of function f , and that is dependent on two variables, the two coordinates x and y . And you can also think of it as the f of a vector of two variables. Now I write it like this. And now I assume that on a certain time interval t , from let's say zero to five hours, I know where I am. So I know the position. So I know that x is a vector, has an x component and a y component. Then you can compute, of course, by composition, what your altitude is at a certain point in time. And that is then h of t like that one, is the f of x t and y t . And you assume you know these. You know exactly where you are at a certain point in time. And if you now want to compute the derivative of h , and that has something to do with how steep the ascent or descent is, then you want to differentiate this with respect to t . In that case, you measure with the time. And that is, in this case, and now I apply for the first time this chain rule, the derivative of f with respect to the first component, and then in the point x t , y t , times the derivative of x with respect to t , let's write that as x' t , plus, but now let's write this dx/dt , otherwise we get confused, plus, and that you should not forget, I have also a second component, and I also have to take the change there, so the partial derivative of f with respect to the second component, in our point where we are at time t , times the velocity at that point in time in the y direction. So you need both these two components. And it is not very difficult to show this. Let's call this for a moment g , because then I have the h still available. What I would do in this case to compute this derivative, I would do, according to the definition, the limit for h going to zero, and this is just an ordinary limit, because g maps from the interval from zero to five to $\mathbb{R}$, so it's just a function of one variable. So this derivative at a certain point in time is the value at time t plus h minus the value at time t divided by h , and then taking the limit for h going to zero. Now, this is, and now I do the trick, the limit for h going to zero, and I simply write what this is, the g t plus h is the x t plus h , y t plus h , minus the f of x t , y t , divided by h . And now nothing has happened, and now I do a trick. I write this as the limit for h going to zero. I put in another term and I add that and subtract that in the numerator of the expression. So I write this as the f of x t plus h , y t plus h , minus the f of x t , y t plus h , divided by h , plus, I've subtracted this one, so I have to add the same term, the f of x t , y t plus h , minus this one, f x t , y t , and that divided by h . So at first it seems that I make it more difficult by adding extra terms, but now I do a trick. So instead of writing here dividing by h , I multiply here with a term where I divide by h , and now I put here and here the same expression, and that is in this case x t plus h minus x t , like this. And I do the same trick over here. So instead of dividing here by h , I divide here by h , and I put here y t plus h minus y t , and I put it also here. So from a simple expression I make this terrible thing, but now I can take the limit for h going to zero in a quite easy way. Because if you look at this term, you see that this y t plus h is here the same, and I look here at the difference in the function x t plus h and x t divided by that difference. If I take the limit for h going to zero, then this will go to the partial derivative of f with respect to x in x t , y t . This will go to the derivative of x with respect to t , simply the normal differential quotient. And similarly here, but now you have, of course, differentiation with respect to y , because now you look only, the first components are the same, and there is this step of h in the y component. So here I get the partial derivative of f with respect to y of x t , y t times dy/dt . Now this is a way of proving this chain rule. I only do it once, because it's much more important that you know how to do this chain rule, and this was the easiest option, the easiest situation. You have to be able to do it in much more difficult situations, but it's not very difficult. That's the good news, I would say. Let's try to explain it. What I do is the following thing. I now look at another example, because in each situation you have to design your own chain rule, but it's always according to the same scheme. I now assume that I have a function f that depends on x and y , but x is a function of two other variables, s and t , and y is also a function of two new variables, s and t . So I'm now looking at a function g depending on s and t , and that is simply in f I plug for x u of st and for y v of st . Later on I will do an example. And then I have a function of s and t . And of course you can do partial differentiation here, but it's related with the partial derivatives of f with respect to x and y . And what I normally do is I write here such a dependency diagram. I call this for a moment z . And z depends on x in this case and on y . And the way how it depends is according to the function f . And the x depends on s and on t , and the y also depends on s and t . And then I have my diagram complete. So I go from this variable that describes the outcome to

the variables that are really free, that are the s and the t . Now in this situation you see two ways how you can get from z to s . That is this road, and it is this path. And what you now say is that the partial derivative of z with respect to s is, in shorthand, the partial derivative of z with respect to x times the partial derivative of x with respect to s . That's part one, plus the partial derivative of z with respect to y times the partial derivative of y with respect to s . If you want to know this for the variable t , and then I think you can guess how you do this, this expression is then the partial derivative of z with respect to x times x with respect to t , z with respect to y , y with respect to t . However, you should be a little bit careful, because the question is in which point should you evaluate these partial derivatives. So it is better to write it this way. So this is just thinking about how does it work for the scheme. But if you then look at the partial derivatives of g with respect to s , that is then, if I just substitute that dz/dx , that is simply the partial derivative of f with respect to x evaluated in u, v, s, t times, and then dx/ds is du/ds evaluated at the point s, t . So here you have to plug in the values of s and t . Here you have to plug in the values of u and v . So you have to plug in s and t in u and v , and that you have to plug in into the partial derivative. That's the same as we had over there in the normal chain rule that you know from calculus. Here also you have this derivative which you have to evaluate in g, x times g' in x . Now here it works the same way. Of course, I also need this term plus the partial derivative of f with respect to y evaluated in the point u, v, s, t times the partial derivative of v with respect to s , and that in s, t . And if you then want to do the partial derivative with respect to the t , I think you can guess how you have to do this. These are called also Jacobian matrices. You can simply do this multiplication. This also always works if you do it like this. I think that the scheme with the, where you show how everything depends on each other, is the easiest option. But in the end, you will find then this. Let's do an example that is not that easy, I would say. I have a function f from $\mathbb{R}^3$ to $\mathbb{R}$. And now I have another function g of x, y is the f of y squared, x, y minus x squared. This looks a little bit, so for the first variable, you plug in y squared, for the second one, x, y , and so on. And now I want to have a second order partial derivative of this g . Now, in this situation, I draw the dependency diagram. So here's my f , and I have here, I write this as variable 1, because it has no name. Here I write variable 2, and here I write variable 3. And variable 1 only depends on y . Variable 2 depends on x and y . And variable 3 only depends on x . Now, what I now am interested in is to compute the partial derivative of g with respect to x . Okay, then you look how do I get for my top variable, so here's my g function, and I have this part, and I have this part, and not more. And so what do I get? I write instead of the partial derivative with respect to variable 1, I write $df/d\text{var } 1$, it's simply the f_1 . I write this notation, f_1 , the partial derivative with respect to the first component, the second component, and the third component. Yeah, still I'm still with you? Yeah, I'm here, but you can still follow? Yeah? Okay. So, now, if I do that with respect to the first component, I can do that, but then I have to multiply with the derivative of y squared with respect to x , and that is of course zero. So that if you take this part, you find nothing because this derivative of this variable with respect to x is simply zero, it doesn't depend on it. So also if you take it into account, you will not get the contribution. In this case, you get the f_2 , the partial derivative of f with respect to the second component, so the f_2 , of course in the point y squared, xy minus x squared, times the derivative of the second component with respect to x . And that is y . Plus, I have to go to all components, so then I get the partial derivative of the original function f with respect to the third component, and then y squared, xy minus x squared, times the derivative of the second component with respect to x . So I get the derivative of minus x squared with respect to x , of course, and that's minus $2x$. Like this. We go? Okay, now I make life more difficult because now I'm going to look at the higher order derivatives and then you see that things really get terrible in these chain rules. Because I'm now going to differentiate this thing, so this expression, with respect to y . And now I'm, okay, I have to differentiate this whole expression by y , and every time I have to look, I get of course now a product rule, derivative of this times that plus derivative of that times that, and so on. And I get lots of terms. So the first one is, so I simply start here, here I get the partial derivative of f_2 with respect to the first component in this point, times the derivative of that y squared, and that's $2y$, so this is times $2y$ squared. But I'm not done with this first term yet. Plus, then I get the f_{22} , this one, y squared, xy minus x squared, times the derivative of this thing with respect to y , and that is x . So I get times xy . Then I get plus the partial derivative of the f_3 , but then I have to multiply with the derivative of minus x squared with respect to y , and that is of course zero, because this third component does not depend on y . So I can save time, simply okay, I have to multiply then that term with zero, so I simply do not write it down. And then I should remember, oh, there's a product rule, I have a third term, that I differentiate this times that, so I get the plus $f_2 y$ squared, xy minus x squared. That is because I have to differentiate that factor y times the other term. Then I've only have done this, and then I still have to do the other one, and there I get plus, now I have to differentiate with respect to y first this one, that's $f_{31} y$ squared, xy , so at this point, times $2y$, so times minus $4xy$. I hope that I've done this okay. Yeah. And then I have to do it with respect to this one, plus f_{32} in that point, derivative of this thing with respect to y is x , times this minus $2x$, is minus $2x$ squared. Then I have to differentiate with respect to the third component, but that is independent of y , so that does not give anything, so that I have, it's simply times zero. And then I have to differentiate this minus $2x$ with respect to y , and it also gives zero times this f_3 , so that also gives no contribution. So at this point I'm ready, then you can do even more partial differentiation, but you see very soon that you get completely crazy out of this, because this goes and goes and goes and goes. So it is possible to do everything, but it's not that nice. No, the f_3 with

respect to the second component. Here? No, because that term here, if you, this is f_2 with respect to the third component, I differentiate with respect to y , that's nothing. And then the other term comes from that I differentiate this thing, it's 1 times this. And that's the strange thing, so it's a good remark. You get here for this higher order partial derivatives, and you see that these first order partial derivatives get into the formula. And you cannot help that, that is part of the deal. Now this was complicated. The important message to you is that you have to simply practice this yourself with help of the exercises, and then you will notice that the bookkeeping is indeed a difficult thing, but if you're precise and you know what to do, it is not really very difficult. It's only time consuming. Okay. Here is some example that is easier on this slide. I have here a temperature profile in a region. So I have a coordinate of a position, so I'm in a balloon, I have three coordinates, x, y, z , so also the height, temperature depends, and I have a time t . So it is the position, I'm in a balloon at a certain time, I have this measurement of the temperature. And of course you make a tour in this balloon, and now you assume here that the balloon rises according to time, where x_t is t , y_t is just a constant, so y is, and z_t is $1 + t^2$. And now you want to know the range of change of the temperature, so if you go up, you expect that it's getting colder. But to know how fast it's getting colder, you are interested in this time derivative. Now in this situation, you have the following thing, this is this temperature, it depends in principle on x, y, z , and t , and the x depends on t , the y does not depend on t , that is simply a constant, and the z also depends on t . And what I'm interested in, and I hope I did this here, yeah, is to know something about the derivative of the temperature with respect to the time. Now this is, then first here is this function f , this function t of x, y, z , and small t that was given, that is xy over $1 + z$ times $1 + t$. And if I take this one, so I have to differentiate this with respect to x , so that's y over $1 + z$ times $1 + t$, times the derivative of x with respect to t , and that is 1 because x is simply t , so it's 1. Plus, now I can also differentiate here partially with respect to y , but y is not dependent on t , that is simply a constant, and the z also depends on t . And where I'm interested in, and I hope I did this here, is to know something about the derivative of the temperature with respect to the time. Now this is, then first here is this function f of, this function t of x, y, z , and small t that was given, that is xy over $1 + z$ times $1 + t$. And if I take this one, so I have to differentiate this with respect to x , so that's y over $1 + z$ times $1 + t$, times the derivative of x with respect to t , and that is 1 because x is simply t , so it's 1. Plus, now I can also differentiate here partially with respect to y , but y is not dependent on t , that is simply a constant, and the z also depends on t . And where I'm interested in, and I hope I did this here, is to know something about the derivative of the temperature with respect to the time. Now this is, then first here is this function f of, this function t of x, y, z , and small t that was given, that is xy over $1 + z$ times $1 + t$. And if I take this one, so I have to differentiate this with respect to x , so that's y over $1 + z$ times $1 + t$, times the two or three variables, but of course if you have more than two or three, everything works the same way. You can also then compute partial derivatives, but the nice interpretations that I often have in the two or three-dimensional case, that is then more difficult. But the technicalities work exactly the same also if you have a function of five variables, and the idea is simply, when you do partial differentiation, you only keep, if you have a function of five variables, you keep four of them constant, and you only look at the change of the function if you take a small step in that only one direction that you do not keep constant. And then you simply look at differential coefficients only in that direction, and that you can then call the partial derivative in that direction. But that can always be done, and the technicalities are just the same as we did for the two- and the three-variable case. It only has more variables. So you do not really have to worry about it, but interpretation is a little bit more difficult, and simply the reason is that to make a graph in four-dimensional space remains a problem for us. Now this is also what I did last time, so that was this condition if you have these mixed second-order partial derivatives, that if the second-order partial derivatives are continuous in a certain point, then it doesn't matter whether you first do partial differentiation with respect to x and then with respect to y , or the other way around. So it's not always true, that was right last time, but it's true for friendly functions, I would say. And you can check that, because you simply check whether these partial second-order derivatives are continuous, whether they are simply friendly functions. So for example, if you have polynomials in several variables, then this is always satisfied. This we already did. Okay, I come now to the important subject, the gradient. And it's so important that I start the next hour with it. So I see it's exactly 9:30. So let's first have a break, be back at quarter to ten, and then we get to the gradient. So, let's start again. As I said already before the break, we now have in this hour a very important subject, and that is the notion of the gradient. And it is easy to define, easy to compute, but the importance is that it has a lot of interpretations, and that you should know, because that is very important. That's why it's interesting. For the moment, we look at a function of two variables. So we have z as a function of x and y . And then, of course, you can compute the partial derivatives with respect to x and with respect to y . And now I say that's the gradient of f in a certain point x_0, y_0 . And they also write it as the gradient $\text{grad } f$ in the point x_0, y_0 . And that is a vector depending on x_0 and y_0 . And it is the partial derivative of function f with respect to the first component evaluated in x_0, y_0 times the first unit vector, plus the partial derivative of f with respect to y in the point x_0, y_0 times the second unit vector. If you have a function of three variables, so if this would be here, comma, z , and you would also have here a point x_0, y_0, z_0 , then you would also have the partial derivative of f with respect to z times x_0, y_0, z_0 times the third unit vector. If you have even more variables, four, and then you're looking in four-dimensional space, so then you

also have a fourth unit vector, and you can continue this. This is simply the definition, but let's now look at what this means. And I hope that, oh, I should be very quick, yeah, that this picture helps you to understand what's going on. So I have here a function of two variables, and it's given by this expression, but that is not very important. The nice thing is you have here two mountains, like this shape. And what we did until now was... like this shape. And what we did until now was we can represent that function, of course, by its function description, and we have here the graph, and we have here the picture of its level curves. And then these level curves that we like that, because then you have the situation that you could depict that in a two-dimensional picture in a much easier way, and making these graphs without a computer is also for me too difficult. At least that I can just say like that. And then, yeah, I hope I'm still able to do this. Now the question is, what is now the gradient? Now, what you do now, you take a point x_0, y_0 , so somewhere a point, and you have a function value, and you are on a certain level curve, and what you now do is you map that to a vector that points exactly in this direction. So the value of the partial derivative in the x direction, and the value of the partial derivative with respect to y in the y direction. So in every point in this plane, you get a vector. And here, in this picture, that is displayed by a red arrow, and if it is a little bit larger, so if the length is a little bit larger, it's depicted by a little bit larger arrow, and if it's very small, then it's a very small arrow, but it has a length and a direction. Do you see something about it? Do you recognize? So if I would depict the gradient on a sort of, in a graphical way, I would have to add to every point in the plane this corresponding arrow, so it's full of arrows, and I only depicted a few, and that is called vector field, because it maps to every point in the two-dimensional space, x_0, y_0 , it adds to that a vector. So not one value, but two, because you have two components, one in the x and one in the y direction. So to every point, I map a vector, and then you call that a vector field. Do you have a question? What happens if there's a point where the... Okay, the question is, what if the partial derivatives do not exist? I now assume that they exist, and if in points they do not exist, you cannot do this. So I only look at, really, the regular situation, and if you compare that with these pictures I showed you, these are my photos, then this holds in such a nice hilly landscape, but it does not hold for these strange walks that are vertical, and then I'm lost. Then you can forget about this. Not completely, but locally. Now, I hope that you see here the following effect. If you think about this landscape, then these are the level curves, so I have my topographical map, and what I now recognize is that these arrows seem to be perpendicular to these level curves. Every point on such a level curve, where is it, this gradient is pointing, is perpendicular to that level curve. That is one of the properties, and that has very interesting consequences, because if you want to go up as steep as possible, you always go in the direction of the gradient. You walk in the direction of the gradient. Let's be more specific, because now this is too much of a riddle. This is the same definition. And here are the properties that are interesting. The gradient is, my claim, always perpendicular to the level sets. So I said to the level curve, but that is for functions from $\mathbb{R}^2$ to $\mathbb{R}$. But if you have a function from $\mathbb{R}^3$ to $\mathbb{R}$, then you do not have level curves, but level surfaces, but they still have a normal vector, and that points in the direction of the gradient. I will show you that the gradient points in the direction in which f increases the fastest, so where you have the strongest ascent. And that the maximum rate of increase is determined by the length of that gradient. So if the length is very small, then you only go very slowly, and if the length of the gradient is very large, you go very steep. Okay, now I have to get my notes. First I do an example to show you that computing such a thing is not very difficult, so that you get a little bit of confidence in your gradient. So I have the following example. I have a function now from $\mathbb{R}^3$ to $\mathbb{R}$, and the variable w is equal to the function value of x, y , and z , and that I take the example that this is $x^2 z + y e^z$ to the z minus y squared times the sine of x . If I then want to know what the gradient is in a point x, y, z , then what I have to do is I have to compute first the partial derivative with respect to x and multiply that with the first unit vector, and then continue. So I get here, if I take the partial derivative with respect to x , I get here $2xz$, here 0 , here minus y squared cosine x times the first unit vector, plus then I take the partial derivative with respect to y . Here I have no contribution. Here I have e^z to the power z minus $2y$ sine x times the second unit vector, plus, and then I have to differentiate with respect to z , then here I get x^2 , and here I get plus $y e^z$ to the power z times the third unit vector. So if I have a point here, a point in three-dimensional space, and I take a, b, c to be a half $\pi, 1, 0$, then the gradient of f in the point a, b, c , so in the point a half $\pi, 1, 0$, is, I just plug in these values for x, y , and z , so here I get a half π times 0 , so 0 , and here I get minus b squared, that's minus 1 , times the cosine of a half π , that's 0 , because this is 0 and the cosine π is 0 , 0 times i , plus here I get e^z to the power z , that's 1 minus 2 times the sine of a half π , so that's 1 minus 2 times j , and then plus, and here I get x^2 , that's a quarter π squared, y times e^z , y is 1 , e^z is also 1 , so plus 1 times k , and you can also write this if you want as a column vector. The first component is 0 , the second component is minus 1 , and the third component is 1 plus a quarter π squared. So in this situation you have a function from $\mathbb{R}^3$ to $\mathbb{R}$, and then the corresponding gradient is a vector field that adds to every point in the three-dimensional space an arrow showing the vector in the $\mathbb{R}^3$ with three components. So it works like this, and of course in every different point in three-dimensional space this gradient vector is pointing in a different direction, or maybe pointing in a different direction, it depends of course on the description of the function. Now, first thing, my claim was the gradient is always perpendicular to the level sets, and we also saw that in that picture that suggested it, that that was the case. Why is that the case? Now, first of all, I want to describe to you the relation with the chain rule, and then I almost have already that

other result. So suppose I'm now in a situation, I do it for the two-dimensional case. So I have a function f depending on x and y , and x and y depend on time t . And what I want to do is I want to differentiate with respect to t the f of $x(t)$, $y(t)$. Then you know from the first hour how we have to do this. This was the easiest way of the chain rule. So this was the partial derivative of f with respect to x in $x(t)$, $y(t)$ times the derivative of x with respect to t , and I write it now as x prime t , plus the partial derivative of f with respect to y in $x(t)$, $y(t)$ times y prime t . This was what we had, but now you can write this as the inner product of two vectors of length 2. Here I write the partial derivative of f with respect to x in this point, and the partial derivative of y in this point, and here a comma for the inner product, or simply the dot product, and you write it as a certain dot. Now I do it as a pairing. I don't know what, what did you do in linear algebra with the dot or with these? Then I also do the dot. x prime t , y prime t . This is exactly the same, because this is this times this, this term, plus this times this, that's that term. So you can see it in this way, but now we have an interpretation of this. The first factor is the gradient of f in the point, the vector x at time t . Now I've white the vector x with these two components, x t and y t . And the second vector here, that is the velocity vector, because it's the time derivative with respect to each of the components. So this is the, I can write it like this, x prime of t or as v of t , the velocity vector of that parameterization. So the chain rule just says that this derivative is just an inner product of the gradient with the velocity vector. Nothing more, nothing less. Now, what do we now do? We take a level curve, so gradient perpendicular to level curve. It's now pretty easy, because I have my function z as a function of x and y , and I take a level curve. Let x t , y t , t is in some interval, be a parameterization of level curve. What do I get? Now I know then that the f of x t , And now I am interested in the question, what is now my ascent if I walk to start hiking in that direction? That you call the directional derivative in the direction of u . Now, if I do that in the point a , b , what I do is the following. I take simply a small step in the direction of u , and I look then at the corresponding differential quotient. So I look, I start in f a plus h u_1 , b plus h u_2 , minus the f of a , b , divided by h . So I look at that, I take a small step in the direction of u , and I simply take the root t times the vector u , that is t u_1 , t u_2 , and then I differentiate that at time t is zero. So I have to do this at time t is zero, and then I get this result, because this is exactly the same. But what is this? We have the chain rule. So this is the derivative of this in t is zero, partial derivative, so I get here the f_1 in a , b , so the derivative of the first component. Let's do it like this. The first component in t is zero, so that's why I get a times the derivative of this with respect to t , so I get u_1 , plus chain rule, the partial derivative of f with respect to the second component, evaluated at t is zero, so in the point a , b , times the derivative of this with respect to t , and that is u_2 . And what you see here is I get again an inner product. I get the inner product of the gradient of f in the point a , b with the vector u_1 , u_2 . And that is what I would say the slope in the direction of u . If I have only these partial derivatives, these are also directional derivatives, partial derivatives, but then you take simply a step in the direction of, I shall do it for you, in the direction of x , that's simply the directional derivative in the direction of x . And the partial derivative with respect to y is simply the directional derivative in the direction of y . And then you simply get that also, because then you get the vector $1, 0$ and $0, 1$, so then you get exactly these components back. So the directional derivative is not that much more difficult than these partial derivatives, but partial derivatives are special directional derivatives in the x and the y direction. This is in any direction. What you see is that you get here the inner product of the gradient with that direction. And now I ask myself, what is now the largest possible value that I get out of this? Because this is a number, of course, it's an inner product, so this is a scalar number. Now what's the largest possible value? That is, of course, the absolute value of that gradient of f in a , b inner product with this vector u , the length of that, the absolute value of this thing. But that inner product can be written as the length of the gradient vector times the length of the vector u times the cosine of the angle, so I have here, for example, this is gradient vector, and here I have the vector u , and I have here an angle θ between it, and you know that this inner product is the length of the first vector times the length of the second vector times the cosine of θ . This length I know, because I said, okay, this u has to be a vector of length 1, so this is just 1. So this is the length of the gradient vector times cosine θ . This is also, if I am in a point a , b , a fixed vector, so it's also a fixed length. How should I take u to get the highest possible value here? Then I should take care that this cosine of θ is equal to 1. So I choose θ to be 0, so where I go in the fastest direction up, where this directional derivative is the largest, is when u exactly points in the direction of the gradient. Because then this is 1. Where do I go down fastest? That's where this is as small as possible. Now I shall not do this, but I do it without this length, but then I also take into account. Then you get, mine in the direction of minus u , if this angle is not 0, but it's 180 degrees, π , then this value is minus 1 and you go down as fast as possible. So in the direction opposite to the gradient, your descent is the largest. You go down as fast as possible. And you also see if θ is equal to 0, the directional derivative is 0, so you do not change altitude in a way, and that is in the direction of your level curve. You see exactly that corresponds with θ is π over 2, 90 degrees, then you do not change. So out of this thing, you see that immediately, and you have to think, okay, if I have to compute such a directional derivative, I simply take a vector u , I take care that it has length 1, because then you have to do otherwise you're measuring in a strange way, you do not know what you measure. And then you simply take the inner product of the gradient with that vector u and you get your directional derivative. Okay. I go back to this thing here, this one. So let's have a look here. How does it work? We had here these level curves. We see that so often, this is a sort of picture of your gradient, but that is a vector

field, because in every point I had to draw a vector, but then I get the completely red figure, so that is not useful. But you see every time here this vector is pointing in a direction where it's going up the fastest. So here, like this. You also see points where the gradient is the zero vector. I see here a point, here a point, but also here a point. They correspond to this top, this top, and the point in between here is the saddle point in between. You are between these two tops. And that's a very special point, because if you walk on this thing and then go down on the other side, then this is the highest point you will be. But if you walk from this top to that top, it will be the lowest point where you are. So this is sort of a solution of a max-min problem. It's the maximum value of where you try to keep as low as possible, and it's the minimum value if you try to keep as high as possible. That's also depicted here, because you see here these, this is this road where you go like this. And you see here these arrows are pointing in the direction of the saddle point, but they go, here they get larger, because there you go steeper, and then it goes a little bit less. Here you also go in this direction. So here you go up, and if you walk in that direction, you go down, because you walk against the gradient. If you look at this situation from the saddle point, you go up here to that maximum. If you walk here, you go up to that maximum. So in a way, if you, and if you want to get, you want to take the steepest ascent here, so you really want to hike with the fastest, with the largest slope, then you always follow the direction of the gradient, and then you go up in the fastest, fastest way. Now, so I hope you are able to make that connection between what is happening in the graph and with this gradient and with these level curves, because that's important. Okay, then I will do, let's do one example, a little bit different. Where do I have it? Okay. Application. I have a cylinder, and that is in three-dimensional space, $x^2 + y^2 = 1$. So this is the, if I would put this in the xy -plane, this is a circle with center 0, 1 and radius 1. So it's this circle, but of course this cylinder is going here into the wall, so you have this tube going into it. And then I also have a sphere, $x^2 + y^2 + z^2 = 4$. You can imagine this sphere. They intersect. You have this big sphere and you put this cylinder through it, and you will have intersection points. So that will be a certain curve. One of the points, so I take the intersection of both, so the intersection curve, contains the point 1, 1, square root of 2. You can easily see that, because if you plug in here x is 1 and y is 1, you get here $1^2 + 1^2 = 2$, so it's on the cylinder. And $1^2 + 1^2 + (\sqrt{2})^2 = 1 + 1 + 2 = 4$, it's also on the sphere. What I want to know is, so I have now a point on this intersection curve, what is the tangent line to that intersection curve in this point? Now one solution would be, try to parametrize this intersection curve, but that might be difficult. But we do not have to do that, because to get that line we know already that the point 1, 1, square root of 2 is on it. And we also know that I have these two surfaces, so my cone and my ball, and I know that in that point 1, 1, square root of 2, I also have tangent planes both to the cylinder and to the ball. And they will be two planes. And the intersection line of these two planes will be the tangent line. You have to think about that tangent line has to be in both of these planes, because it has to be tangent to This one is perpendicular to both gradients. So the tangent line is the line x, y, z . I have a supporting vector, that's of course my vector 1, 1, square root of 2. And then I have my directional vector plus λ times my vector 0, square root of 2, minus 1. I hope I did it okay. Yeah. If you do not believe me that it's possible, then it is also if you have these gradients and you have this point, you can simply get equations of these tangent planes to the cylinder and to the sphere. That's not very difficult. For this one, for the cylinder, you get the tangent plane in 1, 1, square root of 2 to cylinder is $x = 1$. That is this plane here, vertically where x is equal to 1 and y and z are everything that you like, because you have here this cylinder that is getting in. For the tangent plane in 1, 1, square root of 2 to the sphere is, that's also not difficult, because I have my gradient over there that was this. So I get $2x - 1 + 2y - 1 + 2\sqrt{2}z - \sqrt{2} = 0$. Once I have the gradient, it's very easy to compute such a tangent plane, because I've taken care that the point 1, 1, square root of 2 is on this plane, then you get 0. And you have to look what is here the normal vector. That's the vector 2, 2, 2 times the square root of 2. That's exactly the pointing of the gradient. So once you have that gradient, you also have that tangent plane. The equation of the tangent plane. Now you can try at home that if you just compute with normal linear algebra the intersection line of these two planes, you will get exactly the line I just described. So once you have these two tangent planes, and we've now computed them with help of the gradients of a function from $\mathbb{R}^3$ to $\mathbb{R}$, you can do this kind of things. So messages, gradients are very important. You can now make the exercises at least of the section, I think it's 13.5 on the chain rule and also 13.7 on the gradient. And I think both are very important. Next week I will come back to 13.6 and then I will finish chapter 13. Okay. I hope it was a little bit... But we have to stop.

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Good afternoon, welcome to this lecture on the math of the math 2 course. What we're going to do today is to continue in section 13 of the book, so we are studying functions of several variables. And what we want to do is, in this course, mainly how can you generalize differentiation and integration to that kind of functions? That is the main theme. And to be able to do that, we first had to study the notion of a limit that we did last time. I want to repeat it a little because I think it's the most difficult notion that we will have in the course. And then I will continue about how you can generalize differentiation, and that is with so-called partial derivatives. I have to look up my notes here and here. Okay, first, very difficult, what was the notion of a limit for a function of two or more variables? And I try to show that with a picture. I showed this one already last time. And that is this picture. You have here this function that is the sine of the square root of $x^2 + y^2$ divided by the square root of $x^2 + y^2$. In the point $0,0$, that function is not defined, but it tends to 1. And what does that mean? Now, I think that this picture clarifies that a lot. You have to say, okay, that function gets as close to 1 in a neighborhood if you get close enough to the point $0,0$. Now, first of all, for a limit to exist, it has to be possible to walk through the domain in the direction of that point. If you have just an isolated point somewhere and you cannot walk to it through other points, that is also not a notion of a limit because you are interested in the function values not in that point, but very close to it. So it has to be possible to get as close to it as you want. That is important. And if that is possible, then the limit is, in this case, the number 1 because what you say here, this is at height 1, and you say, if I take some small number, I call it epsilon, and I look now not at the equation $f(x,y)$ is equal to 1 because then I do not have the solution as I hoped, but I look where is the function between $1 - \epsilon$, let's say between $1 - 10^{-3}$ and $1 + 10^{-3}$. Then you get, of course, some points in your domain. Now here it is depicted where the function is everywhere larger than 0.9, so $1 - 10^{-1}$. Now then you see that only this top gets out of this plane, and you get here this little circle if you project it on the xy -plane. If you take an xy somewhere in this circle but unequal to $0,0$, then you are sure that the function value is less than 10^{-1} from 1. That is what you are now for sure. Now, what you now, I did now an approximation with epsilon $1/10$, but you can make that as small as you like. Of course, what will happen if you take that plane z is 0.9 and you lift it a little bit up to z is 0.95, then of course you get here a circle over here and here also a smaller circle. And then if you go to z is 0.99, you still get a little circle, but it gets smaller and smaller. And that is the idea then of the definition. So let's now get back to that definition. We need that, so this is x_0 is not an isolated point, so we have to be able to walk in the direction of, to get as close to x_0 as we want through the domain. And what we need is that for every epsilon larger than 0, we took in the previous example epsilon is $1/10$, but that is not enough. It has to work for every accuracy level. There exists a delta, that is the radius of that circle, such that for all x that are less than delta away from x_0 and belong to the domain, the function value in that point is less than epsilon away from the l . So you can get as close as you want to l by starting very close, less than delta away from x_0 . That's the idea. And if that is possible, then the limit is in this case the number 1 because what you say here, this is at height 1, and you say if I take some small number, I call it epsilon, and I look now not at the equation $f(x,y)$ is equal to 1 because then I do not have the solution as I hoped, but I look where is the function between $1 - \epsilon$, let's say between $1 - 10^{-3}$ and $1 + 10^{-3}$. Then you get of course some points in your domain. Now here it is depicted where the function is everywhere larger than 0.9, so $1 - 10^{-1}$. Now then you see that only this top gets out of this plane and you get here this little circle if you project it on the xy -plane. If you take an xy somewhere in this circle but unequal to $0,0$, then you are sure that the function value is less than 10^{-1} from 1. That's what you are now for sure. Now what you now, I did now an approximation with epsilon $1/10$, but you can make that as small as you like. Of course, what will happen if you take that plane z is 0.9 and you lift it a little bit up to z is 0.95, then of course you get here a circle over here and here also a smaller circle. And then if you go to z is 0.99, you still get a little circle, but it gets smaller and smaller. And that is the idea then of the definition. So let's now get back to that definition. We need that, so this is the x_0 is not an isolated point, so we have to be able to walk in the direction of, to get as close to x_0 as we want through the domain. And what we need is that for every epsilon larger than 0, we took in the previous example epsilon is $1/10$, but that is not enough. It has to work for every accuracy level. There exists a delta, that is the radius of that circle, such that for all x that are less than delta away from x_0 and belong to the domain, the function value in that point is less than epsilon away from the l . So you can get as close to it as you want. That is important. And if that is possible, then the limit is in this case the number 1, because what you say here, this is at height 1, and you say if I take some small number, I call it epsilon, and I look now not at the equation $f(x,y)$ is equal to 1, because then I do not have the solution as I hoped, but I look where is the function between $1 - \epsilon$, let's say between $1 - 10^{-3}$ and $1 + 10^{-3}$. Then you get, of course, some points in your domain. Now here it is depicted where the function is everywhere larger than 0.9, so $1 - 10^{-1}$. Now then you see that only this top gets out of this plane, and you get here this little circle if you project it on the xy -plane. If you take an xy somewhere in this circle, but unequal to $0,0$, then you are sure that the function value is less than 10^{-1} from 1. That's what you are now for sure. Now, what you now, I did now an approximation with

epsilon 1 over 10, but you can make that as small as you like. Of course, what will happen if you take that plane z is 0.9 and you lift it a little bit up to z is 0.95, then of course you get here a circle over here and here also a smaller circle. And then if you go to z is 0.99, you still get a little circle, but it gets smaller and smaller. And that is the idea of the definition. So let's now get back to that definition. We need that, so this is x_0 is not an isolated point, so we have to be able to walk in the direction of, to get as close to x_0 as we want through the domain. And what we need is that for every epsilon larger than 0, we took in the previous example epsilon is 1 over 10, but that is not enough. It has to work for every accuracy level. There exists a delta, that is the radius of that circle, such that for all x that are less than delta away from x_0 and belong to the domain, the function value in that point is less than epsilon away from the l . So you can get as close to it as you want. That is important. And if that is possible, then the limit is in this case the number 1, because what you say here, this is of at height 1, and you say if I take some small number, I call it epsilon, and I look now not at the equation $f(x,y)$ is equal to 1, because then I do not have the solution as I hoped, but I look where is the function between 1 minus epsilon, let's say between 1 minus 10 to the power minus 3 and 1 plus 10 to the power also goes to the sine of zero, that's zero, when x, y goes to zero, zero. This is by one, and this is by four. And also by four, the cosine of x plus y will go to one if x, y goes to zero, zero, because x plus y goes to zero, so this cosine of x plus y goes to the cosine of zero, and the cosine of zero is one. Well, then I know that this cosine, this goes to zero, this goes to one, so the sine of, this is first x minus y , I think, but that doesn't matter. That will go to zero divided by one is zero if x, y goes to zero, zero. So in a lot of situations, it's not much more difficult than in the one-dimensional case to prove what this limit is, only in situations where you have this zero divided by zero, infinity divided by infinity, or infinity minus infinity, the cases you know from the difficult cases you know from the calculus course, there you have to, maybe you have to use the definition. But do not worry too much about it, because I think this is quite technical and not, I would say, it's important for you to know that there are some difficulties with the limit, but I have to do some practice, but it will not play a major role in this course. Now, to get a little bit of an idea what can go wrong also in practical situations, I have here some pictures also out of nature, where you see that there limits also, I'll make it a little bit larger. This is also in the Czech Republic, but don't think that I have shares in the tourist sector of the Czech Republic, that's not the case. But they have quite strange landscapes there. And yeah, if you would be over here and you would walk in this direction, then you think that this is the limit, but if you are in that very small valley and you come from that side, you think it will be a completely different value. So if you have these strange shapes, for example these ones, then if you are here on top, you think, oh, I can walk in this direction and nothing will happen, but suddenly you fall down, and if you come from this side, you think that the limit is something completely different. So if you have these very strange shapes that may occur, then limits do not exist, and in that two-dimensional situation you have much more possibilities, because in the one-dimensional case you could only come from the right and from the left, but now you can come from any direction, and that makes life more difficult. Ok, now, so far, so good. I want to go to the next subject, and that is differentiation. You have a question? Yes, but the chance that you have to do that on a final test is very small. The chance that you get a question on limits is quite large, and then you know what you have to do, I think. I think I have been clear enough. Ok, let's, in the homework there will be questions about this epsilon-delta, because it's important to understand the definition. I want to go to the idea of differentiation, and first in the case of a function of two variables. And of course you have then to use the idea of a limit, but this is easier. I look here at partial derivatives, and in a way this mimics the situation of the case from calculus, of the one-dimensional case. So what I do is the following. I have some function depending on x and y , and I have the graph and suppose that that is this pink surface that is lying above the xy -plane. And now what I do is I intersect that with this blue vertical plane. I couldn't find the word. And there the y -value is simply fixed. So that's a plane y is 3 or something. Then they intersect according to a line. Here it's not very visible. Here it's the red line over here, but that's that other plane x is some constant, x is equal to 2, for example. Now suppose you are in this point and you have here this curve that only is in the direction of x because the y is fixed, so you have just a function of one variable, of x . And you would ask, what is the slope there? Then you would do the following thing. You would compute this limit. You would say, okay, I'm in a point xy . I leave the y unchanged and I take a little step h in the direction of x . Then I take the difference of the function value in that x plus h, y minus the function value of xy and divide that by h . Then I get the differential quotient that you know from calculus. And then the derivative there is just the limit that you get if h goes to zero. Now how we call that here, that is called here the partial derivative of f in the direction of x . So that's just pretending that y doesn't change and I only look at change in the x direction. You can do the similar thing in the y direction. Then you take the same point, but now you look only, you keep x completely fixed, so you are in that plane x is constant. You have this intersection curve and you take, okay, I want to know the slope of that tangent line. And then you take from the point xy a little step in the y direction here, y plus k , minus the function value in xy , divide by k , and you let k tend to zero. And then you get the partial derivative of f in the y direction. The notation of that is with this script D . So this special D . In Dutch you would call it chromade, but I'm not sure how that is in English. But that's for the partial derivatives. Now how do you do this in practice? Then it's also easy, because you can use, in the first part this x is simply constant and in the second definition for the partial derivative of y this, no, in the partial derivative with respect to x y is a constant and in

the partial derivative with respect to y is x considered to be a constant. So if I now want, at this example, that function $x^2 + xy + y^2$, and you want to know the partial derivative in a certain point, say, let's say in the point $(2, 3)$, of course I can use that definition, but I don't like it, because it's much easier. I simply get a formula for the partial derivative of f with respect to x in every point (x, y) by doing this. This, of course it depends on (x, y) , is, now what do I do? I differentiate with respect to x , pretending that y is a constant. So x^2 , $2x$. A constant times x , derivative of that constant. A constant squared, derivative with respect to x , zero. So I'm done, because I can simply use the sum rule for differentiation. That means that in the point $(2, 3)$, this value is $4 + 3$, is equal to 7 . Now if I do that with respect to y , then I think x is a constant, and I'm only looking at the situation where y is changing. Now I differentiate with respect to y , this is zero, because it's a constant, differentiated with respect to y . I have the constant x times y . I have to differentiate with respect to y , that's just x . And then I have y^2 , and I have to differentiate that with respect to y , so I get $2y$. And then if you do this in the point $(2, 3)$, you get $2 + 6$, is 8 . So in the end, in these points you simply get values, but for such a formula for the partial derivative in any point works very similar to what you do for differentiation in the one-variable case, only you consider every time the function constant with respect to all the variables to which you do not differentiate. So let's try it together in that other question. If you have $f(x, y) = e^{xy} \sin(x + y)$. If I now want to do the partial derivative with respect to x , what have I to do? Now I think this is a function e^{xy} times a constant times the sine of $x + y$. And then I use the product rule, because the product of two functions. So first the derivative of this with respect to x , that is e^{xy} times y in this situation, because I have to differentiate xy , chain rule with respect to x times, of course, this other function, plus, and I keep this one, and then I have to differentiate the sine of $x + y$ with respect to x . Now this is just the sine of $x + y$ plus a constant, and that derivative is the cosine of $x + y$. So here I get the cosine of $x + y$. So this is how it works. Not really much more difficult, but you have to keep in mind to which variable you differentiate and which variables you consider to be constant, because you are only looking in one specific direction all the time. Okay, now then this, you can compute what it is in $(0, \pi)$. That's not very difficult. I have to look in my notes. Yes. I have to go to these slides. And I look at this example. It's a little bit tricky, because to show you that also nasty things can happen, we look at this situation. And that's the function $f(x, y) = 1 - \sqrt{xy + y^2}$. And I write that as $y \sqrt{x + y}$. This function. Then first you have to find out what is the domain. So for which values of x and y does this expression make sense? And then you just see, oh, I have the square root of something. Now then that something has to be non-negative. So I get the requirements that $y \geq 0$ and $x + y \geq 0$. Four points will be in that interior, so that are here in this part and also in this part. But if I wanted to know what the partial derivative is in the point $(0, 0)$, I cannot do that, because in every neighborhood of this point there are also other points. So this is not an interior point of this domain. So what do you do in this case? What you only can do is use the definition. So you say I'm standing in the origin, and what happens with the function if I take a little step in the direction of x ? So you take the limit for h going to 0 of $\frac{f(h, 0) - f(0, 0)}{h}$. That's the only way what we can do. And that is, in this situation, a limit for h going to 0 . Now, what do we have? $1 - \sqrt{0} = 1$, so $\frac{1 - 1}{h} = 0$. So that's simply the limit for h going to 0 of 0 divided by h . So that is 0 . Yeah, you are with me? Now you have the strange thing. If I now walk here, for example, along $x = 0$ to that point, that's the limit for y going to 0 of $\frac{df}{dy}$ in the point $(0, y)$. And that is the limit for y going to 0 of $\frac{y}{2\sqrt{y^2}} = \frac{1}{2}$. That one doesn't exist. Only the one-sided limits exist, because this is $\frac{1}{2}$ times the absolute value of y . This denominator is always positive. So if you start with a y positive from above, you get something negative divided by something positive, so you get here minus a half. And if you come from y 's that are negative, you get here something positive divided by something positive, and you go to plus a half. So the value here is 0 for that partial derivative if you go from this side, you go to minus a half, and from this side you go to plus a half. So what you can say is that this partial derivative, the value is 0 , because that you have to do with the definition. So the value is 0 , and the partial derivative, so this function, y divided by $2\sqrt{y^2}$ is just $\frac{1}{2}$, the square root of y^2 , that one doesn't exist. Only the one-sided limits exist, because this is $\frac{1}{2}$ times the absolute value of y . This denominator is always positive. So if you start with a y positive from above, you get something negative divided by something positive, so you get here minus a half. And if you come from y 's that are negative, you get here something positive divided by something positive, and you go to plus a half. So the value here is 0 for that partial derivative if you go from this side, you go to minus a half, and from this side you go to plus a half. So what you can say is that this partial derivative, the value is 0 , because that you have to do with the definition. So the value is 0 , and the partial derivative, so this function, y divided by $2\sqrt{y^2}$ is just $\frac{1}{2}$, the square root of y^2 , that one doesn't exist. Only the one-sided limits exist, because this is $\frac{1}{2}$ times the absolute value of y . This denominator is always positive. So if you start with a y positive from above, you get something negative divided by something positive, so you get here minus a half. And if you come from y 's that are negative, you get here something positive divided by something positive, and you go to plus a half. So the value here is 0 for that partial derivative if you go from this side, you'll go to minus a half, and from this side you'll go to plus a half. So what you can say is that this partial derivative, the value is 0 , because that you have to do with the definition. So the value is 0 , and the partial derivative, so this function, y divided by $2\sqrt{y^2}$ is just $\frac{1}{2}$.

[illegible]

nicely, if these partial derivatives exist and are nice, that the slope of that tangent line has the value of the partial derivative. So if I go one unit in the x direction, I go the partial derivative of f with respect to x in the point a, b in the z direction. So this is easier if you go one unit in the x direction, you go the partial derivative of f with respect to x in a, b in the z direction. So that is that vector that points, for example, here it's pointing in this direction, but it can also point in the opposite direction, doesn't matter to me. So this one. I think it's here the right one because x , you have one here. You can of course also do that in the tangent vector, in the y direction. Of course then you take a step in the y direction, so, and not the x remains the same, so you take a step, the x doesn't change to y , you take one step in the y direction. And then what you, what changes in the z direction is then the partial derivative of f with respect to y . So you get here j plus the partial derivative of f with respect to x in a, b . I hope that that is clear, that you can simply say these slopes give tangent vectors to these curves. What you then, what I then would say is that this tangent plane is the following plane. x, y, z is, now I have a supporting vector, that is my vector a, b , function value in a, b that points to that point p . And then I have two directional vectors, one, this one, and this is the other one. And then that plane that they construct together is then the tangent plane. So I would get a parameter description with vectors, so λ times $1, 0, df/dx$ in the point a, b plus μ times the vector $0, 1, df/dy$ in a, b . Is that clear to you? For who is it not clear? Nobody, that's good. And what we want instead of this vector description of that plane, we want an equation. And then we know, okay, we are in three dimensions, so you go back to linear algebra, you have two directional vectors, and now you just want a normal vector. If you have a normal vector of that plane, a vector that's perpendicular to it, then you can derive the equation. So I'm interested in a normal vector, a vector that is perpendicular to both these other vectors. And what you can do is that the normal vector to the tangent plane is then equal to the cross product of these directional vectors. And did you do that with a determinant? I'm not sure whether you did it like this, but this is just, this is bad mathematics, but it is easy to use. But it has no mathematical meaning, it is just a way of remembering the formula for the cross product. So first you put the first one, and then here I will change notation, because for these partial derivatives there are different notations. I didn't say that until now, but you could also call it f_2 , the partial derivative of f with respect to the second component, because it was the y , in the point a, b . It's a little bit shorter. And then here you get $1, 0$, and this partial derivative in the x direction, x was the first component, so then you say f_1 , so the partial derivative with respect to the first component. And what do we get then? Now, this times this times this is $f_1 a b$ times i , this times this times k is 0 , and then you get this times this times this plus $f_2 a b j$ minus k . And this, all these other products are 0 . So the normal vector of z , of the plane, is the vector partial derivative with respect to the first component in a, b , partial derivative with respect to the second component in a, b minus 1 . And if you want to check that this is indeed perpendicular to this one and to this one, then you easily see that that is indeed the case. So that is, and probably it would have been easier to think about that earlier, that you could simply derive it quite easily. This is my normal vector, and I have now the plane with this normal vector, and this is a point on that plane. So the equation of the tangent plane, of an equation of the tangent plane, is I take $f_1 a b$ times x minus a plus $f_2 a b$ times y minus b minus 1 times z minus $f a b$ is equal to 0 . Why is this, this is a linear equation. You know for sure that if you take $a, b, f a b$, then this is 0 , this is 0 , and this is 0 , so then you get 0 is 0 . So that's a point on the plane, and you can also see what the normal vector is, because that is determined by these coefficients before the x, y , and the z , and this is f_1, f_2 , minus 1 if you look at a, x, y , and z . So this is the right plane. You know that for sure. Usually, they write it in a different form. They, you bring this to the other side, this minus z , and then you can write that as z is the function value of f in a, b plus the partial derivative with respect to the first component times x minus a plus the partial derivative with respect to the second component times y minus b . And then this looks very much like how you compute the tangent lines. There you also have the function value plus then the derivative times x minus a , and now you get the function value plus the partial derivative in the first direction times x minus a plus the partial derivative times y minus b . So it's, in the end, you get almost the same thing as you had before for tangent lines. So doing this is not very difficult, I would say. And all formulas are on the slide. So you see this first definition, instead of a, b , here we used x_0, y_0 , but that's not much of a difference. You get this normal vector. And then you have the normal line that is the line through the point that is perpendicular to the surface that describes the graph. That is, of course, the line with supporting vector $a, b, f a, b$ plus, and then λ times this normal vector. And I like that description much better than this description, because here you describe this normal line as the intersection of this plane with, no, yeah, the intersection of three planes that are in a sort of, yeah, that go through one line. But I don't think it is very useful. Because you know in the book they assume that you do not know how to write the parameter description with help of vectors. Now, why, I don't understand. But this description that I gave here, so just make it explicit, so the normal line that is x, y, z is $a, b, f a, b$ plus ρ times $f_1 a b, f_2 a b$ minus 1 , where ρ is real. That's much easier. It says the same thing. Mathematically, it's the same thing. So why not use it? And by the way, also in an exam I will never ask it in that form like the book does, because I think simply it's silly. That other one is much easier. And it's describes the same thing. This is my normal vector. And I have now the plane with this normal vector, and this is a point on that plane. So the equation of the tangent plane, of an equation of the tangent plane, is I take $f_1 a b$ times x minus a plus $f_2 a b$ times y minus b minus 1 times z minus $f a, b$ is equal to 0 . Why is this, this is a linear equation. You know for sure that if you take $a, b, f a, b$ Okay,

this function, so we want to have the tangent plane in the point $(0,1)$. Let's first compute the f in $(0,1)$. That is $4 \text{ minus } 0 \text{ plus } 2 \text{ times } 0 \text{ minus } 2 \text{ times } 1 \text{ squared}$. So that's $4 \text{ minus } 2$ is 2 . Then I compute the partial derivative with respect to the first component, and that is also then called f_{1xy} . That is now, so I pretend that y is constant, so I get $\text{minus } 2x$ from the $\text{minus } x \text{ squared plus } 2y$. And in the point $(0,1)$, this is 0 , it is equal to 2 . And then the partial derivative with respect to the y component is, for the 4 , you get now 0 because it's a constant y . This is also a constant y , so no contribution. Here I get $2x$, and here I get $\text{minus } 4y$. So the f_2 in $(0,1)$ is equal to $\text{minus } 4$. And then you get automatically that the tangent plane in that point, tangent plane to graph f in the point $(0,1,2)$, because if you have x is 0 and y is 1 , then z is equal to 2 , is, now, z is the function value, that is 2 , this one, plus the partial derivative with respect to the first component plus $2 \text{ times } x \text{ minus } 0$, minus 4 , this minus $4 \text{ times } y \text{ minus } 1$. And of course you get these $x \text{ minus } 0$ and $y \text{ minus } 1$ because you have to make sure that in the point $(0,1)$, the value is equal to 2 . And of course you also get different, you can represent by this same plane in all kinds of different ways, but this is the representation that you get from the formula. So that's pretty easy, just computing the partial derivatives. And you also know what's the normal vector is because that is the vector, you bring this to the other side, $2 \text{ minus } 4 \text{ minus } 1$. It's the normal vector because you can of course scale that in the way you prefer. So this is what easy. I now skip something because I want to do the, so I, with this it's also possible, I had an example prepared, but I first want to do the other thing, that with this idea you can also compute the distance from a point to a surface. So if you have time I will do so, but suppose as an example you have the paraboloid z is $x \text{ squared plus } y \text{ squared}$. That's something like, that's a parabola and then that goes circular around it. And now you have the point $(1,1,0)$, I put it over here to make it, and then you want to know the closest point on the surface that is closest to this given point. And what you then do is you project it perpendicularly on that surface. Now with the techniques that you now have, let's still do it, then at least the idea, you can compute that projection point and also the distance. And the idea is as follows, so I take as an example the point, now I have to look in my notes, otherwise I make a mistake, the point distance of the point $(1,1,0)$ to surface z is $x \text{ squared plus } y \text{ squared}$. What I now do is I say, okay, I need a point on the surface that is closest to the point $(1,1,0)$. And $(1,1,0)$ is not on the surface, I know that. How do I do that? I simply say I have two unknowns, x_0 and y_0 , and I take a point, an arbitrary point on the surface, and that has then, take the point x_0, y_0 and then the z is $x_0 \text{ squared plus } y_0 \text{ squared}$ on the surface. An arbitrary point. Then I want to have two things. I have to do one thing. If I now, that this is the closest point, I want to have that the line that points from this point $(1,1,0)$ to that point $x_0, y_0, x_0 \text{ squared plus } y_0 \text{ squared}$, is perpendicular on that surface. Because the closest point always, then you get that the normal line, the normal vector in that point has to point in the direction of point $(1,1,0)$. Now, so what I have on one side is that the vector from, let's say, this point p to q , that's the vector p to q , is the vector $x_0 \text{ minus } 1, y_0 \text{ minus } 1, x_0 \text{ squared plus } y_0 \text{ squared}$. That's the vector that points from $(1,1,0)$ to the other point. On the other hand, I know what the normal vector is. The normal vector to surface in $x_0, y_0, x_0 \text{ squared plus } y_0 \text{ squared}$ is, yeah, that was that normal vector that we have here. So that is the f_1 in x_0, y_0 , the f_2 in x_0, y_0 , and here it's equal to $\text{minus } 1$. That was the normal vector. And what we want to have, and in all cases that's easy because the f_1 of x and y is $2x$ and the f_2 of x, y is $2y$, so this is the vector $2x_0, 2y_0 \text{ minus } 1$ in our particular example. That's the normal vector. Now we want them, these vectors that points from this to this, that it is in the same direction as that normal vector in a way if you scale the normal vector it would point directly to this point. So what you want, the condition becomes that these two vectors are in a way pointing in the same or the opposite direction, so there has to be a λ , there is a λ , such that this first vector is equal to λ times this normal vector. And in three dimensions you get now three equations, nonlinear ones, with three unknowns, λ, x_0 , and y_0 . And all three have to be satisfied. So you get that $x_0 \text{ minus } 1$ is equal to $2x \lambda$ times x_0 , you get that $y_0 \text{ minus } 1$ has to be equal to $2 \lambda y_0$, and you get that $x_0 \text{ squared plus } y_0 \text{ squared}$ is equal to $\text{minus } \lambda$. So λ you know already depending on x_0 and y_0 . So λ is $\text{minus } x_0 \text{ squared minus } y_0 \text{ squared}$. Then you plug that into these two equations, then you get two equations with two unknowns, x_0 and y_0 . $x_0 \text{ minus } 1$ is $\text{minus } 2 \text{ times } x_0 \text{ times } x_0 \text{ squared plus } y_0 \text{ squared}$, and the same thing for $y_0 \text{ minus } 1$ is $\text{minus } 2y_0 x_0 \text{ squared plus } y_0 \text{ squared}$. Now the next thing that you can do, and now I have to be careful because I still have to tell something else, otherwise you cannot continue. The next thing what you have to do is subtract both equations. What you get is that $x_0 \text{ minus } y_0$ is $\text{minus } 2 \text{ times } x_0 \text{ squared plus } y_0 \text{ squared times } x_0 \text{ minus } y_0$. And this gives that $x_0 \text{ minus } y_0 \text{ times } 1 \text{ plus } 2 \text{ times } x_0 \text{ squared plus } y_0 \text{ squared}$ has to be equal to 0 . Now this number is always positive. I add 1 to it, so this factor is never 0 . So the only possibility is that x_0 has to be equal to y_0 . That's the only chance we have. So we get x_0 is equal to y_0 . And λ is therefore equal to $\text{minus } 2x_0 \text{ squared}$. Now I plug everything in this equation and then I get $x_0 \text{ minus } 1$ is $\text{minus } 4x_0 \text{ to the power } 3$. And so I have to solve this cubic equation $4x \text{ to the power } 3 \text{ plus } x_0 \text{ minus } 1$ is 0 . And you can see that x_0 is a half is a solution because then you get $4 \text{ times } 1 \text{ over } 8$ is a half plus a half is 1 , minus 1 is 0 . You can also prove that it's the only 0 that it has, but I skip that part. And what you now see is that that point, this point on the graph, so this was $1, 1, 0$, and this point q is the point a half, a half, and then z is a half squared plus a half squared. That's the accent also a half. So this is the point on the cylindric paraboloid that is closest to the point $(1, 1, 0)$. And you do this simply with help of this normal vector and by projection. But it's more difficult than if you have really a flat plane, then you always, these normal vectors are always the same. And for these strange surfaces, yeah, these normal vectors

every time pointing in a different direction, so that makes life more difficult. Okay, I think this is a more advanced application of what you can do with all this. And then I want to go back now because what you also can do is, like you have for a function of one variable, you have the derivative, but then you can differentiate once more and once more and once more. And for partial derivatives, that's also possible. So you can also have higher order partial derivatives. And in a way it works just in the same way because you describe this partial derivative is again a function of x , y , and maybe other variables. And then you have that expression and you differentiate, try to differentiate once more. So, for example, if we have a function of d variables, then what you do if you want to have the second derivative and then you have to also, you have lots of possibilities because this is two times with respect to x_1 . This is first with respect to x_1 and then with respect to x_2 . Now, what do you do you see immediately that you have lots of others because you have to take these here, in this case, is based on this fact that $3^2 + 4^2 = 5^2$, and you know other combinations that do the same thing. And then you can also find other solutions of this, other functions that satisfy this Laplace equation. This is a very easy option. It's getting more difficult as soon as you have these mixed equations. And I want to show you one thing about it because then I have almost done what I really need to do. And that's the following thing. An example again. $x^3 y^2 + 2y e^x + x^2$. And I'm going to compute the partial derivatives. And first, the partial derivative of f with respect to x , that is $3x^2 y^2 + 2y e^x + 2x$. I simply consider it as a function of x . And then the partial derivative of f with respect to y , and that is $2x^3 y + 2e^x$. That's all. Now, then you can look at the second order and you have four of them. So I have now the second derivative of f with respect to x twice is $6xy^2 + 2y e^x + 2$. And then I have the second derivative of f , and then here $dy dx$, that's I have to differentiate this thing with respect to y . And that is $6x^2 y + 2e^x$. Then I have to take the other one, the partial derivative with respect to y , and differentiate that with respect to x . That is $3x^2 y + 2e^x$. And then I can also do it with y , but I'm now at the point where I see you see that these two mixed partial, higher order partial derivatives are in this example the same. So if I first differentiate with respect to x and then with respect to y , or I do it in the opposite order, first with respect to x and then with respect to y , you get usually the same if you do it in the opposite order, first with respect to y and then with respect to x . And if you want to find a counterexample, then you have to look in, there are counterexamples, but these are, at least for engineers, not very interesting, I would say. So usually, it doesn't matter which order you do, so usually you do that, but if you then get trouble, then you should think, oh, maybe I've done something wrong because I've assumed that they are the same and maybe it's not the case because I'm here in a very strange situation. I don't think you will ever see it in practice. So usually you have this symmetry. Now with these two things, so these high order derivatives, I think you can do all the exercises, certainly from week 2 on limits, but also from week 3 on the sections 15.3 and 15.4. And I'm now again back, I have a backlog because I also had to do 15.5 about the chain rule. I will do that next time. So you can read it yourself, but I will also discuss it next time. Now there is the instruction, but I want, do not forget to make the Sojo exercises. The deadline is, I think, next Tuesday.

Good. So you haven't seen me yet. My name is Rob Mastom. I'm with the Electromagnetics Group. Guys, I'm with the Electromagnetics Group, and I've taught physics for EE. AT students got physics from Patti, I think Patti Stabila. Similar type, of course. And what I'm going to do is I'm going to try to relate things that you are taught in math to to physics. Basically, well, physics for E, physics for AT. but also to, well, already Electromagnetics 1, which you'll have next year. So this is also a primer for that, which will show how to apply mathematics two to physics-based topics. I think these are live streams, so they are also recorded. So that will help you also. Well, please don't leave, but this will help you review the material. So this will recap some of the topics that have been covered by Luke. And specifically, I think he started today with coordinate systems, a bit of spherical coordinates. I'll tap into that as well today. Where needed, I'll cover some relevant new bits and pieces of theory, already as a primer for other courses. And as I said, I will relate this to physics and to electromagnetic. And we will work together in a studio classroom sense, just as we did with the physics course. And I think with Math 1, you also had studio classrooms taught by Schuard Hilsov. this is a similar setup and again we try to I try to nudge you into trying yourself first and then I'll solve the problems on the blackboard we are a bit running behind schedule but I think we can make up for this a bit I will not go over time not past quarter plus five the idea is that I show you that math is relevant and fun at the same time but since you're still here I hope you can enjoy the beauty so there's an incentive for attending these sessions or at least watching them, is how to learn to put math to good use. And similar as last year, there will be some problems in the tests that are related to the pseudoclassrooms. Let's see. So I'll cover mainly the first bullet here, and I think one or two examples of the second bullet, and this relates to what Luke has covered already. But Gordon Systems is really an important topic in physics. So let's start with that. So we're used to Cartesian. We used Cartesian lots, a bit of cylindrical as well in the physics courses. I think the automotive students even got a bit more on cylindrical coordinates than the EE students got, but we'll repair that today. And also spherical coordinates, which is really an important topic. I think Luke briefly mentioned this at the start of his lecture today. And especially also unit vectors in these coordinate systems, which is a weird concept, but I'll talk you through it. So why do we need coordinate systems? we want to describe points in space. We need a unique way of doing this. So Cartesian, we have an X, Y, and Z coordinate, and that uniquely defines a point in space if you choose an origin somewhere. And they have a property that locally at every point in space, so you have an X coordinate, a Y coordinate, and then a Z coordinate. Locally at every point in space, these are perpendicular to each other. So we have orthogonal unit vectors. And we choose them right-handed, which means that they obey the... right-hand screw rule or the right-hand rule that if you rotate this vector to that vector you go up right-handed corkscrew or screw and typical coordinate systems are cartesian the one we're most used to circular cylindrical and spherical so cartesian i'll cover really briefly there's one thing i want to highlight here we have unit vectors i j k a position vector So to a point in space goes from the origin to that point in space. So it has three values here for X, Y and Z. These are the coordinates. And then a vector field can also have three components. So at every point in space, a vector field like electric field, magnetic field, wind velocity, at every point in space that has a magnitude and a direction. And that comes down to three components of a vector in space. If at any point in space you were to say where, what the directions of the unit vectors are, you can wiggle your position vector around a bit. So if you wiggle P around a bit in X direction, it will move like in this direction, then you find out what the unit vector I is. If you wiggle it around a bit in Z direction, you wiggle it upwards a bit. Then you find out what the direction of the unit vector in Z direction is. And that's a concept that also carries over to this. different coordinates.

So let's look at cylindrical coordinates, a cylindrical coordinate system or circular cylindrical coordinates. Typically, if you describe cables, coaxial, cable, or any cylindrical structure, you may need these.

Then you have three different coordinates. Instead of x, y, z, you have a

distance, an angle, and again a distance.

The distance is from the origin outward, well, in a radial direction to a point that is just below point P.

This goes out at an angle θ . So that's why θ comes in. And then there's a height Z . So instead of X, Y , we have R and θ .

These can take values.

They can go from zero to infinity. So R is always positive.

Something wrong? No? R is always positive or large than equal to zero. θ goes, well, there's multiple definitions.

We have to span 2π so we have to go from minus π to π or from 0 to 2π just as you want and that can be well the entire axis so from minus infinity to plus infinity we can convert xyz coordinates to cylindrical coordinates and we have to use these conversion rules and these follow from geometry so a point in the xy plane here at a distance r at an angle θ we can project onto the x-axis so we take the cosine and we get $r \cos \theta$ so that's this x coordinate we can do $r \sin \theta$ which is a projection onto here so that that would be this distance and we have y and z is the same in both coordinates systems right this is something you hopefully that was covered probably in math to lecture one we can also do the inverse so we use pythagoras to calculate r so that's $x^2 + y^2$ plus z^2 and the tangent of θ is y over x so we can calculate well, we can convert from the one to the other or vice versa. Now we get the tricky part. If you have a coordinate system like this, you can also define unit vectors. So any coordinate system needs unit vectors that are locally orthogonal and have a length one. So for cylindrical coordinates, well, as I said for Cartesian, you can wiggle around your coordinates a bit and then you find the direction of the coordinate of the unit vectors. If at this point here, you increase R a bit, you're moving in the... this direction, right?

So the $\hat{R}$ unit vector is in the direction of increasing R . So it would be pointing outward, really outward from here.

$\hat{\theta}$ would be if you increase angle θ just a bit, this point moves over R times θ in that direction, right?

So the $\hat{\theta}$ direction is perpendicular to $\hat{R}$, $\hat{\theta}$ is in this direction. And $\hat{z}$ is the same as we had for Cartesian coordinates, that's just the Z direction. So $\hat{\theta}$ is locally perpendicular to both $\hat{R}$ and $\hat{z}$, which is a property of a coordinates. If you are at a different height, then R is still radially outward because it's defined as, well, the distance from the Z -axis. So also here, it would be outward in the same direction as here, but at a different height. But locally, at every point in the space, there's a unit vector. Depending on where you are, what angle, the radial unit vector can point in the positive in the first quadrant of x and y but can also point backwards towards negative x negative y right that is also possible and that's something well the unit vectors depend on the angle then and that's a well that's an inconvenience but it is inherent to this coordinate system what would a position vector from the origin look like you have any idea we have three unit vectors do we need all three or not?

So if I write a position vector to point P, would that then be something times $\hat{R}$ plus we think this is correct?

What unit does a position factor have? Meters, right? If I have R, θ, Z as the coordinates to describe what happens, R is in meters, Z is in meters, θ is in radian, so that's not meters, right? So I can't put a θ here, and then I don't have something that ends up in meters.

So this is wrong. What we need, actually, is the following. depicted here, I

need a distance out from the Z-axis, denoted by R , in the R -head direction that depends on θ . So depending on what angle I have, $\hat{R}$ points in a different direction, and then I go Z up in K -direction.

So a position vector, always from the origin, does not have an angle, well, an explicit angle contribution like you could write here.

It's hidden inside unit vector $\hat{R}$ that depends on the angle. Is that clear? This is a tricky concept. Yeah, go ahead. $\mathbf{r}$ is written in cylindrical coordinates like this. A position vector is written like this. The position vector is, and that's the tricky part, it's always from the origin. So you can't move in θ head direction because that's locally in this direction. But in the origin, that's not defined. So from the origin outwards at a distance R , at an angle θ , and then you go up Z in K -head direction. The other way around. You can write $\mathbf{r}$. You can write that with an x , y and z coordinate. Or you can write that with an r , θ and z coordinate. And this implies that you have $x\hat{i} + y\hat{j} + z\hat{k}$. And this implies that you have an r . Sorry, r of $\hat{r}$. That depends on θ plus $z\hat{k}$. These are the same. This one is in cylindrical coordinates. With three numbers that you use to go from the origin to that point in space. You can say I move radially outwards. But I need to know in what direction. Otherwise I don't end up in point P . So I need the angle to know in what direction. Are headpoints Cartesian, that's all nice and easy. Because we're used to that. In cylindrical, it's different. Okay, is that concept clear? It gets worse for spherical coordinates. Well, please let it sink in. We'll use this in the pseudo-classroom problems as well. If you have, just a second, if you have a vector field in space, so this describes any point in space where $\mathbf{a}$, let's say, an electric field, magnetic field, wind flow can have a direction and a magnitude, that can have three vector components. This is not a position vector, and this is a description of what at any point in space can happen. And then there can be a radial component, an angular component, and a Z component. So at any point in space, yeah, let's do this example. If you point slanted upwards here, then you need all three components to describe that direction in space. So there you can have three components. That's the main difference between a position vector from the origin and a vector field, which can be at any point in space not referenced to any origin, just reference to a type of coordinate system that we choose. Okay? Does that answer your question or confuse you more? Okay. Yeah? they are, okay, let me answer this after, I think, problem three. Then we have calculated what the unit vectors are. But locally, if you do a top-down picture, top-down picture, then you have, let's see, X , Y . At this point in space, the radial vector is here. That's $\hat{R}$, and θ is in that direction. Because this is the direction of θ . At this point in space, R is in this direction, θ is locally orthogonal to that. So it goes round. If you rotate this, increase the angle, then this set of unit vectors rotates along. And this is something we'll use in one of the studio classroom problem. I think you were first. Yeah, so we don't specify the units of this. The $\hat{R}$ doesn't have a unit. $\hat{k}$ doesn't have a unit. It's dimensional. It's a direction only. For a position vector, we need these numbers in front that have to be in the same dimension as the position in meters. So R is in meters. Z is in meters. θ is in radians, so that's hidden inside our head. For this vector field, if you talk electric field, it's volts per meters. If you talk magnetic field, it's amperes per meters. Wind velocity is meters per second. So the unit goes into these numbers in front of the unit vector. Oh, you had the same question. Okay, good. Two birds with one stone. This is tricky, but there will be examples on this in the studio classrooms. But I'll make it a bit more tricky, and then I think it's already time for a break, but we'll see. We can, well, put the stakes a bit higher even. We can use spherical coordinates as well. Typically, if you have a point source for electric fields and that electric field goes out radially from the origin, it spans out spherically. If you work with space applications, and spherical coordinates are also quite useful. For spherical coordinates, we have two angles. Instead of X , Y , Z , Cartesian. We have a distance from the origin, R , to any point in space. So that's capital R here. Then we have angle ϕ , which is the angle that this line makes with respect to the Z -axis. Then we have θ , which is the same as we had previously in cylindrical course. So to go from any point in space, you go a certain distance from the Z -axis, then you

rotate outward, angle ϕ , and then you rotate, well, in the horizontal plane, angle θ . And then you can reach any point in space. It's also what Luke mentioned in this lecture. With two angles, you can span two angles and a distance you can span any point in space. So instead of X, Y, Z , we now have R, ϕ, θ , where we have two angles and not only one of them need to go to full circle, literally, and we choose θ to be that angle, and ϕ goes from, it's the angle with the Z -axis, it goes from zero to π . And in that way, we can span any point on a sphere. We can reach any point on a sphere, so we don't need both the angles go from 0 to 2π , just one of them. And by convention, we choose the one that is also used in cylindrical coordinates. So θ . So an antenna on a base station, you can often also treat as a point source if you're in the far field. And then a point source in a far field gives radiation in a spherical direction. Sorry? Yeah, if it's not isotropic, then you still describe radiation patterns in deviations from an isotropic pattern. So that's, we use θ and ϕ a lot in antennas. So if you end up with the EEM group, you will, yeah, you will use these a lot. Just one catch only, and that's always mathematics uses I , electrical engineers, use J for complex numbers. Mathematics, like the book of atoms, uses θ and ϕ in physics, in the electromagnetic, they are switched, they are swapped. We have ϕ and θ instead of θ and ϕ . So it's, yeah, that's something we cannot solve. All books are like that. Mathematics book they have five θ physics books have the other way around yes yeah indeed the order makes it a right-handed so it's always first the distance then the angle with respect to the x -axis then the angle with respect to the x -axis that makes it well that that that order is important also these we can convert we can go from cartesian to spherical and I think Luke gave this example with r equals one today actually at the start of his lecture today. To describe any point on the surface of a sphere, of the unit sphere, you need two angles, right?

So I did watch the start of the live streams, and then he described this.

He used R as 1, so you can convert from Cartesian to, no, sorry, from, if you know, ϕ and θ , you can write them into Cartesian, but you can also do the inverse. And then you have, well, the distance is Pythagoras in 3D, so R is this one then it's the same as $r^2 + z^2$ and you can do for the angles you have multiple options θ is still the same but for ϕ you can use cosine or the tangent as you wish it's this angle so it's r over so it's z over r as cosine or tangent of r lowercase r over z this is something that will be on your formula sheet you don't have to learn this my heart yeah that's really tricky if you talk about unit vectors Think about what I said earlier. You wiggle around your coordinate a bit. So if you are at point P , you increase R , just a tiny bit. You are moving in R direction. So R head is radially outward, from the origin outward. If you increase ϕ , you're rotating this down. So locally, this point moves down. So that's the direction of ϕ head. And if you rotate in a circumferential direction, you're not moving in Z direction, but you're just rotating horizontally. then you end up with θ head. If you are at a different point in space, so not P but Q , for instance, then R is rated the outward. ϕ is, well, if you increase the angle, you're rotating downward, so ϕ hat is downward. If you increase θ , you're moving this one to the back. So θ hat is in that direction. It gets more tricky to the picture this, but I hope you get the idea. In the past, we've mainly stuck to cylindrical coordinates because this is hard enough already, for math two at least. In electromagnetic, you will do spherical coordinates. Question so far? A position vector, so from the origin to a point in space, is now described by one distance and two angles. So you go outwards in radial direction, a distance R , but you need two angles to describe this direction, because you need the angle with the Z -axis, and then the angle with the X -axis. Otherwise, you don't reach point P . You end up somewhere else. So again, position vector is in meters. R is in meters here. So that's the number in front of the position, sorry, of the unit vector. And then you have two angles that describe how this unit vector is oriented in space. Is that also clear? I hope after the discussion with cylindrical coordinates, you know, you understand that we need only one distance, but two angles. But the vector field in space can have, at any point in space, three orthogonal components of that vector. So it has a magnitude and direction. So this has the three unit vector. And often you make simplifications due to symmetry, one of

these drops out or things like that. Or there's only a radial component. If you have a point charge in space, we always look radially from it. And then it only is pointing away from that point charge if it's a positive point charge. Okay, let me put this up, and then we have a... Let's do a 10-minute break to catch up a bit of the time lost. I'll put this up. You can work on this and see if you can convert from one coordinate system to the other. and then we'll solve this after the break. So let's take 10 minutes and then we'll continue.

So let's see how we do this. We need the formulas from a slide actually and those who that will be on your formula sheet.

So points A and B are given in cylindrical and spherical coordinates respectively.

So we have to, well the question is to convert them to Cartesian coordinates and we need point B as well.

We can only calculate distances in Cartesian coordinates.

You can't have two different coordinate systems in calculate distances because a distance is really a Cartesian thing. So if we convert this into point A into Cartesian coordinates, we need these, well, X is $R \cos \theta$, so this is X , Y , and this is Z , which is still the same. So we can then calculate what this is, R , so that's 12 , $\cos 3\pi/2$, that's 0 . So this is 0 , i , plus 12 times $\sin$ of $3\pi/2$, $\sin$ of $3\pi/2$, so that's -1 . So I get -12 in j direction plus 2 in k direction. And I have the position vector in Cartesian coordinates. It's just filling in the conversions between the coordinates. For spherical coordinates, I need to do that on 2 because I need to find the distance at question B. I fill in two angles. So I get $r \sin 5$ $\cos \theta$, i head direction plus $r \sin 5 \sin \theta j$ hat, plus $r \cos 5$. k hat so this then becomes two times so sign of two π over three this was minus one no sorry two π over three so this is two π over three sign that's half root three yeah $\cos \pi/4$ that's half root two i had direction then i have two times sorry two times $\sin \phi$ still half root three sign θ that's also half root two j hat plus two times $\cos \phi$, two times $\cos 120$ degrees, that's $-1/2$, k -hat direction. So that gives me two times half times half, that's half root $6 i$ -hat plus half. That gives me this position vector in Cartesian coordinates. You fill in the angles, you hopefully know these cosines of 60 degrees, 30 degrees, 120 degrees. And then you get two position vectors, $R-A$ and $R-B$, and the distance now becomes something we can calculate distance $r_a \cos \theta_a - r_b \cos \theta_b$ so that's $r_a \cos \theta_a - r_b \cos \theta_b$ or $b \cos \theta_b - a \cos \theta_a$ well it doesn't matter so that's a magnitude of $\cos \theta_b$ is 12 yeah it should be $\cos 120$ -12 -12 to calculate the length of this one so it's this one squared plus that one squared plus that one squared and then taking square root so there's one fourth times six six over four plus well then the nasty one to be minus this should be a minus so i can take the plus sign as well doesn't matter plus which you can simplify, I wouldn't call it simplify. You can write this as, that's something you can do yourself. Bottom line here is, I'm not that interested in the mathematical details, the real calculations. You can convert position vectors from one coordinate system to the other, and you can calculate distances now. Just by thinking Cartesian, you have the difference vector, and the length of that vector is the distance between these two points. That's the length of this vector, which is this one squared plus that one squared, does that one squared and then taking square root. This was a warm-up. Let's look at a next one. Let's look at the unit vectors in cartesian and spherical, sorry, in cylindrical and spherical coordinates. And let's express these in Cartesian components. So please make a drawing, for instance, for the cylindrical one, and see what direction that vector would have. I can give you insight in how this relates to Cartesian components. So what I'm asking is to write, continue on a new blackboard in a second, R -Had to express this as something times i -Had plus something else times j -Had, and the same holds for the one in spherical coordinates. So that's what I'm asking here. And there's easy ways to look at this and more difficult ways to look at this, see if you can find an easy one. That makes life really easy. This later, I'll retry it. And at the start of the

lecture today, I think Luke already gave a hint for this one. So is there any of you who has an expression for $\hat{R}$? So the answer to A, basically. That's correct. The function of θ , as we know, this is $\cos \theta$, $\hat{i}$ plus $\sin \theta$, $\hat{j}$. That also follows from the drawing, if you make that. So $\hat{R}$ is radially outward from the origin. If you would decompose that into an x- and y component, then I need to use angle θ . So this is the cosine component. That's the sign. Another way to look at this is to go to a position vector, cylindrical coordinates. If I look in the plane, that's $r \cos \theta$ $\hat{i}$ plus $r \sin \theta$ $\hat{j}$ had, because this is exactly the x coordinate, and that's the y coordinate. And if I divide the position vector by its length, I'm making this one unit length. So I got the unit vector in radial direction. Well, what is the length of this vector? Well, that's exactly r , the radial distance. If I would square this one and that one, I would get $R^2 \cos^2 \theta$ plus $R^2 \sin^2 \theta$.

So that would be one, $\cos^2 \theta$ plus $\sin^2 \theta$ is one. But I also know that the distance, the length of this vector is R . So I have $\cos \theta \hat{i}$ plus $\sin \theta \hat{j}$.

So that's another way of looking at this. I'm not sure which the approach you took.

Sorry, the, sorry? Oh yeah. Yeah. The projection. Yeah, that's this one. But you divide by the length. I hope that gives some insight. If I, another way of looking at it, if you can look at the unit circle, to describe a point on the unit circle, I would need unit vectors. R is one. So that's another way of looking at it. Directly get this one, if R is one, is one into here, I end up with this one.

So then B becomes kind of trivial, right? I divide that vector by the length of that vector, so by capital R . And if I then look at this relation, I could write this $\hat{R}$.

This is $\hat{x}$, plus $\hat{y}$ plus $\hat{z}$ where substitute for x $\hat{i}$ and $\hat{z}$ $\hat{i}$ substitute these this is $\hat{x}$ $\hat{i}$ can write in terms of two angles and a distance for $\hat{y}$ $\hat{i}$ can do the same for $\hat{z}$ as well if I set r to length one I have a unit sphere so the vectors are then unit vector so I get $\sin \phi \cos \theta \hat{i}$ plus $\sin \phi \sin \theta \hat{j}$ plus $\cos \phi \hat{k}$ so these are just the components here with r equal to one so that's a unit vector in radial direction in spherical coordinates the other unit vectors so θ head here and ϕ had θ head here are a bit more difficult to obtain but we'll do the cylindrical one in the next other question so far on this especially for the two d cases I hope you can still construct these you can decompose a vector into two components if you know the angle right 3d it gets a bit more tricky let me go through c and d now and then we will look at the next problem. So it's less interactive than I intended, but it's also due to time pressure. You can still give it some thought on how you would solve this. So what is the question, actually? We are looking at a unit vector that points from here to a point here, $R\theta=0$, so any point on a circle of radius R , where θ I can choose freely. What can we already say about this unit vector? Does it depend on the angle, or is it independent of the angle? Yeah, what I need to describe a point on a circle, So then depending on where I am on the circuit, I'm looking down at a different angle, right? So it should depend on the angle. The unit vector will depend on the angle. But still doing this in cylindrical coordinates allows us to do this. So that's the beauty of cylindrical coordinates. You get a parameterization basically for free because you use a coordinate system that allows for this. So let's first calculate the difference factor because we need, we can do a similar approach as we did here. We have a vector that we divide by its length to find a unit vector, right? and the vector goes from the point, well, H , to a point here. So I need two vectors describing these points. To do this, I need position vectors. R_1 , that's $H\hat{k}$, the position vector from the origin, well, and then H up. So it's a vector from the origin to the point at H . And R_2 is a position vector to a point on the circle, which is $R \cos \theta \hat{i}$, plus $R \sin \theta \hat{j}$. So those are two position vectors, one, two point. point at zero zero $\hat{i}$ I want to any point

on a circle that has radius r and then r_{12} the vector from point 1 to point 2 is r_2 minus r_1 it's always tricky to see it if i go this is r_1 this is r_2 r_1 this i always remember it as follows if i want to go to r_2 then i go from 0 to r_1 and then if i get r_1 plus r_{12} and then And R_{12} is R_2 minus R_1 . So R_{12} from point 1 to 2 is R_2 minus R_1 . So it is $R \cos \theta$ i plus $R \sin \theta$ j minus R_1 minus H . So and then the unit vector is this one divided by its length. So that's, and the length is then this one squared plus that one squared plus that one squared. So that's R^2 . So that's the length. So this is one. so that's r^2 plus h^2 so i get one over r^2 plus h^2 times well that same expression which i'm not going to copy what times the numerator here basically and then you see that this part still depends on θ so i now get a unit vector that is parameterized in cylindrical coordinates because it depends on θ step and that's a weird thing about these different coordinate systems you get parametrizations, well, implicit in this coordinate system. So does the expression obtain at C depend on the angle? Yes, depends on the angle with these $\cos \theta$, $\sin \theta$ terms. So depending on where I'm looking on this circle, I have a different angle from a unit vector. The length doesn't change because the length, well, this one does not depend on θ , and also R and H do not depend on θ . It's just that the angle at which I'm looking, does depend on data.

Go ahead. Yeah, then I'm only interested in the distance. So the direction doesn't matter. In this case, the direction does matter, because that's asked here. I want to go from point, this point here, to that point.

So then I have to take into account the direction. Good point, yeah.

So as long as I'm still in vector form, the direction matters, and I can always check my results. So the difference vectors should have a negative Z component, negative K component, because I'm pointing downwards. And that's indeed here the case. Yeah, that's a good remark. Questions on this one? So this should be easier for the automotive students and for the electrical engineering students. Let's see if this works out for you guys. So we have an expression for the unit vector in cylindrical coordinates, right? The radial unit vector in cylindrical coordinates. For A, you can show, and also by construction, that these relations hold. So I'll give you some time to show this, and then we'll go through this together. Let me see, I can write this one. If we look at what direction this unit vector $\hat{\theta}$ has to be, it needs to be locally perpendicular to $\hat{r}$ head, and it needs to be in the direction of the increasing coordinate θ . If θ increases, we rotate this slightly in that direction. So it should be, $\hat{\theta}$ should be in this direction, and it should be orthogonal to $\hat{r}$. $\hat{\theta}$ head is this one. $\hat{r}$ I could decompose into this component. This component, let's you, this is, sorry, this is $\cos \theta$. $\hat{r}$ head direction this was $\sin \theta$ j head direction and similarly i can decompose $\hat{\theta}$ also into two components if this is angle θ then this is also angle θ so i can decompose this into this component and this component that i align with my xy coordinates because i want to express this cartesian so and this is then minus $\sin \theta$ i hat and this is we're showing $\hat{\theta}$ j hat that's if i decompose this vector $\hat{\theta}$ into cartesian components.

Well, let's see if that follows from the derivation as well.

If I'll calculate $d\hat{\theta}$, the cartesian unit vectors do not depend on angles. So these stay. I just have to calculate the derivatives of the $\cos \theta$ and $\sin \theta$ terms.

$\cos$ becomes minus $\sin$ and $\sin$ becomes $\cos$.

So indeed this works out. This gives minus $\sin \theta$ in i -head direction and $\cos \theta$ in j -head.

And similarly, If I differentiate once more, I'm basically rotating again 90 degrees. That's what differentiation in cylindrical coordinates means. If I differentiate, so this is $\dot{\theta}$, but differentiate $\hat{\theta}$ as a function

of θ with respect to θ . So this is, I differentiate once more, I get minus cosine. This is exactly the opposite of this one, so that's minus $R\ddot{\theta}$ as a function of θ . So hopefully automotive students have seen this, right? In the, I forgot what chapter of Miriam and Craig this was, but you should have seen it. This approach clearer. So this is something we need if you calculate, and this is where I want to relate it to physics, if you want to calculate velocities and accelerations in cylindrical coordinates, then we need these expressions. So if we have a position vector in cylindrical coordinates, and we want to calculate the velocity, which can also be noted by a dot, this is $\dot{\mathbf{r}}$, we have to take it to account that R depends on θ , which can depend on time. So we have to apply the chain rule for this. So there's $\dot{R}$, $\dot{\theta}$, $\dot{\theta}$, $\dot{\theta}$, $\dot{\theta}$. And for the acceleration, we have to do one step more. But this is something we have to take into account. $\ddot{\mathbf{r}}$, so that's the vector part, that gives me θ -head, θ . That's what we got here. The $R\ddot{\theta}$ is θ -head times $\ddot{\theta}$, so that's θ -dot. If the angle changes, if I prescribe an angle over time, I know how the angle changes over time and then the velocity depends on how the position vector changes as a function of time as well as how the angle changes as a function of time and similarly for the acceleration so it's $\frac{d}{dt} \frac{d}{dt} \theta$ and the $\theta \frac{d}{dt}$ because that also depends on angle θ let me correct this I have to wipe it because there's a mistake a shortcut not allowed let me fix this this was for a constant r this is not true a position vector in cylindrical coordinates is r times $\hat{r}$ function of θ both r and $\hat{r}$ can depend on θ so let me correct this the $\frac{d}{dt}$ can be I need to apply both the chain rule and the product rule r can depend on time and θ can depend on time so I have to take that into account I got $\dot{r}$ $\hat{r}$ which is $\dot{r} \hat{r} + r \dot{\hat{r}}$ so no plus r times $\dot{\hat{r}}$ for this one I have to apply the chain rule so this becomes the $\dot{r} \hat{r} + r \dot{\hat{r}}$ plus r times the $\hat{\theta}$ θ $\frac{d}{dt}$ so this gives me $\dot{r} \hat{r} + r \dot{\hat{r}}$ plus the $r \hat{\theta} \dot{\theta}$ that was given here that's θ head times $\dot{\theta}$ $\frac{d}{dt}$ there this is correct so let me correct that let me at least be correct now so a position vector in cylindrical coordinates is written as r times $\hat{r}$ so a distance from the z -axis at a in the direction of the unit vector the radial unit vector so the derivative of that is a derivative of that expression if I insert this this can depend on time and this can depend on time so I have to take derivatives of both the parts with the product rule in mind yes no I'm not missing an r here here oh yes sure this are yeah $r \dot{\theta}$ yeah it should be in meter per second this is in one per second this is in meters yes I'm missing an r sorry, so I get $r \ddot{\theta}$. $R\ddot{\theta}$. So I have to take this into account also if I do the accelerations. This is where I get messy, $\ddot{\mathbf{r}}$, $\ddot{\theta}$ of this whole expression, $\ddot{r} \hat{r} + 2\dot{r} \dot{\hat{r}} + r \ddot{\hat{r}}$ plus $R\ddot{\theta}$. So here I have to apply chain rule, a product rule and chain rule. I got $\ddot{r}$, $\ddot{\theta}$ plus $\ddot{r}$. So that's the differentiation of this one, and then for this I have three terms so I get three time derivatives $\ddot{r} \hat{r} + 2\dot{r} \dot{\hat{r}} + r \ddot{\hat{r}}$ plus $r \ddot{\theta}$ so if I then gather the terms we get $\ddot{r} \hat{r}$ plus let's see this one gives minus $r \dot{\theta}^2 \hat{r}$ and this is another θ dot squared I get minus $r \dot{\theta}^2 \hat{r}$ plus $\ddot{r} \hat{r}$, $\ddot{\theta}$, and then I have this term, this term, plus the last term, this one, plus $R \ddot{\theta}$, double dot, θ , θ , so let's clean this up a bit, plus I get a very nasty expression that has all the derivatives, and this for sure the automotive students have seen, E.E. students haven't.

And this, I'll link this to circular motion in a second.

If for an arbitrary motion in space, both the position and the angle, so the distance from the origin and the angle that I used to describe the position are function of time. I have to take into account all the derivatives in this.

If we use uniform circular motion, we have an angle that increases with a constant angle of velocity in time.

And uniform circular motion also means that R is constant.

The fixed radius A . If I insert this, so R is constant, then $\dot{R}$ is zero, $\ddot{R}$ is zero if I insert that here and in the

velocity there I will find the velocity equals so $\dot{r}$ is zero this term drops out only that term remains so that's $r\omega\hat{\theta}$ head towards the angle of velocity in the direction along the path and the acceleration becomes $\ddot{\theta}$ double dot is zero every derivative of r is zero so this one drops out this one remains this one drops out, that one drops out because this one is zero and that one is zero. So I get minus a ω^2 our head.

The acceleration towards the center of the circular motion and the acceleration along the trajectory. If you draw that, go around in uniform circular motion, the constant velocity, then the velocity is in this direction, the radial acceleration, sorry, towards the origin is in that direction. And this gives exactly their magnitude. Yes, go ahead. Yeah, we ignore this. a Z component. This is a plain problem, a 2D problem, which I use circular coordinates to do this. But you can do this in 3D as well, and there could be Z components to this. But these, the unit vector in Z direction does not depend on the angle. It's just, in cylindrical coordinates, it's just a z-axis.

Spherical, this would become impossible to do, at least not something we can ask at an exam, or to any sane person.

So the complexity arises from the fact that we have time derivatives of vectors that depend on coordinates implicitly. And if this coordinate is also a function of time, so if θ is a function of time, then all of these combined terms, like we got here, this one will not vanish. This one here will also not vanish, because θ is a function of time. And then the derivative of the angular, sorry, of the tangential component of the velocity will give a component that's pointing towards the center. So in general, a non-uniform circular motion, we can prescribe in this sense. And even the radius could change. You can use any coordinates to describe it. Is the reasoning clear here? This is a challenging subject, I have to admit. But just as we can use Cartesian coordinates, we can use also a different coordinate system to describe motion of objects. It might not always be the convenient one, but for circular motion, it sure is. So if an object goes through an arbitrary path, in space, instead of Cartesian coordinates, X and Y, I can also use θ and R to describe the position of this object in space. I just have to take into account that these vectors also depend on the angle. And if the angle then depends on time, I get these dependencies.

Both the distance and the angle can depend on time. So the unit vectors as well, and that's the derivatives I calculated here.

Well, actually, up on there and this one. The way to see this is that if you have a unit vector $\hat{R}$ in a certain direction, and I change the angle a bit. So I have $\hat{R}$ and $\hat{\theta}$, which are locally orthogonal. So if this is X and Y, the point here, there's a local $\hat{R}$, $\hat{\theta}$ perpendicular to each other. If I increase the angle a bit, then this comes down to a slight rotation of these two unit vectors. So a small change, this gives you an idea how $\hat{R}$ changes if I increase the angle. It moves perpendicular to the vector itself. So $d\hat{R}$, a change of $\hat{R}$ is in a direction of $\hat{\theta}$. And similarly, a change in $\hat{\theta}$ head from this to that point is in a direction of minus $\hat{R}$. So that gives, well, some intuitive explanation for how these unit vectors change as a function of the angle. And if that angle changes as a function of time, then also the unit vectors change as a function of time because they depend on the angle. Is that clear?

I think. So do the unit vectors. So let me conclude. two days to the classroom with this slide. I think this is... I'll shift the other problems to the next session. It did not help that Luke ran out of time, but I don't want to take you longer than necessary. I will cover the rest of the material, but I'll do it next student classroom, which is after the midterm test. So there will be quite some time without student classroom. The scheduling this quartel is really wonky. There's so many days off. Next week, there's a day off again, and there's Ascension again on a Thursday, So it's really messy. But we try to make the best of it. You should not be, well, this should not cause problems for you. It

should be our problem. And we try to make the best of it. I hope this gave you insight in how coordinate systems work. And I'll do the remaining student classroom problems that relate to Chapter 12. Actually, I'll do those next to your classroom. So see you in three weeks time.

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So, good afternoon to all of you. Welcome to this lecture on Mathematics 2. I'm glad you're here and that you, let's try to understand a few things about the mathematics course. I'm a little bit behind schedule, so I will do a little part of what I had to do last week, and then I hurry up to try to cover the material for this lecture. Last time, we were talking about functions, functions f , from an interval to a higher dimensional space, to $\mathbb{R}^2$ or $\mathbb{R}^3$, let's put it here, $\mathbb{R}^3$. And what the main theme of today is, is exactly the opposite. So, after you finish this, we will look at functions from, for example, $\mathbb{R}^2$ or $\mathbb{R}^3$ that map to $\mathbb{R}$. And that is quite a bit different, but let's first look back to last week. What did we do? We were looking at curves in two or three-dimensional space, and we defined a vector depending on a parameter t , and you can think about this as time, but it not necessarily has to be that. But r is a mapping from an interval from a to b to, let's say, $\mathbb{R}^3$, and it is of the form that at a certain value of t in that interval, you have a function x of t times the first unit vector plus y of t times the second unit vector plus z of t times the third unit vector. And we assume here that this function r of t is continuously differentiable. So, we assume that the function is continuous, otherwise you would get maybe single isolated points, and we want to have really a curve. So, if you change the t a little bit, you only move a little bit on the curve. That's the idea, so it has to be continuous. It has to be differentiable, and the derivative has to be continuous. Now, such an, this is called an arc, and such an arc is non-singular if for all t in a, b , I assume that the velocity vector is non-zero. It's the problem that it is zero, not that much, but in points where the velocity vector is zero, strange things can happen. So, for example, things like this, that you have a curve and you go on like this, and at this point the velocity gets the zero vector, and then once in a sudden you go like that. Now, this is possible, but, and it's also still a curve, but you don't call it an arc, because there you want to have that everything is nicely continuously differentiable. Here you get a strange thing, because you move here from a zero velocity and then exactly in the opposite direction, and that's a little bit strange. This is still a contour, because we just say that contour or path is just a finite union of arcs that fit to each other, such that the end point of arc, let's say with number i , is the begin point of the next one, arc i plus 1. So we get nice continuously differentiable arcs, so that are nicely like this, and you have this on a finite interval like this, nicely smooth, and then it may happen that at that end point here, so this is the begin point, the other end point, that here a new arc begins and then you may have that it goes like, for example, this. Then this is still fine, but this is a special point, because here the velocity changes very abruptly, because you turn on the angle immediately, so you, and that's why you have to, that are special points where you have to take care of, but you can simply think of it, okay, I consider this one arc and another arc, and they are simply, they follow each other. Now, the point is where we looked at mainly is how you can parameterize such arcs, and if you can do that, you can also parameterize these things, because simply a succession of these arcs. And what we've seen is that this parameterization to describe all these points is not unique, because you can walk, if you have here a path, I can walk very slowly on that path, or I can take the same path and start running, and if then the time would be the parameterization, yeah, then I follow the same path, but with a different parameterization. Okay, now that is, I think, quite obvious. What you also can do is parameterize not only these arcs, but you can even parameterize surfaces, so two-dimensional objects in three-dimensional space. But then I need not one parameter, but two parameters, and the easiest example is the unit sphere. If I want to describe that with a parameter description, the easiest way to do that is use what we know about spherical coordinates. And now I look at the unit sphere, so the radius of that sphere is one, so the capital R is one. That's the equation. And then you have that x is equal to $\sin \phi \cos \theta$, y is $\sin \phi \sin \theta$, and z is equal to $\cos \phi$. So if you remember the circle coordinates, I had the capital R in front of it, but now you have to think about this, oh, this is just a function of two angles, θ and ϕ , and also the y and the z . And if you then take all corresponding points in three-dimensional space where θ is between zero and 2π and ϕ is between zero and π , then you describe simply all the points on the unit sphere. And then you also have other possibilities, also with these surfaces. For example, if you look at this ellipsoid, $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$. That is now a very easy job to do. I simply take that parameterization of the unit sphere and I tilt a little bit to get that ellipsoid, and then I have to multiply this with a , that's a fixed number, this, the y , with b , and this with c , and then you can check if you sum up $x^2/a^2 + y^2/b^2 + z^2/c^2$ that you get indeed one, that you get all the points on that ellipsoid. So, for two-dimensional objects in three-dimensional space, you need two parameters. Okay, now, you can get strange things. And here are some pictures, so it is possible. We call always, this is this arc, and I call then the points that correspond to t is a , the begin point, and the points that correspond that the point rb is the endpoint. But it can happen that begin point and endpoint are the same point, and I call that a closed curve. Quite straightforward. But it's also possible that you have double points, so that you start over here and that you make this loop and then end here, and you visit this point twice. That's not a problem. It is allowed. So, and also things like this are allowed. So you have to, it is much more general than you maybe think at the first time you see it. Creating these parameterizations, so as already said, they are not unique. You have infinitely many different choices, but you have even more choices that are incorrect. But it's a sort of art how to do it, and you have to look a little bit to understand what is going on and find a nice way to do it. And this is an example of that. I look now at this red curve here. It's a closed curve. And it's the intersection of an

elliptic cylinder, $x^2 + 4y^2 = 4$ is equal to four, and plane $x + 2y + 4z = 4$ is equal to four. And I want to describe that instead as a parameter description, for example, to compute how long this ellipse is. Now, I see that $x^2 + 4y^2 = 4$ is equal to four, so the first equation does not contain the z . z is there completely free for that elliptic cylinder. And I think, okay, then it is with an ellipse I can do that as follows. If I say that x of t is equal to two cosine t and y of t is the sine of t and z is still free, then this description gives sort of, not a sort of, gives simply an ellipse in the two-dimensional space, in the plane, and then the z value is free, so you get this elliptic cylinder in that way. But now I have also a relationship with the z . I know, so this takes care, all these points satisfy $x^2 + 4y^2 = 4$. Now if I now take this other equation, $x + 2y + 4z = 4$, then I can express z in terms of x and y . So I get that $4z = 4 - x - 2y$. And if I then divide by four, I get this. And now we have already x and y in terms of the parameter t , so the height is then z of t is equal to $1 - \frac{x}{4}$, that is two cosine t over 4, so minus a half cosine t , and then minus a half, minus two over four, sine t . And this for t between zero and 2π . And then you go simply around in that ellipse. So it is not a unique way of doing it, but you look a little bit how does this look like, and then you try to find such a parameterization. Now, last week we already looked at also the length of a curve, and there I said, okay, idea is very simple. You just imagine driving a car, and you have your speedometer. If you integrate your speed over the time interval you're driving, then the outcome will be the distance that you have covered in that time. So that will be the length of the road you've driven, you've passed. Yes, so that is a question on the previous thing. I thought about if I have the unit sphere, $x^2 + y^2 = 1$. This is just the speed then, because that is very locally this limit of a difference of two positions divided by Δt , and so in the limit that will go to the velocity vector. And then I take the length of the velocity vector and that's the speed, so I get that that is the integral in the end of the speed from a to b , $\int_a^b v \, dt$. So I get exactly the same thing, what I had before. But then you can also understand it with help of the picture, and that's maybe more convincing. Now that formula is important because you can do a lot of lengths, compute a lot of lengths with that, and the easiest example is this one. If you just have the length of the graph of a function, so here I have the function $y = f(x)$ on the interval for x from a to b . And then it's always very easy to find a parameterization, and one would be I take t in the interval from a to b , and I take x to be t and y to be $f(t)$, that has to be the f of x , that is then the f of t . Then I simply, I have this graph of this function, I simply walk over that curve. And I want to know how long it is, but now I know how to do that because I have my formula, the speed here, so here I have the $x'(t) = 1$, the $y'(t) = f'(t)$, so the speed is the square root, the length of this vector is $\sqrt{1 + f'(t)^2}$, that is the velocity. So the speed is 1, the length of that velocity vector is $\sqrt{1 + f'(t)^2}$, and you see here this formula just says, okay, if I want to have the length of that curve, I integrate from a to b the square root of $1 + f'(x)^2$. That's a formula you, maybe you know it from high school, but it's just the same thing. The problem with this kind of formulas is that it contains this square root, and to evaluate these things is often very difficult, so to really compute the integral. I want to give an easy example. I look at this function, a quarter x^2 minus a half times the logarithm of x , and I look at it for x on the interval from 1 to e . So the begin point is when x is 1, your $f(x)$ is $\frac{1}{4} - \frac{1}{2} \ln 1$ is a quarter minus 0, the end point is then the x is equal to e , and then the corresponding function value is a quarter times e^2 minus the logarithm of e , so minus a half. So I have that begin point and that end point, but of course I do not go there with a straight line, I go there with a certain curve, and you want to know how long that curve is, the length of the graph. Now that is now not difficult anymore with our formula, because you get here that the derivative of f is equal to $\frac{1}{2}x - \frac{1}{2} \cdot \frac{1}{x}$, so $\frac{1}{2}(x - \frac{1}{x})$. And now I have to compute $1 + f'(x)^2$, and a later one I have to take the square root of course, but let's save some time by not writing every time that square root. And then you get $1 + \frac{1}{4}(x - \frac{1}{x})^2$, now that is in this situation $1 + \frac{1}{4}(x^2 - 2 + \frac{1}{x^2})$, but you get that twice, and then x divided by x is 1, and then you get a half times a half is a quarter, but you get it twice, you get minus a half, plus $\frac{1}{4x^2}$. And now the trick is that in this situation you can replace that $1 - \frac{1}{2}$ by $\frac{1}{2}$, so this is $\frac{1}{4}x^2 + \frac{1}{4} + \frac{1}{4x^2}$. And then I think, okay, this looks very much what I computed here, but instead of a minus I have a plus. And then you think, okay, this I can rewrite as $(\frac{x}{2} + \frac{1}{2x})^2$. So, by accident here, it's not an accident, it's because I took this example on purpose, you see that the expression that you get for this $1 + f'(x)^2$ is itself a square. And then the square root of this x is here always positive, so that is $\frac{x}{2} + \frac{1}{2x}$, and then the length that you want to compute is now the integral from 1 to e of $\frac{x}{2} + \frac{1}{2x}$ dx , square root of that $1 + f'(x)^2$. And that is in this case, I have to take the perimeter, so I get here a quarter x^2 plus a half times the logarithm of x , upper bound e , lower bound 1, and then I get a quarter e^2 plus a half minus a quarter plus the logarithm of 1 is 0, so plus 0. So I get a quarter e^2 plus a quarter. Not very difficult, I would say, in this situation. If you do the general case, you may get awful integrals. And I want to give one example of that. If you have this curve, $t \cos t$ times i plus $t \sin t$ times j plus t times k , that was that spiraling out over a cone. So I start in the point 0, 0 and then I get this cone and I spiral around and now my arm is too short. You have a question? Wait a minute, I have to get back to my formula. Now you are able... Yes, because that's because of the parameterization that I took. It was not necessary, but I gave that information to give you

a little bit more impression of what's going on. But you do not need it for the length. I want to go to that other situation where you have this cone. So this cone where you have your arm and that is going around this cone. And I hope you understand what's going on here. It spirals around this cone. And if you want to know in this case how far this arm, the distance that this arm passes in a certain time interval, you of course compute the velocity vector and that is in this case, I have to differentiate that, $\cos t - t \sin t \mathbf{i} + \sin t + t \cos t \mathbf{j} + 1 \mathbf{k}$. And now I have to compute the length of that velocity vector, so the speed, and that is now the square root of this squared, $\cos^2 t - 2t \cos t \sin t + t^2 \sin^2 t + \sin^2 t + 2t \cos t \sin t + t^2 \cos^2 t + 1$. And then you have to write out what these squares are and you get $\cos^2 t - 2t \cos t \sin t + t^2 \sin^2 t$. And this one is $\sin^2 t + 2t \cos t \sin t + t^2 \cos^2 t$. And then you see that these parts with the plus and the minus sign, they cancel out and you get the sum of $\cos^2 t + \sin^2 t$, that's just 1. And you get here the $t^2 \sin^2 t + \cos^2 t$, so you get t^2 . So what you get here in the end is the square root of $1 + t^2$. And then you have to integrate, for example, from if you look at the first turnaround, the integral from 0 to 2π of the square root of $2 + t^2$ dt. Nice formula, you know for sure that it has an outcome because this square root of $2 + t^2$ is not very dangerous. But to compute it analytically is quite a hell of a job. If I have time, later on I will explain how you can do this with a sort of trick. But in general, it is nice to have this formula for the length of a curve, but in practice you get very often very difficult integrals to compute analytically. And then you often have to use a numerical approach to get an approximation of that, so that you cannot do it exactly. In this case, it is possible to do it exactly, but I will hopefully have time later on to explain that. Now, then there is, to end this first part, there is this idea of a special parameterization, and that is called the arc length parameterization. And you have to think about it in the following way. So, where shall I... Let's take this blackboard. Suppose I have this arc, and it's parameterized by a parameter description r of t with t from a to b . And then this point corresponds, that's the r of a , and this point is the r of b . And I walk from begin point to end point, because I know that the velocity is never zero, I know I cannot return. Always... Because if it would be zero, if I would return, then the velocity would be zero somewhere, that's not the case. So I simply walk there, and then after, at point b , you have had the distance that you've passed is of course the length of that arc. Now you can think about the following situation. Here in a point in between, in a point t hat, I can also look what's the length is from the begin point to this point r t hat along the curve. Now that length at until t hat is of course the integral from a to t hat of the velocity vector, I now write v τ d τ , velocity now, the length of the velocity vector, so the speed, v τ d τ . And this v is always non-negative. Okay. Now I can also say, okay, I can parameterize, I can characterize this point, that is the point after having a distance Now I call this simply s is the L of t . And this function L is always increasing and therefore invertible, so t is the L inverse of s . The inverse of L exists. And then I get a new description r , a part of description r of this normalized parameter according to the length s , and that is simply the old one where for t you plug in this. So it's always possible to get this, what is called the arc length parameterization. Sometimes that's easy, so if the speed happens to be constant, it can also be a hell of a job. But for some cases this is easy to know that this exists and that is not that difficult to understand, I think. OK, now I had three quarters of an hour about parameterizations. I hope you now know everything, not everything, but enough about it. Let's first have a break until 2:30 and then we start with the material of this week. So let's start again. Please take your seats. The next subject, so next I will start with, I think it's chapter 13 of the book about functions of several variables, and we will go to a large part of that chapter within a few lectures. And the idea is now different from the first hour. I now look at functions where I have a domain, let's call it D , that is for example contained in three-dimensional space. And I look at functions that map from $\mathbb{R}^3$ to $\mathbb{R}$, for example, or from $\mathbb{R}^2$ to $\mathbb{R}$. And it is about how to think about this and what is a graph of such a function. Now here you see the same thing. So you have a point x_1 until x_d in D . And now I assume that D is equal to 2, so I have just a function of x_1 and x_2 . And then if you have a point in the x_1, x_2 plane, then there is a function value connected to it, and then you can take the height, the corresponding height, the corresponding function value. And then you can connect these points, and what you then see in this situation of functions from $\mathbb{R}^2$ to $\mathbb{R}$, that this graph will be a certain surface, because for any point xy , you have the corresponding z value f_{xy} . Now these functions occur very often in lots of practical examples. Here is this planet of the earth. You can think about the following situation. For example, we are on the earth and we have all kinds of measurements during the year of the annual, the amount of rain that has fallen at a certain position on the earth. Now then you can make a function that maps for every coordinate on the earth the corresponding number of millimeters of rain that has fallen, for example, in 2025. That is then a function from $\mathbb{R}^2$ or part of $\mathbb{R}^2$ to $\mathbb{R}$. For every point on the earth, it maps to that number of millimeters. But you can also have a function from $\mathbb{R}^3$ to $\mathbb{R}$ if you think about the following. f_{xyt} , so now it's different. t is time, and I now think what is f_{xyt} , that is the temperature on the earth in position coordinates x, y at time t . And the temperature is then just again a scalar value. That's important. So you map always to a scalar value. The total rainfall was also a given scalar number. And that's what we always think. To a point in space, we map such a value. And here it's a little bit different because I have here a function of three variables, x and y determine the position on earth, and t the time. And then the value will be the temperature, for example. Now, you have to think how you would want to graph this, to depict this. That's very difficult

because if this is the temperature, if the temperature is t of $fxyt$, then I need four, a four-dimensional picture to get a sort of picture of this. And I cannot imagine that. Maybe you can, but for me that's too difficult. So in this situation, this graph still exists because it's simply the set of all points x, y, t in capital T for which x, y is an element of earth, t is larger or equal to zero, and capital T is the f of xyt . That is the temperature on position xy at time t . That is still the graph. You can still say, OK, this set of vectors in four-dimensional space, but how to understand this? What do we... but how to understand this, how, what do we want to get out? Now, this is an example of such a situation, well, how you could be able to depict it, and if, for example, you use colors, and they have, I do not have a movie here, but you should have a movie then. Then you have the position on Earth, you have here colors for the temperature, and then you can play on a movie that simply shows how temperature changes, for example, over one day on Earth, and then you see these colors changing. But that would then be a sort of, yeah, depiction of, because I need the x, y , so I need the plane, and I need the time, that is then just the time of a movie, and then I need this fourth value and I depict it here with a color, then. So for a function from $\mathbb{R}^2$ to $\mathbb{R}$, you can still have these graphs like in the left part. In the higher dimensional case, it's getting more difficult, but there are some ways how to simplify this. Now, first, some examples. First about the domain. If the domain is not given explicitly, it is assumed to be the largest set for which the expression, the function description, has a meaning. So mathematicians should give that, but you're often too lazy to describe exactly what the domain is, and then you simply think about, okay, for all x, y for which this expression has a meaning, that is the domain. Now, here you have the graph of the function $f(x, y)$, so z is $f(x, y)$, and this is minus six times y divided by two plus x squared plus y squared. And you see here that the denominator is two plus the sum of squares, so that's always positive. So you never divide by zero, so the domain is here everything, because you can always, for every x, y , this expression has a meaning. You see, for example, if y is equal to zero, z is zero, that is this line on level zero. If x is zero, you get z is minus six times y divided by two plus y squared, and you get something like this. And in principle, you can show that the graph in this case is this sort of hill and this lake. And for the rest, it's very flat because if x and y become very large, then you divide by a very large number compared to the numerator, and that will become small, close to zero. This one is much more difficult. What is this? Now, this is the graph of the following function. So I have my notes here. The following function. G of x and y is equal to the logarithm of one minus the square root of one minus x times y . And on purpose, I didn't describe the domain because I'm going to find out what the domain is. And then I have to find out for which pairs x, y can I interpret or can I read this expression. Now, then, first of all, you have your square root, and under the square root, you should have a non-negative number. So the first condition is that one minus x, y should be larger or equal to zero. That is, otherwise, I cannot do this. So I get that x times y is smaller or equal to one. But you have another thing because the logarithm of a number can only be taken for positive numbers. I do not know what the logarithm of minus five is. So I get here that one minus the square root of one minus x, y should be strictly positive. So I get that the square root of one minus x, y should be smaller than one. And that means that one minus x, y should be smaller than one. So x times y should be positive. And that means that x and y should both be positive, or they should both be negative. That's easy to solve. So what I now get is the following situation. I now, so this is a picture of the domain. I first look at the situation over here that I need that x times y has to be larger than zero. So I have to be in the first quadrant over here or in the third where both everything is negative. But I cannot be on the x and the y axis because there I get trouble. If x is zero, I get a logarithm of zero, and I don't know. So I get this. And then I have to take care that x times y is small or equal to one. So I get here that y is smaller or equal to one over x . Now then you recognize the form of the hyperbola. So I get here this graph. And your y should be smaller or equal to one over x . So then I get this part. And if x is negative, I get exactly this part. So that's my domain. So the domain of G is the set of all points in $\mathbb{R}^2$ such that x times y is larger than zero and smaller or equal to one. And then the next thing is you have then there on that domain, our expression makes sense. With this information, I hope you can understand a little bit why this is the graph of this. You have here the x axis in this direction, y axis in that direction. This is the first quadrant, and this is the third one. And you get the following situation. If you get very close to one of these two, to the x axis or the y axis, your function ceases to exist, and you get here a curve like this that goes to minus infinity. Here you see this other little bit, this other hyperbola. And of course, when you get there, so if y is equal to one, so if you're on this spot, this will be the logarithm of one, so you get zero. So here the value will be zero, and then you get such a surface that goes down but has a very limited domain. Because for other x and y , the expression has no meaning. So to find out what domain is and how this graph looks like is not at all so easy. And sometimes it is easy. So in this situation, that's the easiest example. If you want to have the sphere with radius 3 or the hemisphere, the northern hemisphere of that sphere, you simply say I have the function $f(x, y)$ is the square root of 9 minus x squared minus y squared. You simply know that x squared plus y squared plus z squared has to be equal to 9. And then I make $z \geq 3$. But if you then use the northern hemisphere with the positive z values, you get this. And in this case, the domain of f is, of course, the set of points x, y in $\mathbb{R}^2$ for which x squared plus y squared is smaller or equal to 9. Because otherwise, I would take the square root of a negative number. And we have no complex numbers here, so we cannot do that. So sometimes it's easy, then you can also get that picture. But to depict that yourself is still quite difficult to do that. Now you can go one dimension lower, and that is by using level curves. So in this situation, I have a

paraboloid. So that's the graph of my function. z is, for example, $x^2 + y^2$. And what I can do now, so this is difficult to draw. I say, okay, let's do it, let's take level curves. What I do, I have this paraboloid, and I intersect that simply with the plane z is some constant. For example, z is equal to 1. What do I get on the height z is equal to 1? I get a circle that is described by $x^2 + y^2 = 1$. And z is also equal to 1. Now, what you now do for these level curves, you forget about the z coordinate. So you depict this level curve, so you know, okay, this corresponds to z is equal to 1. You simply depict this as a circle in the x, y plane. And you say, put to it, oh, this is at height z is equal to 1. And this is exactly the way you know from topographical maps. If you go hiking in the mountains, you certainly have seen this. You know how the altitude changes because you have on your map these lines of the same altitude that are level curves. So I simply, my map describes the height of the landscape as a function of the position, but not for every point. It depicts these lines, these curves that are on the same height. And then from that you can conclude a little bit whether you have to go walk upward or that the road goes nicely down. Now, that is here depicted for these height lines for Eindhoven, but that is not very spectacular. So that we won't look at. What is important for you is that you are able to make the connection between the graph for a function from $\mathbb{R}^2$ to $\mathbb{R}$, which is a two-dimensional object living in $\mathbb{R}^3$, for example, a sphere, that is a two-dimensional object in three-dimensional space, and the picture of the level curves. And these are just in $\mathbb{R}^2$, so these you can draw on paper very easily because these are just curves. Now, what do you then do? The following thing is important. If I, now let's, I prepare an example for that. If I look at this function f_{xy} is the square root of $x^2 + y^2$, and I want to compute some level curves and draw them, then I take, for example, f_{xy} is equal to 1. Then I get a circle around the origin with radius 1. So this corresponds to level curve on height 1. And then if I look at f_{xy} is equal to 2, then I get the square root of $x^2 + y^2$ is 2. So I get the circle with radius 2, Holiday, two years ago. I made this picture on purpose to show what a saddle point is. Here you have a saddle. Can you see the photo? Yeah, good enough, because for me it's quite fake. You see here, this is on the border of the Czech Republic and Poland, quite close to Germany, by the way, so in the western part of Poland. And you come in from a mountain, it's called Snezka. And you see, if you walk along this road and you get here, you are at a local minimum, and then here it goes up again. The road I just showed you is approximately the border between the Czech Republic, that is on the left side, and Poland, that's on the right side of the picture. But if you want to go from Poland to the Czech Republic, what you do is you want to go there, but you want to pass through a point where the height, where you have to pass these mountains, is the lowest possible. So you take this walking route, because here you have the maximum value has a minimum. And then you walk down on the other side. So this is this effect where you have, so these saddle points are interesting, because that is where you want to build a road, if you want to look back at the situation. This is at the saddle point. That other picture was taken probably somewhere here, or maybe here. And this is that mountain. And now the Czech Republic is on the right-hand side, and these people are looking into Poland. But you see that it's then different. So these saddle points are interesting to, have interesting properties, and you will look at them later on. Okay, now I skip that example. I have here a question for you. Now I have here this situation. I have this topographical map and some questions. I have here the point A, here the point B, and here the point C. So where is the hill the steepest, in point A or in point B? Yeah, everything, well, they say B. That's, yeah, the level curves, they are equidistant, the 100, 200, 300, 400 meters. So it is, of course, over there, because they are the closest. Where do I have peaks, mountaintops? Yeah, over here, I think, and over there. And what is the point C? A saddle point, yeah, because you're there. If you go here from this top to that top, it's a minimum, but if you go like this, it's a maximum. So I had planned to bring the hiking map on the border of the Czech Republic and Poland, but I forgot. So that was stupid, but yesterday evening I forgot. Okay, so you have to be able to read these level curves and to understand how then such a shape is. Okay, now, I don't think that that is too difficult. What we do next is much more difficult, and that's what is a limit, a limit for functions of two variables. And I have here the first example, the function $x^2 y / (x^2 + y^2)$. And I want to look at that point in the neighborhood of the point $(0, 0)$. You see that in the point $(0, 0)$, the function is not defined, and I want to know if the function is going to a certain value around that point, or is it maybe it doesn't exist. And then you have to ask yourself the question, what does a limit mean in this situation? And this is an easy situation. This is the graph, and I think you will notice, okay, this will go to 0, probably. But what do you need for this, for a limit? You have to do some quite technical parts of the definition. The first part says that if I take the limit for x going to x_0 , so I want to know this, I get the definition that when is the limit of a function for x going to a fixed vector x_0 equal to a certain number L . That is the question. Now first it has to be possible to go to this point x_0 in a way that it is in the domain. And I also have to do it in such a way that there are points in that neighborhood that I can get there. So the first condition is x_0 is not an isolated point, but in any small neighborhood of that point, there is a point that belongs to the domain that is in that little circle. So if you would depict it here, you have this point 0 . And here it's very easy. What you need is that if I take a small number δ and I draw here a circle with radius δ around this x_0 , so here's my x_0 , then there has to be a point of the domain inside this circle and it should not be equal to this x_0 . So for example, this one. But then you say, this has to hold for every positive δ . So if I now change the situation a little bit and I'm going to this little circle δ , also in that one has to be a point of the domain that is unequal to x_0 . So I get, for example, this point. But by doing so, I

can make this circle smaller and smaller and smaller all the time. There has to be a point in the domain within that little circle that is not equal to that center x_0 . So what I then automatically get is a sequence of points from the domain that go to that point x_0 . So said more properly, it has to be possible to go, to walk through the domain to that limit point. Otherwise, if it's an isolated point, you just have to think about a little island in the ocean that we do not take into account. So you just have one specific, and there is this way how to go there. Now the difficulty is that for functions of two or more variables, there are lots of different ways how to approach such a limit point. And the situation that you had in calculus with limits for functions of one variable, that was the easy situation because for a limit of a function of one variable, you can only come from the right-hand side or from the left-hand side. You can have this one-sided limit and that other one-sided limit. That's two possibilities. And now you have infinitely many because here if I want to approach this, and in our example, everything outside the point $0, 0$ is here, then you can go like this or like this or like this. And you have to... How do you handle all these possibilities? Now that is the second part of the definition that is even more difficult. And that says, okay, what I want to have, so the limits, you have to be able to approach x_0 to the domain, but then what does it mean it has a limit? That's the corresponding function values on that approximation gets arbitrarily close to this limit value L . But how to express that? Then you say, somebody gives me an accuracy level ϵ , and I will show you a picture quite soon, for which there has to exist a δ , such that if I have an x in the domain and within a distance δ from x_0 , the function value of f_x minus L has to be smaller than ϵ . Now this is still too difficult, so let's first look at the picture because then it gets more, it gets clear, I hope. I have to go here. Yes, finally. Okay. I'm looking now at a different situation, but to make clear how the definition works. And that is this function H , the sine of the square root of x squared plus y squared divided by the square root of x squared plus y squared. The point $0, 0$ does not belong to the domain, but it's the only point that does not belong to the domain. Yeah? What does it now mean? My claim is the limit of that function for x going to $0, 0$ is 1 . And what does that mean? Now I have here, this blue is the graph of my function. And now I take, here is the level 1 , and now I take an ϵ and I look at the height, the yellow plane is the plane z minus ϵ . z is 1 minus ϵ , so a little bit below the z , horizontal plane. Now, what you see is then in the graph you intersect with this little circle, and this part is above this yellow plane and the rest is below. Now, if I now project this little circle down on the xy -plane, it means that if I take an xy pair in this part of the domain, in this little red circle, then the corresponding function value will be here on this top. So the corresponding function value will be larger than 1 minus ϵ and also smaller than 1 . So it will be within this ϵ margin of the value 1 . And what does this definition now say? If I now take this plane, so that is for a certain ϵ , and you see, okay, for this ϵ you have this little circle here, and if you are close here, then you are close to your limit value everywhere. Now, the situation is that if you now take ϵ smaller, so this yellow plane goes up, then you can imagine then you get here a little circle and then you get also here a little circle, and then if you are in that little circle, you are even closer to your limit value everywhere. And you can see here from this picture, I think, that if you let ϵ get to 0 , that your red circle is getting smaller and smaller and smaller, but has a positive radius, and then within that little circle, you still guarantee that your function is then very close to this L . So if you keep this picture in mind, we read again the definition. Over here, what you say here is, so for every ϵ positive, so that is the ϵ that I subtract from the limit value But for any of these points, this expression is always between 0 and 1 . I know that for sure, because x squared plus y squared is always more than x squared. Maybe it's equal if y is equal to zero. So this is smaller or equal to the absolute value of y . But this point is in this circle radius δ , so this y , the largest value of y that occurs in that circle, that's over here, that is smaller than δ . This is the largest y that occurs. But it's on a distance δ from zero. So this is smaller than δ , but δ was chosen to be ϵ . So I've now said, okay, if I start in a point on a distance less than ϵ from the origin, and I take the corresponding function value, then I know for sure, I don't know what the function value is, but I know for sure that the corresponding function value is less than ϵ away from zero. For sure, because think of the picture, you had that yellow plane with this 1 minus ϵ , and then you had this little circle, and if I get closer to that limit, my red circle gets smaller, but it has a positive radius. And I can get as close as I want to my limit value if I take that radius of that circle small enough. And that's what I've proven here. And that's why you say that in this situation, the limit of this function, so the limit for xy to $0, 0$ of f_{xy} , is equal to zero. And now I take the freedom to... May I take the freedom to continue for five minutes? Because I won't be able to finish everything, but to depict the difficulty. I now am going to look at the limit for xy going to $0, 0$, not of that function, but of xy divided by x squared plus y squared. This one. And what you now should see is the following. Of course, I can try to get this proof done in the way it is written in the previous case, but I will fail because it turns out that this limit does not exist. Simply not there. And to make that clear, there is a very easy argument. Because what you do here is you go here to the point $0, 0$, but I have to show that in a little circle around this... So if I'm getting close enough to this point, I get close enough to a certain limit value. Now what I now do is I go along a certain line, a certain way to that point. So this is the line y is λx . So I call this function g of xy . g of xy is xy over x squared plus y squared. But now I am not considering the whole function, but I'm considering the function g where y is λ times x . So I'm only considering it on this line. And I go to x is 0 to this point. Now what is now the function description? Now y is equal to... So I get the x times λ , so λx squared divided by x

squared plus $\lambda^2 x^2$. So that is $\frac{\lambda}{1 + \lambda^2 x^2}$. But I take the limit for x going to zero, but then x is not zero. It's very close to zero, but it can be as close as it is. As long as it's not zero, $\frac{x^2}{x^2}$ is equal to 1. So the limit of this expression will be $\frac{\lambda}{1 + \lambda^2}$ for x going to zero. Now what does that mean in the picture? In this point, that does not belong to the domain, the origin. But if I go, if I take λ is 1, so here the line y is equal to x , my function value is then $\frac{1}{1 + 1}$ squared. So on the line y is equal to x , I have that the g of x, x is always a half. Here by accident, it is enough that it will go to a half, but here it's by accident everywhere a half. If I take λ is minus 1, I get that the g of $x, -x$ is minus 1 divided by 2. That's along this line. So on this orange line, the function value is always a half, except of course in this point. And on this line, the function value is always minus a half, except in this point. And now I ask, is there a value L such that in any neighborhood of that point, you're getting close to L ? And then the answer is, of course, yeah. And can you also explain it? Who said no? Yeah, yeah, you cannot approach it by different values. Here it goes to minus half, here it goes to half, and the reason in the definition is you can draw a circle as small as you like, but always the function value a half occurs and also minus a half. And you can make this circle smaller and smaller and smaller, still you have points on both these lines. So in the neighborhood of that point, any function value a half occurs and also minus a half occurs. So there is no value where in the whole neighborhood you are very close to. That's the reason. But your point of if you can go to a certain point along one road, one part, and to another part, and the limits are different, then you know for sure that the limit does not exist. And that you have to keep in mind. And it is much easier to prove that the limit does not exist just by a little bit of trying than to prove that the limit exists and to prove that. And the good news is that I want you, yeah, you have to be able to understand how the definition works, but I will not put that much, I do not expect you to give very difficult proofs in a set. You have to understand what this limit is.

Mathematics 2. I noticed that there was a lot to do because last Monday I was a little bit behind schedule.

But I also got some questions after my lecture so I have to, I want to give an answer to them because I think it's important for all of you because they're very good questions. And it was mainly, these questions were on the notion of open and closed sets and all the notion of a boundary. And there I want to start to make that hopefully a little bit more clear.

So we, last time we were looking at let's say three-dimensional space and then we looked at what's called the neighborhoods of a point in three-dimensional space here.

And that are all the points in the ball, all the points y in the ball around x on a distance less than r away from x . So you have a point in space and then you have this ball around this point with radius r and everything within the ball is then, that are the points described by this set. And now you call a set open if for any point in that set you can find a distance r , a positive r , and maybe different for every x , such that a little ball with radius r is fully contained in s . So if you have a point in s also a little bit, you say that the set is closed if its complement is open. Now this looked a little bit difficult, at least there was some, I got some questions, but maybe I should have told it in this way. If you have a set s , you call a point x , an interior point of s . So if there exists a distance r such that not only x is in s , but the whole ball around x with radius r is in s .

And then you can say a set s is open if all its points are interior points.

Now you can also say what an exterior point of s is and then it just works the other way around. You also look at that point and there exists now a positive radius r such that this little ball around x , so not only x with also its neighborhood, fully belongs to the complement of s . So it's fully outside of s and it's called an exterior point. Not only the point is outside s , but also its neighborhood.

Now when you know this you can define what a boundary point of s is, that is, and now you get mathematically the correct definition, but it looks a little bit awkward maybe.

This is the case if a point is not an interior point and not an exterior point, so that means if for all r positive the ball $B_r(x)$ contains elements of an element from s and an element from s and an element of the complement of s .

So a point is a boundary point. If you can you look at a small neighborhood and you can take it as small as you want, doesn't matter, and maybe if r is big you get a big ball, which usually you think about r small, which you get a small ball, but in such a small ball there's always a point in s and always a point outside s , and that's typically the situation if you have a situation like this, where for example in $\mathbb{R}^2$ I have this line, so I look at the sets of points in $\mathbb{R}^2$ where $x + y$ is small or equal to 1, so that is this line and everything over here, then the interior of that set, so the set of interior points, that's called the interior of s , that is the set of all interior points of s , that is then in this picture, in this example, you're not on that line, but you are really below that line, so everything what is over here are interior points, because if you're really below the line, so after your point below the line, then the distance to that line is positive, now you take simply half that distance and then you take, now in this case you get a little disk around that point and that's fully below the line, this line $x + y$ is equal to 1, and if you're not on the line it's always a positive distance, so you take that r simply smaller than this positive distance, however if

you are on the line, then it's impossible because any circle that you take here, you always, you can take it as small as you like, but always you have points below the line and points above the line, and you can, as long as this radius is positive, you cannot prevent that, so here for every, how small you take r doesn't matter, but positive, you always have in that neighborhood points inside as and outside as, so this is typically such a boundary point, now in the points over here, these are exterior points because there you can again make a little circle around it in such a way that everything within the circle is still in the set, so this is the meaning of the boundary, so you have, you look for neighborhoods that are completely in as or completely outside as, then you have interior, exterior points, and then what is remaining?

That are the boundary points, question, usually it is if you have a non-strict inequality like this, you usually have closed sets because the opposite complement there you have a non-strict, there you have the strict inequality and that is usually open, but it's not, you have to think it this way, I got questions because I gave this example that in $\mathbb{R}$ the interval from minus infinity to five is open, and then I got questions about minus infinity, why is this not, that is, why is that not a boundary point? Now it is simply what is this interval, this is the set of all x 's in $\mathbb{R}$ that are smaller than five, that is this set, and if I dare take a point and it doesn't matter which point it is smaller than five, then you can always get a little interval around that point that's fully contained in that set, and also if you go to, you never reach minus infinity, but if you are at minus 100,000 then you can still do that, and even further, so that there is no bounds there in mathematics because you look at, you can always get smaller wherever you are, so that minus infinity such that it's not really a boundary point, the only boundary point in this case is this point five, because there, in that there if you look around five, an open interval that contains five will have points lower than five and larger than five, so that's the idea of boundary, and usually if the boundary of a set does not belong to it, so if the boundary does, if no boundary points belong to S , then S is equal to its interior, and then it is an open set, but if S contains both interior and the boundary, that will be a closed set, so that is the idea, but you have to think in these little neighborhoods, think about these little circles or balls around a certain point, whether they stay inside or not, in the electronic homework there will be exercises on this, so keep this in mind.

Second thing that I didn't do last time, we looked at these quadric surfaces, and one of them was the cone, and if it's centered around zero, you had an equation of the form, for example, z^2 over c^2 is x^2 over a^2 plus y^2 over b^2 , and then you get a shape like this, so here's a square in z and it's a sum of the squares x and y , of course you can switch the walls of x and y , then you get this cone orientated along a different axis, this is along the z -axis.

Now, what happens now in the following situation, because that are the two we didn't discuss yet, that are the two forms of an hyperboloid, that's almost the same thing, but here you add here minus one, so I put here minus one, and then you get an hyperboloid, but here you can see that it is a hyperboloid of one sheet, everything is connected, and the reason is, if you look at this equation and you bring the one to the other side, you get here one plus z^2 over c^2 , and here you get a sum of two squares, so if somebody of you gives you z either negative or positive, I have to add one to it, take the square root and then I get a radius, and then on that height I have to take this ellipse with that arc, and then I get a higher ellipse, and then I get a shape like this, every intersection with the plane z is constant, will be an ellipse, but I can solve this all the time. Now you have a different situation, and that is when you have a hyperboloid with two sheets, and then you look at this equation, it's almost the same, and here I get plus one plus a constant, and so in this case you have a sum of squares, but

no, not a sum, well you have x squared, y squared, and z squared, in two of them the sign is the same, and then the other one the sign is the opposite, so here you have, you can write this as x squared plus a squared plus y squared over b squared minus z squared over c squared plus one is equal to zero.

Now in that case you get a hyperboloid, and in this situation you get a hyperboloid of two sheets, and the reason is you have two separate paths, now I bring this plus one to the other side, and you get your z squared over c squared minus one, and if then z is very close to zero you will get your negative number, because you subtract from a very small number minus one, and that means that this sum of squares has to be negative, and of course you do not find any solutions for that, that is that empty area here, here z is too small and then you cannot solve for x and y , that's this in between, if that z gets large enough then you get your ellipses, and then you get this hyperboloid of two sheets, because you get one on the positive side and one on the negative side, because for plus z and minus z you get of course the same thing, everything is symmetric here, so you see here that you then get that situation, and you might think about it in the following way, if you think about a hyperbola simply in the x - y plane, so not a three-dimensional case x - y , and now I have here two asymptotes, and I have such a hyperbola, and it can be either over here and here, or but let's now assume that it's over here, and over here, then this is everything in the plane, and now I put my third direction, so I put here my z -axis, a positive direction, in that direction you know it because of the white hand rule, and what I'm going to do is I'm going to take this part of the hyperbola, and I'm rotating it around this y -axis, so then you get a shape here, and then behind the blackboard, and then you get this shape, however it is not around the z -axis, but now around this y -axis because you have this this goes here around like this, and you get what I would call a cooling tower of such an electricity of a power plant, let's say. If you take the same hyperbola, and now I put again here my z -axis, I can also take this shape and rotate it along the x -axis, so this way around, and then I get, I have to do the same thing, then I get such a hyperboloid of two sheets, so you see this difference, it's just a sort of idea of rotation, but what is important for you is that you recognize from the equations what kind of object you have, and that you can play a little bit with it, because later on we use these equations quite a lot, and then you have oh this looks a little bit like that, because otherwise that is much easier to do all kinds of computations.

Okay, now so far for the quadratic surfaces, and then there is one thing I really need to tell, and that is the two other possibilities of, there are many other possibilities, but two possibilities of describing points in different coordinate systems, and there are instantly many options, but there are three very important, that was the one we already looked at with x , y , z coordinates, the Cartesian one, and then the cylindrical and spherical coordinates are also very important, and with this picture you can understand what this, in this case the cylindrical coordinates are, so I want to describe a point in space, and what I did before is I said okay, I take this point, I project it to the xy -plane orthogonally, then I get here, and this distance I call z , that will be my z coordinate, and then what I did in Cartesian coordinates, I said okay, now at this point I project orthogonally on, for example, the x -axis, and then this distance is the y , and this distance is the x , so I get this x , y , z in these three directions.

Now I'm going to do it a little bit different, I look at the same point, and my first step is still the same as before, I simply project this down, and I call this length z , and that will be my last coordinate z , but to describe this point in the xy -plane, I do not use xy -coordinates, but polar coordinates instead, so what I do is I compute the distance of this point in the xy -plane to the origin, and that's called here small r , and I take the angle between the x -axis and this vector from the origin to this projection point, that's θ , then you get automatically that this x is equal to r times the cosine of θ , and this y is r times the sine of

theta, so you get here a different coordinate system with coordinates r , θ , and z , z is just the one we knew before from the Cartesian coordinates, this r is always largely equal to zero, because it's distance, and θ is an angle, it's always between zero and two π , and then your vector x , note here that on the slides, this x is fat, and then it's a vector, and if x is non-fat, it's a scalar, then it's the x coordinate of that vector, so this vector x has the, in the x direction, it is $r \cos \theta$, then $r \sin \theta$, and then z , and that way I can, for every point in space, I can find these coordinates, and also an opposite way, if I have these coordinates, I can find the point. Now it's a good thing to look what is happening with the shapes, if I take one of these coordinates constant, so if I take r to be constant, then my x and y are on a circle with radius r , and my z is free, so what I get is this blue circular cylinder that you get when r is simply constant r zero, so then you get, you know I'm on the cylinder.

Then you look what happens if z is constant, oh that is easy because then you have a constant horizontal plane, z is some constant, that is this, this red plane.

Then what happens if θ is constant? I have to be careful because, now I'm here with my, here's the origin, and this is the z axis, the x axis, the y axis, no, and now I take, I'm in that xy plane, so I take just this θ , for example like this, and then I get here a plane, a half plane, not a full plane, because it's only in that direction, but not behind my back, because that angle would be by more, so if I have this θ , if that's six, I get such a half plane, that stretches out third one, and that is a yellow plane. My point is just the intersection of all these three things, so the intersection of the cylinder, this plane, and this half plane, but it's good to see what is happening if you keep one of these coordinates constant, what is then the shape.

Now it is possible to go from one coordinate system to the other, and that is sometimes pretty obvious if you want to know the small r that is the square root of $x^2 + y^2$, and things like that, you can, you can compute that yourself because you worked away with polar coordinates. This is just polar coordinates and the z in the k direction. A slightly more difficult one, but that you know very well already for a very long time are the spherical coordinates, and there you again want to describe a point, but again in a different coordinate system, so with three different numbers, and what you now do is the following. You have this point p , and the first thing that you do is to determine the distance from this point p to the origin, and you call that capital R . That means if you keep capital R constant, that you are on such on this blue sphere, because you get an sphere, that's why it's called spherical coordinates. If R is constant, you are on a sphere with radius capital R . Now there are two other coordinates that describe where you are on this sphere, and it is done with two angles, and the vector k , θ θ 1, and the vector that points from the origin to p .

I call that ϕ . If you know that, then if you project this point p again down to the xy plane, you know that this length of z will be equal to capital R times $\cos \phi$, because you have here this triangle with also here the same angle ϕ .

So z is capital $R \cos \phi$. Now I again look at this triangle, and then I know what is distance from this point to the origin is. That is here called small r , that is the name in the polar coordinates, but of course that distance is capital R times the sine of ϕ .

So we have here z is $R \cos \phi$, and I have r , small r , is $R \sin \phi$. Now to determine now the coordinates, the relationship with the normal coordinates x , y , z , the Cartesian ones. Okay, I know how to do that, because this was just this r from that we also saw in the cylindrical coordinates. So if I want to know what is x is, I have to project it down over here.

So this θ is the angle between the x -axis and this vector from the origin to this projection point, that's θ , and then you know that x is equal to this small r , as this, times $\cos \theta$, and the y is the small r times the sine of θ .

So if you give the distance from the origin, capital R , the ϕ , and now you have to determine what ϕ , in which interval this ϕ should be. Now if the z -axis over there, and here is my origin, then this angle ϕ can only go like this until here, because if I have to do their point, when I have a point there, I later on make such an with that other angle, I can go around. So this ϕ is the, I always remember, this is the angle with the positive z -axis, and for that reason, ϕ is always, so r is always positive, ϕ is always between zero and π , and then my θ , once you have this, if r and ϕ are fixed, then you have to go around on that circle, so θ is between zero and 2π , and then you get the spherical coordinates. If you look in that situation to these surfaces where one coordinate is fixed, then that was easy for the r , because if r is fixed, you are on the blue sphere, is if ϕ is fixed, I have this angle, but I have to go around, so then I get, for a fixed ϕ , I get such a cone, a half cone, because it's not on the lower side only, it's only on one side, depending on whether ϕ is smaller than π over two or larger than π over two, so I get a half cone, and then for the θ , I have again this half space that we also had for the cylindrical points, and then you see if you are on a point here where r , ϕ , and θ are given, you know you are on this blue sphere, which you know exactly where you are, because this angle describes on which circle you are, and then the θ describes where you are on this circle, and you know this very well from geography, because on earth, to give a positioning on earth, you always have to give two coordinates only, because the arc of the earth is assumed to be fixed, and then you only have to give, if you have to describe a point on this blue sphere, if you give the ϕ and the θ , you know what it is, now we know this, this ϕ , we know this as, for example, northern latitude, if you give a point, then you say there's so many degrees, northern latitude, so many degrees, eastern longitude, and that is this other angle, so this is just the coordinate system that you use on maps with these degrees, nothing more, nothing less, and r is in that case on earth that's fixed, so you don't have to give that, it's not changing, so with these two angles that you know, okay, for earth positioning, you get, in a way, that's already the first application of spherical coordinates, and that's why I said to you, you already know this, now I hope this is clear, you also see that you have to be careful about, okay, what happens if one of these coordinates is fixed, that you get a certain shape, keep that in mind, because we will use that also later on. I only will now go to the second part, after, to the real lecture of the second week, so I'm really now behind schedule, okay, this is, let's go to which, what I, where we now look at is, so the main topic of today, I only get to this later, is functions from arc to r^3 , and the easiest way to think about this is that you think about a particle that is moving around in space, and you assume that it cannot jump from one place to the other, so that movement of that particle will be something that is continuous, and you want to describe the movement of that particle, now what you then can do is that you say, okay, I simply describe what at time t , the position of that particle is, and I call that the position vector $\mathbf{r}(t)$, and that has three components, $x(t)$ times the first unit vector plus $y(t)$ times the second unit vector plus $z(t)$ times the third unit vector, and what I assume is that these functions $x(t)$, $y(t)$ and $z(t)$ are just functions from $\mathbb{R}$ to $\mathbb{R}$, and I assume that they are continuously differentiable, x, y, z . Okay, now what you now see on this picture is if you have a cylindrical helix, you have a question? No, sorry for the people at home, but I stop here because it's not doable here, I will try to get another one after the break, 10 minutes. Hello, hello, storing. So, can you hear me? Yes, is it now okay?

If otherwise, please complain. So, I hope for me it's also a lot easier in this way.

So, let's quickly start. We were talking about a point, a particle moving in three-dimensional space where we have a position vector, and if you then determine the derivative of that position vector, you get the velocity vector. Of course, you can also determine the change in the velocity, and that goes in the same way, and then you get, of course, the acceleration of that point.

So, acceleration a of t at instantaneous, that is the derivative of the velocity vector, the time derivative, or it is the second derivative of the position vector. So, then you get, of course, that the acceleration vector is, and then you get this representation with these unit vectors i , j , and k , and here you get the second derivative of the x component, the second derivative of the y component, and the second derivative of the z component. So, that is the acceleration. Now, if you do that just in one direction, for example, only in the x direction, and you assume that y and z always remain the same, then you get a picture like this. So, if you have, for example, the acceleration is like this, which, so first you go up, you have an acceleration of four meters per second squared, then you go up like this, and the velocity is, in this case, going up linearly to a certain value.

Then the acceleration stops, and if the acceleration gets zero, what is happening is that the velocity will remain the same. So, the velocity will remain the same, and you, the distance traveled, it still goes up, but now in a linear way, of course.

Then you decelerate, so then the acceleration gets negative, and then what you are, the velocity will go down. It first is still positive, it then becomes negative. It means that first you still go up, but slower and slower and slower, and then you go down, and here you can just see this relationship between velocity, acceleration, and position, and to get from one to the other, you have to differentiate, or you have to integrate, of course.

Okay, now, of course you can go further, but usually you do not do that in physics, and that is just because of Newton's law, that you know that the force is mass times acceleration, and that is why you are especially interested in this R of t , its derivative, and its second derivative, but the third derivative, yeah, if there is a physical meaning, then it is useful to do that, but otherwise, why would you do it? Now, I want to look at, oh, I already did some things, I already did this question, the conical helix, what is the velocity speed at acceleration? I did not do the acceleration of that particle that goes along this conical helix. Now, this is very easy, what is the position factor and velocity of a particle having constant acceleration? I call it a , because on the slides, you think that it is the gravity, the acceleration because of gravity, but just if you have an acceleration that is fixed, so the acceleration is always a , then you know that the speed factor Vt is equal to, yeah, the speed at the begin time, let us say V_0 , plus a times t , because if you differentiate this thing, you get the abet.

If you then want to know what the position is, then you have to say, you have to integrate once more, you have to know what the initial position is, if you do not know where you start, you also do not know where you end, so that is the begin position plus the initial velocity times t plus $\frac{1}{2}at^2$.

Note that this is different from what I think you learned in high school when you looked at these, when you were shooting, when you were throwing balls, let us put it that way, because here automatically, because these are factors in three dimensions, so this is the initial velocity in three dimensions, and this is the initial position as a factor in three components, so I have all components the same, and if you just look at the gravity field of the earth, then this a will be the vector minus g times k , where g is 9.8 meters per second squared, and then you get, of course, that the components of a in the x and the y direction are just zero, so you have to read this in a vector way and not in a way of that it are

scalars, and that's, I think, what you're used to from high school.

Okay, now we get a different question, it has to do with our other helix that is depicted over here.

We have an ant, and it works on this helix, it starts over here, and it walks around along our cylinder in this way, and now it is given that its speed is constantly five centimeters per second, and its speed along the z-axis is three centimeters per second, and the cylinder has a radius of two centimeters, so it's going at five centimeters per second around, and in one second it goes up three centimeters.

This change in z-coordinate is three in one second, then it's a pretty fast ant.

Now, what you want is to describe the position of the ant in terms of a formula, and then you have all the information that is asked here. How can you do this? So, it's important to remember that the speed here is constant, but the velocity is not, of course, because every time the direction in which this ant is going is changing, so the velocity will not be constant, otherwise it drops off the of the cylinder.

Okay, so let's assume that this is of the form x_t in the x direction, y_t in the j direction, and z_t in the k direction. What I then know is what the derivative, what the speed vector is, v of t , so I told that in the break this, if I use this underscore, then I mean a vector, if I use it without this, so then it's the velocity, if I do not use this, then it's the speed, and on the slides a v underscore is just a fat v .

Now, this is, of course, x prime t .

I now write it as in coordinates instead of with these i, j, k, so I have these components, and then my speed, v of t , is, of course, the square root of x prime t squared plus y prime t squared plus z prime t squared.

But I know that's the, and this is always equal to 5 centimeters per second.

This is a constant, and I know that the speed along the z-axis is constant 3, so this z prime t is always 3.

So I get here that this is equal to 9. Now, if you then square up and bring the 9 to the other side, you will find that x squared t squared plus y prime t squared is equal to 25 minus 9 is 16.

Okay, now, I also know how this aren't is moving in such a spiral, but I do not know how fast it's moving. But I know, of course, that this r of t also has to be something like the following thing, because the speed is the same. I have the angular velocity, so where you go around, is also constant, because otherwise, if that would be different, my speed would, if that would change, my speed would also change. But I do not know what this angular velocity is. So, but I know, of course, the path is around it, so it should be something like 2 times the cosine of ωt to sine ωt , and then here c times t . But in the z direction, I know that z prime is equal to 3, so I know that this is equal to 3 times t to follow the path.

Now, I have to, I also differentiate this, and now, this is my x_t and this is my y_t , so what I now get is x prime t is equal to minus 2 ω cosine, no, sine ωt , and y prime t is 2 ω cosine ωt . I plug this into this equation, and then I get, this squared I get here, 4 ω squared times sine squared ωt plus 4 ω squared times cosine squared ωt is equal to 16.

So, I find here that ω^2 has to be equal to 4, so ω has to be 2 or minus 2, and that is not much of a difference, but it just describes in which direction it's going, whether this arm is going like this or like that. But that's, of course, with the same speed, but it's from a sort of different perspective.

Now, so ω is plus or minus 2, but we take, there's not much real of a difference, it's only this direction which it's going up, so take ω is equal to 2, and then we get for this that you get $2 \cos 2t$, $2 \sin 2t$, $3t$ as the position of the arm, and if at time t , and if you then want to know all these things that over there, it starts at 2, 0, 0 at time 0, and you can compute the velocity and the speed and so on. Speed is all the time 5.

You can check that over here. If you differentiate this, you get here this 4, and this 4 you get again, and a 3, and you get again square root of $4^2 + 3^2$ squared is the square root of 25 is 5, so everything works. So in this case, you simply have an this arm going around this part with a constant speed of 5, but then the velocity is time dependent, and also the acceleration will be time dependent.

Now, this looks like a very difficult slide, but these are some computational rules that you have to know for the differentiation of factors that depend on one variable, and let's say that's the time, and they look very much like the normal rules you have for a differentiation, but to prove them is a little bit more complicated. Here, I have two functions from $\mathbb{R}$ to $\mathbb{R}^3$. Let's put it like this, so this position factor u depending on t and the position factor v depending on t . If you now add up and then differentiate, that's the same as differentiating the first, differentiating the second, and then adding up. That is pretty obvious because what you get if you add up the first component, so if u of t is equal to $u_1 t$, $u_2 t$, and $u_3 t$, and v of t is $v_1 t$, $v_2 t$, $v_3 t$.

If you first add up, you add this up and then differentiate, but the derivative of $u_1 t + v_1 t$ is of course the derivative of u_1 plus the news here.

This is if you not have an addition, but you have λ of t is just a function from $\mathbb{R}$ to $\mathbb{R}$, and u of t is such a vector, and if you have done this product, this scalar product, you have to use the product rule.

In this case, you multiply one such a vector, everything with λt , so if you then look at the first components, then you get a first component, $\lambda' t$, $u_1 t + \lambda t$, $u_1' t$, that is of the first component of that product. In the second component, you get just the same. In the third component, you get just a similar thing, and then it turns out that you can connect, collect these terms in a different way, and you see this as the derivative of the λ times the vector u , that is by this part, plus λt times the derivative of u , and that is this part. The only thing that might be confusing is that I only do it for the first component, but I should do it for also for the second and the third component, but everything is the same, so you get a little bit tired of me, and also I get tired of myself, but I hope you understand you have to do it for all three components separately with every time it works in the same way.

Now something a little more tricky, if I have the inner product of two vectors, u and v , then that inner product will be just a $\mathbb{R}$, because the inner product of two vectors is a number. So if I take inner product of two vectors that both depend on t , I get a function, a scalar function depending on t . Now this is the formula that describes this, and that is also not difficult to understand, because what is this inner product of u , the stock product with v , that is just $u_1 t v_1 t + u_2 t v_2 t + u_3 t v_3 t$. And then you have to differentiate this.

Now what do you get then? Quite obviously I have to use the product rule, so I get

here as the derivative, so the ddt, is I get here u_1 prime times v_1 plus $u_1 t$ times v_1 prime.

Then I do the same thing for this term, so that's the u_2 prime $v_2 t$, and I get also $u_2 t$, this is v_2 prime t , and I have to also do it for the third component plus the derivative of u_3 times v_3 plus u_3 times the derivative of v_3 .

And I have to adopt these six components, but I've written them already in a nice form, because if you look at this part, you will see that this is just the inner product of u prime t dot product with vt plus, and here you get this u_1 , u_2 , u_3 , and v_1 prime, v_2 prime, v_3 prime, so this is the inner part of ut dot v prime t , you just have to add it up, and then you see that this is the same thing. Then we have for the cross product, it works the same, but you have to be careful, I will not do that, you can try this yourself, also you just have to compute this cross product, so you get also these terms with these products, so it's not very much different of the other one. There is a difference because you know that the cross product is not commutative, if you change the v and the u , you would get a minus sign, that means also in this situation, so you have to have this u stays at the first component across the derivative of v plus the derivative of u in the states at the first position across the v , and you cannot change these factors from one side to the other, because the cross product is not commutative, so you have to be careful, but it's easy to remember, you just keep your factors at the same spot, that is the situation, and you differentiate one, and you do not differentiate the other, and you add up. Okay, then this is a chain rule, if you have u of λ of t , you can differentiate it just with the speed factor, no this is no, this is the length of, no it's not the speed factor, this is the length of the factor ut to the power s , let's assume that s is an integer, if you differentiate that you get this expression s times the length of ut to the power s minus two times the inner part of ut with its derivative, now this looks very complicated, and I'm nevertheless going to do this on the blackboard somewhere, what I do here is the following, I write this in this way, I look at the length of the vector ut squared, and this raised to the power s over two, and now I'm differentiating with help of the, so this is just a scalar thing, if you look at these lengths, it's just a scalar thing, so what I now do is I differentiate with help of the chain rule, so this is s over two times this function to the power s over two minus one, simply if you differentiate the power, times the derivative of this function length of ut squared, which was respect to t , but that one we know, because if you look at this function ut is like this, everywhere put here this vector symbol, if you look at this, the ut squared, so note that the length of ut squared is simply the inner product of ut with itself, it's simply the inner product of it, because what was the length, it was the square root of the inner part of the factor with itself, now so if you differentiate this, you get, according to our rules, ut inner product with u prime t plus u prime t inner product with ut , but in this case, because you get two times the prime t , so if I now continue here, so I now go from here further, you get that this is equal to, I copy this, times two times ut , inner point of ut with u prime t , and then with this two, I multiply with this s over two, you get s , and then I get ut , this two I multiply here, so I get here s minus two, times the inner product of ut with u prime t , and that's exactly the formula that is on the slide, and the interesting case is, s is equal to one, so if s is one, so the d/dt of the length of a vector is now, s is one, one, so I get the length of ut to the power minus one, so I divide by the length of ut , and here I just get the inner part of ut with u prime t , so it's the inner part of u with its derivative divided by the length of u , it's a little bit tricky, now it is for general s , but s is one is easier, but s is one is not much easier than the general s , that's the point, and I hope that I show that. Okay, important thing here, show that the speed of a particle remains constant if the acceleration and the velocity factor, only if the effect are constant, that's now easy to see, so you want to know the speed of a particle, I now do this trick for the vector v , so for the velocity vector, and then I know the following thing, that the d/dt of the length

of the velocity vector, and that is the derivative simply of the speed, that is equal to this, but now with the u replaced by the v , so that is the v of t in a product with v prime t divided by the speed, and now if the speed is constant, the derivative has to be zero all the time, so this expression has to be zero, that means that this inner part of v with v prime, so this is the velocity factor, this acceleration factor, that the inner product has to be zero, so it means that the speed is constant, that is if and only if the inner product of the velocity factor and the acceleration factor is zero, and that means that the acceleration factor and the velocity factor are perpendicular, inner product of two factors is zero if they are perpendicular to each other, and this is exactly what you get in situations where you have for example circular movement, where you know that the velocity factor and the acceleration factor are perpendicular to each other, okay, now I cannot do this also, I've done section 12.1 I think completely, in section 12.3 they are talking about general parameterizations, and that is not much more difficult than what I did, but they are thinking then about these paths, so in my picture from Georgia, it was this gray spiral and here it has a red point that was moving on it, now if you just are interested in this path, this spiral, then you can also describe that as okay all the points that are passed by the red point, however that is not a unique parameterization of this spiral, because if I have this ant that is climbing my normal circular helix, it was with constant speed, but maybe I have another ant that is sometimes a little bit slow and then a little bit faster and so on, but if he climbs the same thing, he describes the same path, only he does that as a function of time, it's different, but if you're interested in what is the path that this ant travels, it's of course just the same, and that means in a way that if you just look at the path, then there are many different parameterizations to follow that path, that's just if you make a walking tour and everybody who's making the same walking tour will do that in a different amount of time and will be on a different place and will have a different parameterization, that is simple, but you all walk the same distance, so the length of such a path is the same.

Now in section 12.3 it is about this idea that if you have such a path, you can also describe it by thinking okay I just walk on this path by a particle goes through this path and I parameterize this as a function of time, but you don't have to do it as function of time, you can do it of anything because that time doesn't have to play the role of time, because the time for if I'm walking with four kilometers an hour my hiking tour or I do it with five kilometers an hour I make the same hiking tour but my parameterization will be different. So to have the same path you can have many different parameterizations and you know that for example, this example shows that if you look at the upper part of the circle, so the the part of the unit circle that is above the x -axis, you can describe it by this $\sin t$ i plus $\cos t$ j where t works from minus π over two to π over two and the other one is here a much more difficult one t times the square root y coordinate.

You can easily check that it's the same thing because if you take the square of this plus the square of that you will find that's equal to one. So these points are indeed on the circle and also on the upper part. So you just describe the same arc but in a different parameterization and what you have to think about is okay if I have such a parameterization so let's do it with the help of another parameter. So I have a function r of let's say τ that's not the t and I want to know and τ runs from an interval from a to b and I want to know the length of the path that is now traveled when you go from r a to r b not the distance between these vectors but the path that is traveled by doing so then you get the same thing then you have to integrate you have to compute the velocity vector that is v τ is r prime τ and you have to integrate the speed so the length of that velocity vector over the integral from a to b . So this is just the same part but with a different parameterization but the length will be the same. In this way if you have a parameterization of a curve you can compute its length and that is one of the main messages of section 12.3. Please read it yourself if necessary I will come back to

it next time.

So, good morning to all of you. Welcome to this first lecture of the course, mathematics two.

I first have to introduce myself. My name is Luke Harbats, and I'm working here in the department of mathematics and computer science. If you want to reach me or you're in the meta forum, fifth floor, room 106, and the easiest way to approach me is by either here in the lecture or by email, and then you have to use all my initials and my last name.

name at TUE. What are we going to do? First of all, I teach this course for the first time. Last year it was another professor, Mitra. I use his slides. The course didn't change, but the teacher changed. What are we going to do? In a way, we are going to combine the things you learned in calculus and the things you learned in math one. Calculus that was on integration differentiation mainly and about limits.

In linear algebra, you learned about how to work with lots of variables.

Maybe it's usually three. What we are going to do in this course is that we are going to extend this idea of how to do integration and differentiation calculus for functions of one variable to the situation where you have several variables.

The main emphasis will be on the three-dimensional case, but in principle, the theory works also in higher dimensions, but then it's much more difficult to understand what's going on. In the three-dimensional case, you have a sort of three-dimensional world where you can see what things are happening. I first want to tell you something about the organization of the course.

I've given that here. I want to discuss these three topics. What do you need for material? Now, you need the book of course, and then you need the study guide of the course. That is easily obtained. That's a PDF file, and you find that on the conference page of the course. I've made this study guide in such a way that in principle, I think that all the information you need is there. You have a sort of overview of every week, every week what are the educational activities, so the lectures, the instructions, and so on. There is a list of exercises that you have to make from the book. The situation is a bit complicated in this fourth quarter because we have many holidays.

Today we meet on Monday, but the next Monday that we meet will be, I think, in three weeks or so. So the schedule is rather irregular.

So normally we meet on the Monday morning, but it's cancelled, I think, three times, and we meet on the Thursday afternoon, and then we always start with a lecture. That gives in total 14 lectures, but I will take care that in the last week, so that is in the ninth week, that I want to be through all the material and that we can prepare for the... So there will be a lecture, but also as a preparation for the test. I have to use that last week because of all these holidays. This is the minimum of time I need to try to explain everything to you.

Now, after the lecture, there's usually an instruction, but there are in total only 10 instructions, and the reason for that is that three times there will be a study or classroom, so my co-lecturer, Rob Mestrom, will do three times after my lecture on Thursday, it's always on Thursday afternoon, he will give a special lecture on how to use the techniques that you learn in the mathematics part of the course in examples from physics and mainly, of course, of electrical engineering, so that you see how these things are applied. And I have to say that what we are doing in this course is really applied a lot everywhere in science, so it's in that sense an important subject. I will show you some pictures later on. So I think that's very useful, and then there is one Thursday afternoon, and that's on the 21st of May. There will be no instruction, no study or classroom, but we will have an

intermediate test. And in that way, I come also to the assessment of the course, so what are the tests and what do you have to do?

Now, it consists of three parts. First of all, there will be five electronic homework tests, and this works in the same way as you had for part one, so we use the same system, Zoejo, and approximately every two weeks there will be a test available on Zoejo, and it works in the, I think in the same way as in part one, I'm not completely sure, I do it in the following way. If I open the test, I will put an announcement on Canvas that you know it. There's also a deadline for the test, about two weeks, and you have to make a test within that period. But my main intention is that you just learn from the test, so it's not that much a test, it's more learning by doing.

You have to make exercises, and then during the test you cannot ask for help, you just have to make the exercises, but once you submit the test, then you get of course your grade, then you can also see what exercise you made wrong. At that moment, everything is open, so when you see, okay, that exercise was wrong, you can click on it, and then you can see how you should have done it, so you get an explanation how you can make a test, or you can make an exercise, and if you then are not satisfied with your grade, you can redo the test. Then you get in principle the same exercises, but with different numbers, and then I hope you learned how you should do that question, and then you can practice it once more. The grade is every time per test, so you have to redo the whole test, but the highest grade counts, so if you do it five times or five times, it counts.

That is the idea. So this is more a way of practicing than of testing, but nevertheless, 10% of your grade, the final grade, is determined by that electronic test, by these five.

Then there is an intermediate test, that's a serious one. That's halfway, and that is comparable in level to the final test, and that is the level that I wanted to do, and that test should be a sort of alarm bell, if necessary. Oh, this is going warm, or hopefully a green light, oh, I'm doing quite well, but that should be a warning. And then there is a final test that counts for 80%, so the other ones for 10% of the grade, but the final test you have to score at least a 5.0, otherwise you fail, and that final test is on the 22nd of June, so hopefully you pass, then you have a nice summertime, otherwise if you do not pass, there is in the middle of the holidays, on the 10th of August, a reset. You do not like it, I also do not like it, but it's the way how this university is organized. Okay, if you need information, I think you can find everything on the conference page of the course. Additionally, you will get, at least I hope that everything is now recorded, it should be. There will be UDR videos on the lectures of the course, because the room would have been too small, but there was plenty of seating room, so what I would suggest to the students who are at home, come to the lecture, because I think it's a much nice experience to study with your fellow students, then sitting at home and looking at the screen. Then I come also to the instruction part.

The instruction part of the course starts already today, so after these hours, I will put a schedule on the blackboard in the break, and everything is in one instruction group, there will be four different rooms. I think that instruction is one of the most important parts of the course, because mathematics you can only learn by doing it, so you have to practice, you have to make these exercises. I know there are a lot of students who make exercises using AI, that is not the idea, it is simply cheating yourself, because I do not give these exercises because I'm interested in the answers, I give these exercises for you to practice with the material and to learn it, so they are just a means of learning, the answers are not interesting, and in the homework if you make mistakes, that's also not a disaster or so, you just have to do it by learning, also by trying and also by making mistakes.

However, then you should learn from these mistakes, and for that these instructions are very important, so there there will be some exercises that will be made on the blackboard that you can see how everything works, which you can also ask a lot of questions, so the idea is that you work there on the exercises and at home, and if you have at home some problems or you don't know how to continue or you do not get the correct answer, just ask questions, so the instructors and the student assistants are there to help you, and it's no problem of making mistakes, but it's important to get a practice for doing so, and you know it from football, these players they play much better football than I do, but still they have to train every day, they still have to do it, and otherwise they also cannot perform well, and that is the same in a way for mathematics, you have to do it to learn it. Okay, now I think I told you now quite a lot about the organization, are there questions at this moment on the organization of the course?

If not, then we start with the contents, so log in again, oh of course I make a mistake, okay, a nice picture here, spiral, and you see here these lines, and this has something to do with our course, what we are, I already said a few words about it, we are going to look at functions of several variables, but not only functions that map let's say $\mathbb{R}^3$ to $\mathbb{R}$, so function from the three-dimensional space to the real numbers, but also a mapping from for example $\mathbb{R}^3$ to $\mathbb{R}^3$, and these things are called vector fields, and you have to imagine why it's called a vector field, you have a point in vector, and then you can think about that vector placed at that position, and that is then you have a sort of, if you try to imagine it, you could think oh there is a, the whole space is filled with arrows, and you see that quite often if you see these pictures from the weather forecast, they also see this wind direction everywhere, and that is the easiest example I think of a vector field of wind, how the wind is blowing, yeah, on the world, but in electric engineering it is very important, they play a really huge role, although you can see it because you have electric fields and magnetic fields, and the next course you will have, now next not first in the second year, will be on that subject, and they will use all the things we do here will be used in that course, so this is indispensable for if you do this electromagnetics course, and also if you want to go further in that direction, but there are also other things, so here is for example automotive example, you see here this car in this wind tunnel, and there you can of course think okay that is that also that same idea of a vector field, because you have this sort of wind blowing and it's how these streamlines, you want to study how these streamlines go to know how much resistance this car has when it drives, now other examples are for example in signal processing, which you can also think about very difficult situations, so this picture should show a kind of nuclear fusion reactor, and of course people are studying a lot about this, but the physics that's involved there requires a lot of mathematics, and the type of mathematics that is required there is typically the type of mathematics that we are starting to do in this course, of course that is much more difficult, but we can, so I would say everywhere in physics these functions of and vector fields play an important role, and what we are going to do is we are going to look at them in a mathematical way of course, but mainly about how to do differentiation and integration with them, and that is turns out to be also in these applications, mainly these integration turns out to be very important, because you want to know what fluxes are and how much, for example if you have a water stream, how much water goes to a certain, let's say a circle and or sort of bounded region, and the same thing, but then invisible for us holds them for electric and magnetic fields. Okay, now in principle many variables involved, so the theory works mathematically also in a more dimensional case, emphasis is on the three dimensional case, because that is what we do in this case, and now what we want to do is describing points in three dimensional space, and of course you know already how to do that, then I need three coordinate axes. I want to look a little bit in more detail to that, and you can see that on this picture. I assume for a moment that I have an orthonormal axis, so an orthonormal coordinate system, and I look here first at the Cartesian

one, and that's the one you're used to, that you say simply oh I'm going that many units in the x direction, that many units in the y direction, and that many units in the z direction, but why is this, it's important that these axes are perpendicular to each other, and it's also important that you choose the right orientation that you should keep in mind, so usually you see here in this picture this x on this side and y there, and then always the z axis is a positive direction upward.

Now the situation is like this, if you have this x here, and this y axis here, then what we choose is what's called the right-handed, and for you you look left, but it's my right hand, so let's look at like this, and this right hand determines the direction of your z axis, and what I do is I place my hand on the x axis, then I point with my fingers to the y axis like this, and then my thumb is coming out over here, so I have to do it with my right hand, and that means that the z, the positive z axis is now pointing into the classroom, so this is now the positive z axis, and if you do it like here on this picture, it's a little bit difficult, but I have to put my hand on this x axis, I have to point to the y axis, and then like this, and then my thumb is moving upward, so you have this right-handed rule size. A different way of thinking, I saw people already do this, that's also fine, I do not have a second stick, I take this triangle, you can also do it in this way, suppose this triangle is pointing in that direction, that's the z axis, and going to stand where they meet, and if I now take the shortest angle with my head from the x axis to the y axis, I turn to the left, I look now out of me, so you see me turning, you think I'm turning to the right, but I'm turning to the left because I have to look at, so if you stand there, and then my head is in the direction of the positive z axis, and there is a difference because if you put it the other way around, you get a negatively oriented coordinate system, and you get troubles later on with minus signs, so you have to do it this way, and then you get, and that comes, you know I think what cross products are, yes? Do you also know what this i, j, k are?

Okay, now then you know this is the unit vector in the x and the y in the z direction, and then you know that if you ditch the thing with the right hand rule, that if you take the cross product of i and j, you get k, $k \times y$ is j, and this is the last rule, but you also know then that the cross product is not commutative, that means if I switch the order of these vectors, then I do not get the same answer, and in particular you get that j cross y, that's not y cross product j, this is equal to minus k, so you should be careful with that.

Now, if I have these vectors, then it is possible to characterize a point in three dimensions of space, and what do I do? Suppose I have this point p, I first take the point in the xy plane that is close to it, so I project orthogonally down to the xy plane, so if I have this thing, my wiper, then I just drop it here on the floor, and then this point at my feet is the orthogonal projection of this point on the floor. Then I have this point q, and for that point q, I project an orthogonally on the x and the y-axis, just as you do in two-dimensional space, if you have a point in two-dimensional space. You also say, okay, I look at the point closest to it on the x-axis and the closest point on the y-axis, you do orthogonal projection, and then here you get your point x_0, y_0 . Now, the idea is now that you get then three coordinates, of course x, y, z as you used, and you can also think about this as a vector that points from the origin to that point, and in that picture you get that the vector can be written as x times the vector i plus y times the vector j plus z times the vector k, and then x, y, z are the coordinates. Later on we will look at different coordinate systems at the end of the lecture. Now, once we have this coordinate system, you also know what distances are, and the easiest way to do is with this picture, so I have the distance from the origin, I look again at this point and I want to know what the distance is from this point to the origin. Now, you just use rectangular triangles and Pythagoras rule, we projected this down here to this point q, and I first determined this distance s

from the origin to q , now that I know because that is simply the sides of this triangle here side x , here side y , so the length of s is the square root of x squared plus y squared, and then here I have again a rectangular triangle, this one with this one, and this is the other side of that triangle, so the length of that is r is the square root of s squared plus z squared, and then you see s squared is just x squared plus y squared, so you get that this is x squared plus y squared plus z squared, so distances from the origin you can immediately see from the coordinates. If you want to know what the distance is between two points, then you also get a formula that you know, I have here two points $x_1 y_1 z_1$ and $x_2 y_2 z_2$, here is a picture, so you want to know the distance from p_1 to p_2 , you look at this set plane, so you look in what direction do I have to go, this in the x direction, no this in the x direction, this in the y direction, and this in the z direction, and then you see that this is simply the difference of the coordinates squared, added up, and then taking a square root, and that is the Euclidean distance in three-dimensional space. Now some questions, very easy questions, I have now a z axis and a y axis, so here's my y is pointing in that direction, and the z is pointing in that direction, in what direction is the x axis pointing into the room or into the wall, into the blackboard, and y , so into the wall, who agrees, who thinks that's going into the wall, who's thinking that's going into the lecture room, now I think that's going into the wall, because if you have here this into the wall, and then, we don't do it right, set, y times z is x , for me it's difficult to do it the wrong way, I put this upside down, so I rotated everything, and then if I take this unit factor times that one, I get this direction, but this is the wrong one, because I first have to take the y and then the z , because I said, so the people who said it had to go into the wall, they are right, but sometimes it's difficult to rotate everything, and therefore it's much easier to work with the algebra, then you can keep your hands, okay, now the second question, the length of the vector of the vector that points from the origin to 3, 4, 12, that is of course easy to do, because then you have just to compute that distance, that is the square root of 3 squared plus 4 squared plus 12 squared, so that's the square root of 9 plus 16 plus 144, that's the square root of 169, and that's 13, so that is that well-known number in this situation, but the question is what angle does it make with the x -axis, so that you need to enter product, do you remember? Because I think you know this, you just know this, and I think I have the factor that points to my points, so I have this factor that points from the origin to p , and that is 3 times the vector y plus 4 times the vector j plus 12 times the vector k , and I have my x -axis, so I have the factor that points in the direction of the x -axis, that's just the i , now you can so call this factor v and this factor w , then you know that the inner product of v with w is equal to the length of v , that's denoted in this way, times the length of w times the cosine of the angle between them, now in this case the inner product of this factor 3, 4, 12 with the factor 1, 0, 0, that's clearly 3, the length of the vector v we just computed, the length of the vector w , yeah that's 1, so you know that the cosine of α is in this case 3 over 13, so α is the $\arccos$ of 3 over 13, so in this way you can also compute angles between vectors in the way you know, okay I want to go to equations in three-dimensional space and the first one, the first thing to look at is the situation that we know from linear algebra, and that is over here, that's why you have just a plane, so the equation for plane in three-dimensional space you know this, you have four numbers a , b , c and d , real numbers, and then a plane is described by the equation of the form, let's say 2 times x plus 5 times y minus 2 times z is 24, so I now used some numbers for these a , b , c and d , but it is just that case, that way, it is easy, so here you see the plane x plus y plus z is equal to 1, that's an easy one because there you know for sure that if one of these coordinates is 1 and the rest is 0, you are on the plane, so you know that this points 1 0 0 0 1 0 and 0 0 1 are on the plane, if you then, if you know that two points are on the plane, the line connecting them is also in the plane, so you have this line is in the plane, this line is in the plane and this line is in the plane, so you get such a plane that is looking like this, and if you can do the same here, if a , b and c are non-zero, all non-zero, then it's very easy to see that the point

d over a 0 0 , the point d over 0 , d over b 0 , and also for the set chord you can do the same thing, they are on that plane and then you can find that plane, if one of the coefficients is 0 , then the, in that direction you are the, it is free, so that is in this picture, this is the plane x is equal to y and this isn't, and people, students get often confused if they say the equation x is equal to y , and if I then ask what does that represent, then everybody says a line, and I say I don't know, depends on the situation, because yeah, where do I consider this equation x is y , in the two-dimensional situation in the plane, yeah, then you're right and it's just a line y is equal to x , but if I do it in R^3 , then it is this plane, because here I have this in the xy plane, this line is x , but the set is completely free, so if you have just an equation, so the same equation can describe several things, it depends in the way how you interpret it in 2D or in 3D or maybe even more difficult.

Okay, if you know that you have this equation a times x plus b times y plus c times z is d , then, and that this is important for this course, although you know this, but I repeat it, the normal vector of that plane, so the vector that is perpendicular to it, is then described by a times the vector i plus b times the vector j plus c times the vector k , so you can read off the normal vector simply by looking at these coefficients. Now, why is that the case?

If you look at this equation, a x plus, now, let's do it with an example, $2x$ plus $3y$ minus $5z$ is 24 , then you can think about it in the following way, this is the inner product of the vector $2, 3$ minus 5 with the vector x, y, z , and this should be equal to 24 . Now, I assume that I have two points in the plane, p_1 , that's x_1, y_1, z_1 , and p_2 , you can guess what it is. What do I then know? Both points are in the plane, so I know that the inner product of the normal vector with the vector x_1, y_1, z_1 is equal to 24 , but also, if I look at that factor with x_2, y_2, z_2 , is also equal to 24 , because that other point was also in there. Now, I look at the points, at the vector that points from p_1 to p_2 , and that is the vector x_2 minus x_1, y_2 minus y_1 , and z_2 minus z_1 .

Now, if I now subtract both equations here, you get here for three, that's the inner part of n with x_2 minus x_1, y_2 minus y_1 , and z_2 minus z_1 , that's simply this inner product minus that inner product, so that's 24 minus 24 , this is equal to 0 , and you see that this normal vector, that I claimed that was the normal vector, is indeed perpendicular to this vector that points in the plane, so that is one of these directional vectors of the plane, because it points from a point in the plane to another point in the plane. If you take two other points, it will be the same, so the factors that are in the plane are all perpendicular to that normal vector, so you get this information away for free.

Now, you already know quite a lot about this, because you did that in linear algebra, what is the normal vector to the plane that is, and it's a unit normal vector, so it has to have length one, so the vector of length one that is perpendicular to the plane, what would it be?

Then somebody gives me a normal vector of this plane, you will be able to see that here, so this is simply one, two, three, and what do I do if I don't have to make this vector length one?

You divide by its length, now the length of n is the square root of one plus two squared plus three squared is the square root of 14 , I hope, so I get that this new than the normalize factor, so $\tilde{n}$ is simply one of the square root of 14 times this one, two, three.

Okay, now, something else, yeah, it's called the geometry, but just what is the intersection, what is, well, it depicts between this plane, x plus two, y plus three, z is four, and the plane x plus y plus z is equal to one, what will it be?

A line, yeah, and do you also know how to compute it? Very good, you take the, there is, I will do it in two different ways, I just do it the first linear algebra way from math one, I have this equation $x + y + z = 1$, and the after equation $x + 2y + 3z = 4$, I just do row reduction, so I subtract the first row from the second, and I get zero, two minus one is one, three minus one is two, four minus one is three.

What I do next is I subtract the second row from the first one, and then you get this here, zero, one, two, three, and here I get a one, minus one plus one is zero, minus two plus one is minus one, and minus three plus one is minus two, and then you see that you have a pivot here, and a pivot here, I cleaned everything up, you have a non-pivot column going to the z-coordinate, so the z-coordinate is three, $z = 3$, z is equal to λ , and then you read off what the x is, $x - z = -2$, so $x = -2 + z$, so $x = -2 + \lambda$, and $y + 2z = 3$, so $y = 3 - 2\lambda$, and then you get the description x, y, z is $-2 + \lambda, 3 - 2\lambda, \lambda$, so that is the linear algebra way, that was your suggestion with cross products, that's also okay if you look at the normal vector of the first plane, that is one, one, one, and I look at the normal vector of the second plane, that is one, two, three, and now you have to think about the directional vector of the line, the line of intersection, and that line of intersection is in the plane, the first plane, so that directional vector is perpendicular to normal vector and one, but the same reasoning, it is also perpendicular to the normal vector of plane two, because it's to find the vector that is perpendicular on both, and one of the possibilities is doing them the cross product, so in this situation you get $n_1 \times n_2$ is, and now I do it a little bit in a stupid way, but I want to show you that it also works with these two j times three, okay, I have to do this cross product, and now I have to write out all the possible products, so never do this in this way, just use this trick with the determinant, but this is the way how you do these cross products, you get the i cross i plus i cross j plus three times i cross k , oh yeah I forgot the two, and then you get j cross i plus two times j cross j plus three times j cross k , and then the final one k cross i plus two times k cross j plus three times k cross k , you get a little bit sick and tired of this, but so you do that only once, now i cross i and j cross j and k cross k are just zero vectors, because they point in the same direction, and then you get that this is i cross j is k , i cross k that is minus j , because k cross i is equal to j , and then here I get j cross i that is minus k , here I get three times j cross k that is plus three i , k cross i is j , and k cross j is minus two i , and then what you get is three i minus two i is i , then minus three j plus j is minus two j , and I get here three i minus two i is plus k , minus k plus two k is plus k , so I get in these this one minus two one, one minus two one like this, just exactly the same thing as you're used to, okay, now I didn't look at a, you should have warned me, it is really time for a break, after break we will look at new things where you look at more difficult equations, and then it gets more interesting, so welcome back, let's start again, still a lot to do, one of your fellow students telling me on a very stupid mistake of mine in that has to be plus, so there is only you have this factor one one one, cross product one two three, and I did the computations okay but I wrote a cross instead of a plus and that was stupid, if I make stupid mistakes I do that all the time please warn me, um then something about instruction in the third and fourth hour, I made a scheme, so there are four groups one two three four, I uh depending on the the letter which for your last name to start you belong to group one two three or four, now if you want to be in the same room as your friends and it's impossible here, I don't mind if the if you're in a somewhat different group, I do mind that all the groups should be of approximately the same size, so do not take this as a complete law but if there is one group of 70 people and one of 10 we will change that because that's not the that's not the intention, on Monday so today all these groups are in atlas but on different floors, on Thursday this is the usual uh room setting, however this is not fixed, so there are sometimes changes, so please have

a look in the study guide I for every date I have written down what the correct rooms are, but usually it's this but it's not always this, so please check, um okay now let's continue with mathematics, you've seen what the plane is and now you can also uh see what the half space is, so in uh if you have a plane $a \leq b \leq c$ and d are just given numbers, so for example one times x plus two times y plus three times z is equal to four, if you now change the equality sign in a larger than or smaller than sign then you do not get a plane but what's called a half space and this is just an example y has to be larger than x then you have here in the $x \ y$ plane there's line you get that plane that is over here I would say and then uh the you have to be at the side where the y is larger than x so you have to be on this side, so if you get such an inequality where the equation looks like a plane you get something like this a half space and if you look at the intersection of half spaces for example here x larger than zero y larger than zero and z larger than zero of course know what that means then what they call that intersection the first octant so you know for the $x \ y$ plane you have these uh uh you have also one two three four if you know this is quadrants but now you have octants and uh the where all the uh so this is not a half space but this intersection of half spaces where all coordinates are positive they are called the first octant and then you have also the second one and the third one and so on uh but half space is not much more difficult than spaces uh we now get something more difficult and that is because it's a little bit technical and mathematical first why is this important it turns out to be important in mathematics and what you have to think about is just what you learned from the real line there you have these intervals and if you have the interval from two to five that's called an open interval because the boundaries do not count so you have to add the the number of the real numbers that satisfies the condition that x should be larger than two and smaller than five and this is an open interval and you also know what is closed then for example if you look at this set x in $\mathbb{R}$ such that x is larger equal to minus one and small equal to one so that is the interval from minus one to one but then with uh these brackets that you call a closed interval and it's very easy for intervals because you have simply this highest number and the smallest number and it is in that interval or not it does count or does not count and if they both these boundaries count then it's called a closed interval and if both these boundary points do not belong to it then it's an open interval and if you have this uh then it is not open and not closed this interval we want to have the same idea for sets in two and three dimensional space that is the problem because these notions of open and closed have important consequences later on also for mainly for differentiation and integration theory so we have to know what how you can generalize this idea of open and closed intervals to 2d and 3d sets now what you do for this is the following idea um we go to $\mathbb{R}^n$ so think n is two and three but if you want to do n is 100 be my guest and we know what the distance is because we use the euclidean distance now we look at this set $B_r(x)$ so x is a factor given r is a positive number given and then we look at all the points y in $\mathbb{R}^n$ such that the distance from y to x is smaller than r so if i put it in the my drawings in three-dimensional space are not very good but in two dimensions i can do this so you have a a vector x and you have a positive number r and then you look at all the points that are all distance less than r from that point x but really less than so this circle does not count it does not belong to that set $B_r(x)$ and if you do it in three dimensions of course you get in what's called an open ball you get a point and then all the points that are this is less than a half for example from that point and then you get three-dimensional object and it will just be a ball but the sphere where the point this is equal to a half does not belong to that that's the idea where this boundary does not belong to it or does not belong to it that's the idea and that's the same thing as for these intervals now this is the so i hope that you understand what such a ball is now what do you now call an open set an open set is an uh so a subset of let's say $\mathbb{R}^2$ $\mathbb{R}^3$ is called open if around any point x in that set you can find a positive r such that everything that is within that ball around x with radius r is in S and that generalizes this situation of open sets because if you look at this situation so of this open interval this one if you take there a point on that set does not matter

where let's say here then you can always find a little neighborhood of that point because these p up is called neighborhood such that it's completely contained in that interval this part is contained in the open interval in the whole interval of course it gets more difficult as more as i get to the boundary because then i have to make that r very small but there's always some space left in this situation now in such then you call it open so always if you have a point in the set then there's also is a little neighborhood of that point that's also still in the set that's the idea now what do you uh uh when do you say that a set is closed if it's complement so everything that is not in the set everything else except that set is open so why is uh this set from minus one to one closed you look at the complement that is everything that is smaller than strictly smaller than minus one and everything is strictly larger

than one and if you now take a point over there in one of these two parts you get of course that property that if you take such a point in its complement then there is this neighborhood if you take r small enough of that point that it still belongs to that complement so open means around every point in the set there is a neighborhood that's also in the set and closed means the complement is open now question to you i look at the set of all points in three-dimensional space such that x plus two y plus three z is large or equal to four who thinks that that is open who thinks that that is closed who thinks it's neither of them because some tests are not open and not closed now there there is a light majority for the right answer in this situation uh and it's often the case it's often the case not always but often the case that it's about strict and non-strict inequalities you have to look here at this uh non-strict inequality so in that set i look now at the complement of that set that is the set of so i look here so this is that's my set s as complement is the set of all points where x plus two y plus three z is smaller than four strictly smaller and now if you now look at that plane x plus two y plus three z is equal to four now i get a sort of half space but you have to be really strictly on one side of in this situation you always know so if you have a point here if p is an element of this complement and p is x zero y zero z zero then you know that x zero plus two y zero plus three z zero is really strictly smaller than four so it's not on that plane so it was positive distance to the plane what you do is you take you simply project that point on that plane octagonally then you compute the distance yeah you don't have to do this really but you do it in your in the reasoning so you do it in your head and say okay there's a positive distance i take half that distance that i take as r and then that's all these points that are are on less than a half of this distance from that plane they are all in that and in in that complement it does not work for that other set here that's uh because there if you i hope it's is it uh yes if you would start here on a point over here where the x zero plus two y zero plus three z zero is equal to four then around that point i cannot make a little neighborhood that remains in that set that's impossible so the points on that plane x plus two y plus three z is equal to four they are so-called boundary points if you take a point there and you make a little ball around it you can make it you can as small as you take it you always have points inside this set s and outside the set s in the complement you always have that you are on both sides of that plane because you're exactly in the plane so if you take a little ball around it you get always points on one side and on the other side these points where around where every neighborhood is in both parts in s and in this complement they are called boundary points and roughly speaking it is that in an open set these boundary points do not belong to the set and in closed sets they do belong to the set that's in a way how you but this is not completely correct but it's almost correct uh question to you this set x squared plus y squared smaller than one who says this is open who says this is closed luckily nobody who says this is neither of them also no no it's open because yeah here you get this non-struct inequality so you are inside let's say the unit disk so there's always a positive distance to that boundary and then you can make this r a little bit smaller than the distance and you can stay the points in a neighborhood of such a point are still in that open disk then this one you have here x squared plus y squared is smaller than one and y is larger equal to x who thinks that this is open

an open set who thinks that it is a closed set and who thinks it is neither open or closed yeah they luckily the majority so i always make the remark sets are no doors the door is open or it's closed but sets can be neither open nor closed and they can even be open and closed at the same time so if you look at the whole for example three-dimensional space that set is open because about every point yeah that's it's clear but it's also closed because the complement of the whole space is the empty set and the empty set has any property you want because there are no elements in that empty set so you can say all elements in the empty set are yellow but they're at the same time also purple that is true because there are no elements in the empty set so you can make any claim on it because it's simply you do not say anything in that way now in this situation you get that you have to everything within the unit circle so here's the x axis and here's the y axis y should be larger than x larger equal so you get this is y is equal to x and then you see that it is indeed not open not closed because it is not open because if you are on a point here any small neighborhood around that point will have points inside or outside that set so that set is of course this one but then if you look at the complement that's everything outside but the complement contains this half circle and if you take a point on that half circle of course you get points inside so in a neighborhood always contains also points inside this inside that circle and then you get outside the complement of s so it's not open and not closed uh later on in the xr in the exercise you get a lot of examples where you have to decide on this and reason with it now i have to look where i am okay now we get to the next subject and that is quadric surfaces and where we are this is of interest for a lot of situations what i what my goal is is that if you see an equation that is not too difficult that you can almost recognize what shape it describes so we know if we have in three-dimensional space and the equation of the form $x^2 + 2y - 4z = 7$ then we know okay this will be a plane and it will not go to the origin because otherwise then it should be is equal to zero and then you can think about the total vector and then you have some idea okay this looks like this but i want to that you also can recognize that in more difficult situations for the so-called quadric surfaces so this is in three-dimensional space and these capital letters a b and so on are just constants so we look at equations of this form we have this $g x^2 + h y^2 + i z^2$ so if this is not there these are just planes but now we also allow second order terms so we also allow $a x^2 + b y^2 + c z^2$ and also the mixtures $d x y + e x z + f y z$ but they are all of degree at most two and the question is if we have an equation like this what does it describe in three-dimensional space now of course you can think okay this is too difficult for us and let's throw away one variable so i throw away the z and i just look in the two-dimensional plane to this situation so what can be described in this situation $a x^2 + b y^2 + d x y + g x + h y + j = 0$ and what you then can show is that these are the so-called conic intersection sections so i hope you know what a cone is so a cone is so you take for example the point is fixed and i have to i will give it a try but this is fixed and now so this is the i point this in the along the z axis and then you have a fixed away with a fixed angle and you rotate that around that point and then you get a cone and that is simply the shape of your ice cream of the waffle of your ice cream that's a cone it's also called a cone so you get that so you go out like this but on the same way you also go you do the same thing in the other direction so you get also a cone there i never get that when i buy an ice cream that i get two if i ask for one but in a way they should give me two one negative one one negative ice cream um now what you what can be shown is that the the shapes you can get with expressions like this are the so-called conic intersections and that are you have here this cone and you uh intersect that with the plane and what can you get you can get a circle if this plane is horizontal you get an ellipse like this but you can also get a parabola then this shape is in such a way that it does not intersect my negative in this case my negative part of my cone but only the positive part so here you get a parabola you get an hyperbola that's also possible if it takes out a part of the positive cone of the negative cone like here this one and that one but you can also get a point you can also get

a line and you can also get cross lines these are called the conic intersections but the main the most important one are the these are the general situations but these are the usual on the circle $x^2 + y^2 = \text{constant}$ and ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the parabola and the hyperbola you will know usually hyperbola you get $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ now i want to look at the situation where this is where you look where you have to one more variable set what can i get in that way and how do you recognize that and then you get the so-called quadric surfaces i have a little bit more i take my chalk uh the first situation is this one and that's a degenerate situation so it's described by an equation of this form and that is of the form that i have given so you get a constant set $2x + 3y + 5z = 17$ and here you have another uh this should be another number this should not be a_1 this should be a_2, b_2, c_2 and here d_2 but this should also be a different number and then what you get you get two planes that are crossing that's what you get here because if this has to be zero you get a product has to be zero so one of the factors has

to be zero so you either have that $a_1 x + b_1 y + c_1 z = d_1$ or it is equal to d_2 in this case so you get two planes that are crossing in uh in this way uh and this picture here this really should be different um picture looks like this but this is the very unusual situation just as if you do a contact intersection you go exactly to the point of the the center of the cone that's also unlikely um now i look at the next ones but i make the auto maybe a little bit different okay no here i look at uh what's called a cylinder along the z-axis and the the thing is that i assume here that the z-coordinate is not involved so the z is not there so for example i get $x^2 + y^2 = 4$ you get a circle of radius two but we are looking at this in three-dimensional space and then we know from the beginning that's the positive z-axis is pointing into the room so we have here this cylinder that is pointing uh yeah yeah you're in the middle of it um so it's just this uh the shape of a cylinder but you have also different situations if you would have the situation $y = x^2$ this equation then in the you know of course this is a parabola x, y and then something like this but in three-dimensional space you have this third coordinate so you have this shape sort of uh yeah it's intersection is a parabola but it goes into the room that's also called this is called a parabolic cylinder and this is called a circular cylinder so a cylinder means just that one of these coordinates is completely free i've now chosen that the z but of course you can also do that along a different direction then one of these other directions is not involved you have also of course an hyperbolic cylinder then you have an equation where you have for example a shape like of an intersection like this and that goes into the room because the z-axis is uh uh uh z is free so that is what you have to think about the cylinders are not always the cylinders you think about the circle of cylinders but they may be of a different shape okay now we go to the uh next one and now all three of those are described and this is the risk but this is the cone and the cone is described by integration of the form let's say $z^2 = x^2 + y^2$ and if you take a and b one you get really a cone that is uh circular so where these intersections horizontal intersections with that cone are just circles and it just describes here if you take z so if you take for example z is equal to zero yeah x and y have to be equal to zero so you get that point and if uh z is for example two or minus two you get here $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 4$ you get if a and b are not equal you get an ellipse and otherwise you get a circle so this these cones do not have to be these nice cones you get for your ice cream maybe a little bit you have a little bit of different shape of an ellipse this one square is the sum of the two other squares and in general this is the general situation this does not have to be in the points around the points the origin here it is oriented this point is in the points $x=0, y=0, z=0$ so you can move these cones around simply by there you get an equation of the following form let's erase something so we had the cone so the cone was of the form $z^2 = x^2 + y^2$ and now

you get an ellipsoid and that is uh of the form uh $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ and if a , b and c are just one or at least at the same then you get a ball or not a ball you get a sphere because it's really is equal to one so you get you get not what's in the ball but you only get the surface of the ball and that's the sphere that's called a sphere so this is a sphere if a , b and c are the same if they are not the same this sphere is deformed a little bit so for example if uh b and c are one and a is two then you can the x coordinate will be between minus two and two and uh you're sure that the y and z are between minus one and one so then you have pulled it out so you get in what's called an ellipsoid which you can with this a and b you can tear and wall you can change the shape of that ellipsoid i will show you later uh i have a nice application of that often a nice uh thing how you can see what's happening um now we get the more difficult ones at least uh and it depends on all times i have these squares and it depends on a plus or minus sign what is happening um in this situation you have an uh what's called an elliptic paraboloid so that is and uh so paraboloids you have you have two cases one is elliptic and then you get an equation of the form z is $x^2/a^2 + y^2/b^2$ so if you take a negative z then you're lost because there will be no points there because $x^2/a^2 + y^2/b^2$ will be positive so negative z you have nothing if you now look at the situation that you only look in the x direction then you get z is x^2/a^2 you get something like this and in the other direction y^2/b^2 you get something you could also parabola like this but you see every time when the z is fixed you get an ellipse so you get a shape like this here i think a and b were the same well maybe not but you get sort of uh a parabola in one direction in the other direction and you get this elliptic four yeah there's these ellipses around it that is this form but of course the other situation is that uh uh this is hyperbolic and then you get z is $x^2/a^2 - y^2/b^2$ and then one of these power walls is like this but the other ones is like this goes down because you get a negative and then you get a shape like this which looks like a subtle point then there are also hyperboloids but before i you can experiment with that and also read that but otherwise i will do that next time but i want to show you the following thing um this is not um it the link to this application is also on the canvas page of the course it's not made by me but by our student assistants who support me also for the showy show part so it's not my and uh it's a very nice thing because uh it's uh shows you how the shape of such an equation is and uh the nice thing is that on the left hand side you can choose yourself what these uh um what these fellows are and at this point a , b and c are chosen to be one and uh uh the j the right hand side is also one so what happens if you now make this j a little bit larger then you blow up the sphere that's happening you see that immediately the nice thing is is that you can also change a little bit with these other parameters so for example if i uh take the g the yeah this term and i move that a little bit then you see that this ball is moving a little bit away and it's sometimes also it's also growing and the reason is if you have this $x^2/a^2 + y^2/b^2 + z^2/c^2 = 4$ and i replace that with $x^2/a^2 + y^2/b^2 + z^2/c^2 + 2x = 4$ because that's i ended a couple of times what you can do here is that you write this as $(x+1)^2/a^2 + y^2/b^2 + z^2/c^2 = 3$ so completely square $x^2/a^2 + 2x/a^2 + 1/a^2 + y^2/b^2 + z^2/c^2 = 4$ and then you bring this minus one to this side and you see that you get now a sphere with center minus one zero zero but radius is square root of five so you see the the center is moving and it's the radius is going up that's exactly what we saw so you can understand that this is happening now the interesting things occur if you are changing here the uh so if i go here to uh i think a is zero no uh i should pull this back to zero so and then you get your uh silent yeah if so now we can look through it and it's uh getting nicer if i make that uh that zero now a little bit negative then you get this you get this hyperboloid where you can also look through and if you want to know what's happening if a parameter is changing all the time you can also do it you can play around with it now it's every time going from one side to the other so this was a

large and larger ball and now it's it goes like this um now what my intention is for this uh quadric surfaces is that you recognize if you see an equation that is the goal that it's not just an equation but that you see a little bit okay this is a surface that looks a little bit like this that that you should know that's important um so in section 10.5 of the book you will find questions about that but on the uh conference page of the course you will also find this it's just a link to it you can just click on it and and and play with this yourself so where you see okay if i change a little bit with these uh play a little bit with these coefficients you can see what is happening in the picture and to get a better understanding what these parameters mean that's that's the idea so you can also play around a little bit with that um i did uh section 10.1 10.5 maybe your instructors will also want to do 10.6 that i also planned but i have to do that next time they also have to wait i hope to see you on thursday